

Design Documentation

Version 1.0

For

eMaap Platform

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Department of Consumer Affairs

Submitted by:



National Informatics Center

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1. Introduction

This design describes how the functional and non-functional requirements recorded in the Requirement Document, the preliminary user-oriented functional design recorded in the High –level technical Design concept/Alternative Document, and the preliminary data design documented in the Logical Data model transform into more technical design specification from which the system is built. It describes the design goals and considerations, provides an overview of the system-architecture, and describes the data designed associated with the system and operation scenarios. The high –level system design is decomposed into low-level design specification for each system component, including hardware, internal communications, and software, system integrity controls, and external interfaces, serves as input to the Detailed Design Review (DDR).

1.1 Purpose of Low-Level Design Documentation (LLD)

The Design document tracks the necessary information required to effectively define the architecture and system design in order to give development team guidance on the architecture of the system to be developed. Design document is incrementally and iteratively produced during the system development Life cycle, based on the particular circumstances of the Information Technology Project and the system development methodology used for developing the system. Its intended audience is the project manager, project team and development team.

2. General Overview and Design Guidelines

2.1 General Overview

The *eMaap* (Electronic Metrology Administration and Automation Portal) is a unified digital platform developed under the Legal Metrology Division of the Department of Consumer Affairs, aiming to streamline, digitize, and standardize the enforcement and administrative processes concerning weights and measures across India. The platform’s architecture is designed to automate end-to-end services including registration, licensing, verification, inspection, and grievance redressal, thereby replacing legacy manual processes with a robust, scalable, and secure digital ecosystem.

At a high level, eMaap enables users such as manufacturers, dealers, repairers, packers, importers, and legal officers to access and manage services across various legal metrology functions. The system integrates workflows for stakeholders at both the Centre and State levels, enabling real-time document verification, license processing, inspection tracking, and analytics. A major design goal is to ensure compliance with the Legal Metrology Act, 2009 and associated state-specific rules while aligning with the vision of the *Digital India* initiative.

The platform adopts a modular design that accommodates multiple functional modules—registration, license management, packaged commodity handling, verification and re-verification, enforcement, and analytics. The backend employs API-based integration with external systems such as UIDAI, GSTN, PAN verification services, Digi Locker, and state legal metrology portals for seamless interoperability and real-time validation.

A key architectural decision is the implementation of *Role-Based Access Control (RBAC)*, which ensures that system users—from applicants to inspectors to administrative reviewers—have appropriate permissions tied to their roles. The frontend is designed to be mobile-responsive and multilingual,

offering accessibility and ease of use across regions and user demographics. Workflow automation reduces turnaround time, minimizes human error, and enhances accountability by tracking every user action in an immutable audit trail.

From a software architecture standpoint, the system follows an N-tier architecture separating concerns across the presentation layer, application logic, and data access layers. The application also supports notification services (via SMS and email), configurable approval hierarchies, geo-tagged inspections, secure document storage, and real-time dashboard views.

2.2 Design Guideline

The LLD document and its underlying components follow these key design principles and guidelines:

- 1. Modularity and Reusability**
Each feature—such as user registration, document upload, fee payment, or license approval—is implemented as a discrete module. Common components like OTP verification, audit logging, and file uploads are designed to be reusable across modules.
- 2. Scalability and Performance**
The system must support high concurrency across 36 States and Union Territories. Design choices prioritize stateless REST APIs, database indexing, and asynchronous job handling (e.g., for notifications or report generation) to improve scalability.
- 3. Security and Compliance**
Adherence to MeitY and NIC standards for encryption, authentication (2FA), and data protection is ensured. All sensitive data is encrypted at rest and in transit. Authentication follows OTP and password-based validation, while authorization is handled via RBAC.
- 4. API-First Integration**
System interactions with external services such as Aadhaar, GST, and state metrology systems are handled via secure, version-controlled APIs with standard response structures and error codes.
- 5. User Experience (UX)**
All forms are dynamic, responsive, and context sensitive. Tooltips, real-time field validation, and draft-saving features help reduce user friction and improve data accuracy.
- 6. Data Traceability and Auditability**
Every transaction, application update, review, and communication are logged in a centralized audit trail. This enhances traceability and ensures accountability, particularly in enforcement-related modules.
- 7. Configurable Workflow Engine**
Approval and escalation workflows are not hardcoded but defined using configurable metadata. This allows administrative flexibility to accommodate state-specific rules without modifying core code.
- 8. Disaster Recovery and High Availability**
The system is designed to support backup and disaster recovery protocols with minimum RTO/RPO. Load balancing and failover strategies are implemented for high availability.

9. Accessibility and Inclusiveness

The platform supports Indian language scripts, assistive technology for differently abled users, and mobile device access to ensure inclusive participation across all demographics.

2.3 Design strategies and Approaches

1. Modular Architecture

- **Approach:** Develop each functional unit (registration, licensing, inspection, etc.) as an independent module.
- **Strategy:** Use a component-based design to isolate responsibilities and allow for independent development, testing, and deployment.
- **Benefit:** Enhances maintainability and allows easier scaling or updating of specific features.

2. Service-Oriented & API-First Design

- **Approach:** Every business process is exposed via RESTful APIs.
- **Strategy:** Use an API gateway to manage access, authentication, rate-limiting, and monitoring.
- **Benefit:** Enables integration with external systems (UIDAI, GSTN, state portals) and supports future microservices evolution.

3. Role-Based Access Control (RBAC)

- **Approach:** Define clear user roles (Admin, Inspector, Stakeholder, Legal Officer).
- **Strategy:** Implement access control logic in middleware or security service layer.
- **Benefit:** Ensures secure, permission-based interaction with modules, minimizing misuse and data exposure.

4. Data-Driven Workflow Configuration

- **Approach:** Use configuration tables or rule engines to define approval chains, form behaviors, and conditional logic.
- **Strategy:** Store process flows and user role actions in the database instead of hardcoding them.
- **Benefit:** Allows quick adjustments to workflows without needing source code changes, supporting state-specific rules.

5. Responsive & Accessible UI/UX

- **Approach:** Design mobile-friendly, accessible front-end interfaces.
- **Strategy:** Use responsive web design techniques, accessibility standards (WCAG 2.1), and progressive disclosure for complex forms.
- **Benefit:** Enhances usability across devices and improves citizen participation.

6. Security by Design

- **Approach:** Apply security controls at all system layers from the beginning.
- **Strategy:** Use HTTPS, data encryption, input sanitization, OTP-based authentication, and audit trails.
- **Benefit:** Prevents unauthorized access, tampering, and protects sensitive personal and business data.

7. Scalable Backend with Stateless Services

- **Approach:** Design backend services to be stateless and horizontally scalable.
- **Strategy:** Use load balancers, caching mechanisms (e.g., Redis), and queuing (e.g., RabbitMQ) for performance.
- **Benefit:** Supports high availability and seamless expansion as the user base grows across 36 States/UTs.

8. Cloud-Native Deployment Strategy

- **Approach:** Host the application in a cloud environment (e.g., NIC cloud, MeitY data center).
- **Strategy:** Use containerization (Docker), orchestration (Kubernetes), and automated CI/CD pipelines.
- **Benefit:** Enables resilience, faster updates, and efficient resource use.

9. Auditability and Traceability

- **Approach:** Record all user/system activities and decisions for legal and compliance purposes.
- **Strategy:** Implement structured logging and audit tables for critical actions (e.g., approvals, logins, file uploads).
- **Benefit:** Supports RTI queries, forensic audits, and improves transparency.

10. Test-Driven and Version-Controlled Development

- **Approach:** Use TDD practices and maintain version control for all APIs and components.
- **Strategy:** Implement automated testing (unit, integration, regression) and Git-based versioning with branching strategies.
- **Benefit:** Reduces defects and ensures quality and maintainability.

11. Disaster Recovery and High Availability

- **Approach:** Architect for resilience and failover.
- **Strategy:** Use database replication, backup strategies, and failover clusters.
- **Benefit:** Ensures data continuity and system availability during outages or failures.

12. Localization and Multilingual Support

- **Approach:** Make the platform usable in multiple Indian languages.
- **Strategy:** Externalize all UI text and notifications and use Unicode-compliant storage and rendering.
- **Benefit:** Enhances usability for non-English-speaking users and improves adoption at the grassroots level.

2.4 Assumption/Constraints/Risk

2.4.1 Assumptions

1. User Access Assumptions

- Users (stakeholders, inspectors, admins) will have access to smartphones, desktops, or laptops with an internet connection.
- Users will possess valid mobile numbers and email addresses for OTP-based authentication.

2. System Integration Assumptions

- All required external APIs (UIDAI, GSTN, PAN, state portals) will be available, functional, and responsive within agreed SLAs.
- States will cooperate in integrating their systems with the central eMaap platform.

3. Operational Assumptions

- All stakeholder departments (central and state) will allocate reviewers/approvers for license processing, inspection, and grievance handling.
- The hosting environment will provide scalable infrastructure to handle peak user loads.

4. Security and Compliance Assumptions

- All data will be encrypted in transit and at rest as per NIC/MeitY guidelines.
- The application will comply with the Legal Metrology Act, 2009 and future amendments.

5. Data Quality Assumptions

- Users will provide accurate and complete data/documents during application.
- Uploaded documents will be clear, legible, and in supported formats (PDF/JPG/PNG).

2.4.2 Constraints

The Design and implementation of the eMaap system are subject to several global limitations and conditions that impact software behavior, hardware compatibility, and integration workflows. These constraints affect the choices made during architecture, development, testing, and deployment phases.

Constraints can	Description	Impact
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Imposed		
Hardware or Software Environment	Must run on NIC Cloud with Linux OS, Docker containers, and Kubernetes.	Restricts technology stack; proprietary cloud-specific services cannot be used.
End User Environment	Users may access on older browsers, low-end mobile devices, or low-bandwidth networks.	UI must be lightweight, responsive, and optimized for slow connections.
Availability of Resources	Shared hosting environments on NIC Cloud may fluctuate based on national usage.	System must support auto-scaling and caching to handle load spikes.
Interoperability Requirements	Must integrate with UIDAI, GSTN, PAN, Digi Locker, and state portals using external APIs.	Requires strict API standardization, schema compliance, and version management.
Interface/Protocol Requirements	Communication must occur via HTTPS, using REST APIs with authentication tokens (JWT/API keys).	Increases complexity in security, session/token management, and endpoint validation.
Licensing Requirements	Only open-source tools with NIC/MeitY-compliant licenses (e.g., Apache 2.0, MIT) can be used.	Restricts use of commercial/proprietary libraries unless explicitly approved.
Data Repository Requirements	Data must reside within centralized NIC data centers with secure backup and redundancy.	Affects data design; must support indexing, geo-tagging, backup scheduling, and disaster recovery.
Security Requirements	Must comply with AES-256 encryption, OWASP Top 10, 2FA, and MeitY cybersecurity norms.	Adds encryption overhead, secure login/session handling, and stricter compliance checks.
Memory/Capacity Limitations	File uploads restricted to max 5MB; only PDF, JPEG, and PNG formats allowed.	Frontend and backend validations needed; affects upload, parsing, and storage logic.
Performance Requirements	System must ensure 99.5% uptime with response time under 3 seconds for key services.	Backend must be highly optimized using caching, indexing, and asynchronous workflows.
Network Communication	Users in remote areas may face poor connectivity.	Requires retry mechanisms, session timeout handling, and lightweight UI behavior.
Verification & Validation	Dual OTP verification (email/SMS), multi-level approval flows, and dynamic validations are required.	Increases workflow complexity and requires robust input validation and session tracking.
Quality Assurance Measures	SLA enforcement, audit trails, and user feedback must be built into all modules.	Mandatory logging, status monitoring, and real-time alert generation become integral to design.
Legal/Compliance Requirements	Must comply with Legal Metrology Act, 2009 and adapt to central/state legal procedures.	Requires dynamic workflow configurability, rule-based routing, and state-specific overrides.

2.4.3 Risk

1. Integration Failures

- APIs of external systems (UIDAI, PAN, GSTN, state systems) may fail, causing delays in verification or approvals.

2. User Adoption Risk

- Users unfamiliar with digital systems may struggle to navigate forms, leading to incomplete or incorrect submissions.

3. Data Inconsistency

- Inconsistent data entered by users may lead to errors in license generation or processing.

4. Infrastructure and Load Risk

- During peak application seasons or pan-India rollout, system may face performance bottlenecks or downtime.

5. Security Breaches

- Unauthorized access attempts, phishing, or data leaks can compromise user trust and system integrity if not mitigated by strong controls.

6. Regulatory Changes

- Amendments in the Legal Metrology Act or state-specific rules may require major changes to workflows or modules.

7. Dependency on State Participation

- Lack of cooperation or delays by some states in onboarding their workflows may affect end-to-end service delivery.

8. Training and Change Management Risk

- Without proper user training, department staff may not fully utilize the features or follow new digital workflows properly.

3. Design Consideration

Heading	Description	Impact
Inter-State and Central Integration	The system integrates workflows of 36 States/UTs and the Central Legal Metrology Division through API-based architecture.	Requires standardized API interfaces, version control, and configurable support for diverse state-level rules and workflows.
Role-Based Access and Security	Access is granted based on predefined user roles (e.g., Admin, Inspector, Stakeholder), secured by OTP and	Role-Based Access Control (RBAC) must be implemented, along with secure session handling and user credential

	document verification.	management.
Regulatory and Legal Compliance	The system must adhere to Legal Metrology Act, IT Act, GST guidelines, UIDAI standards, and e-signature norms.	Incorporate secure data handling, encryption, audit trail, and integration with Digi Locker, PAN, and Aadhaar systems.
Document Upload and Management	Users need to upload mandatory documents (e.g., Trade License, Authorization Letters) in specified formats and sizes.	Design should include secure cloud storage, format and size validation, re-upload options, and linkage with user records.
High Volume Usage Scalability	Nationwide platform usage may result in high-volume simultaneous transactions, especially during peak periods.	Backend architecture must support horizontal scalability, caching mechanisms, and asynchronous processing queues.
Dynamic Form Rendering	Form fields and workflow logic vary based on the user type and selected service (e.g., License, Nomination, Inspection).	UI must support dynamic field rendering, form metadata mapping, and configurable validation rules for maintainability.
Audit Trail and System Logging	All user actions and system events must be traceable for compliance, monitoring, and dispute resolution.	Design must include a centralized logging framework capturing user ID, timestamp, event type, and status.
Notification and Alert System	Users should receive alerts (email/SMS) at each key stage—registration, submission, approval, etc.	Integrate with robust email/SMS gateways, ensure retry logic, and maintain logs for each delivery attempt.
Data Validation and Input Accuracy	Inputs such as PAN, GSTIN, Aadhaar, email, and mobile numbers require strict format and content validation.	Centralized validation service must enforce input formats using regular expressions and logic rules before submission.
Geo-Tagging and Mobile Support	Inspections and field-level activities must be geo-tagged and time-stamped with device-level data.	Mobile modules must capture GPS coordinates and upload metadata in real time to the backend securely.

3.1 Goals and Guidelines

Goals:

- Ensure **clarity and consistency** in the design by using uniform terminology and formats across all modules.
- Maintain a **modular architecture** to promote reusability of components like authentication, notification, and document management.
- Support **performance and scalability** through asynchronous processing, caching, and horizontal scaling capabilities.
- Comply with all applicable **regulatory and legal requirements**, including the Legal Metrology Act, IT Act, UIDAI, GSTN, and digital signature standards.
- Design for **maintainability and extensibility**, allowing easy configuration of new states, workflows, or user roles.
- Enable **interoperability** through well-defined APIs to integrate with state portals, UIDAI, GST, Digi Locker, and more.
- Ensure **reliability and resilience** using retry mechanisms, fallback strategies, and circuit breakers for external dependencies.
- Deliver a **user-friendly experience** via dynamic UIs, tooltips, multi-language support, and responsive design (mobile/web).
- Promote **testability** by isolating components and services for unit, integration, and user acceptance testing (UAT).
- Guarantee **auditability** by logging all critical user actions, data changes, and system activities securely and tamper-proof.

Guidelines:

- **Architecture**
 - Use a layered architecture (Presentation, Business, Data Access, Integration).
 - Separate configuration and environment variables from code.
- **Naming & Standards**
 - Adopt consistent naming conventions for services, classes, APIs, database objects, and configurations.
 - Use versioning for APIs (e.g., /Api/v1/licenses).
- **Security**
 - Enforce OAuth2/JWT authentication for APIs.
 - Encrypt sensitive data at rest and in transit.
 - Implement Role-Based Access Control (RBAC).
 - Sanitize and validate all user inputs to prevent injection attacks.
- **Error Handling & Logging**
 - Implement global exception handling.
 - Use standardized error responses with codes and messages.
 - Include correlation IDs for request tracking.
 - Log actions at appropriate levels (ERROR, INFO, DEBUG) and store securely.
- **Document Handling**
 - Allow only secure file types (.pdf, .jpeg, .png) and limit size (≤5MB).
 - Store files in structured folders by user and type.
 - Link document metadata with application records.

- **Notifications**
 - Use a central notification service with support for SMS, email, and future channels.
 - Include delivery logging, template support, and retry logic.
- **Database**
 - Enforce referential integrity, use indexes on frequent search fields.
 - Partition high-volume tables such as logs or audits if needed.
- **Dynamic UI**
 - Use metadata-driven forms and workflows for different roles and license types.
 - Provide field-level validations and tooltips to assist users.
- **Geo-Tagging & Mobile**
 - Capture GPS coordinates during inspections with time and device metadata.
 - Ensure consent-based location access in mobile apps.
- **Testing**
 - Implement unit, integration, regression, and load testing.
 - Ensure all API contracts are testable and documented.
- **Documentation**
 - Maintain API documentation (e.g., Swagger/Open API).
 - Include data dictionaries, system diagrams, and deployment/configuration guides.

3.2 Development methods and contingencies

Development Methods:

- **Technology Stack**
 - **Backend:** Java/J2EE (Spring Boot framework) for service-layer development.
 - **Frontend:** Angular/React for dynamic and role-based UI rendering.
 - **Database:** PostgreSQL or Oracle for RDBMS, optimized with indexes and partitioning.
 - **Mobile:** Android (Kotlin/Java) using REST APIs for field inspection apps with geo-tagging support.
- **Design & Modeling Standards**
 - UML (Unified Modeling Language): For class diagrams, activity diagrams, and sequence diagrams to represent workflows and modules.
 - Entity-Relationship Diagrams (ERD): For database schema design.
 - BPMN (Business Process Model and Notation): For modeling business workflows and approval hierarchies.
 - RESTful APIs with OpenAPI/Swagger: For defining service contracts and ensuring interoperability with state systems and external APIs (GSTN, UIDAI, Digi Locker, etc.).
- **Document & Data Standards**
 - XML/JSON: For data interchange formats, especially in API communication.
 - PDF/JPEG/PNG: Standard formats for document upload and download (identity proofs, licenses, inspection reports).
 - ISO 8601: Standard for date/time representation.
- **Security Protocols**
 - **OAuth2/JWT:** For secure authentication and authorization.

- **HTTPS/TLS:** For secure data transmission.
- **AES Encryption:** For sensitive data at rest (e.g., Aadhaar, PAN).
- **DevOps and Deployment**
 - **CI/CD Pipelines:** Using Jenkins/GitLab CI for automated testing and deployment.
 - **Containerization:** With Docker and orchestration via Kubernetes.
 - **Cloud Deployment:** NIC Cloud/MeghRaj for scalable and secure hosting.

Contingencies:

To address potential risks during development and ensure design resilience, the following contingencies are planned:

- **API Downtime/Latency:** Implement retry mechanisms, caching for reference data (e.g., LGD codes), and fallback UIs for offline states.
- **High Load Conditions:** Enable horizontal scaling of services and use load balancers for optimal resource distribution.
- **Security Breaches:** Enforce data encryption, access audits, and vulnerability scanning using OWASP guidelines.
- **State-Specific Rule Conflicts:** Use configuration-driven workflows and metadata files to adapt the system without code-level changes.
- **Integration Failures:** Maintain asynchronous queues for external APIs and isolate third-party calls through abstraction layers.
- **Document Upload Failures:** Provide retry options, real-time format checks, and support for re-upload with version control.
- **Mobile GPS/Connectivity Issues:** Allow local storage and deferred syncing for geo-tagged inspections when internet connectivity is restored.

3.3 Architectural strategies

The architectural strategy for the **eMaap (Electronic Metrology Administration and Automation Portal)** platform is guided by the objectives of scalability, interoperability, security, and regulatory compliance. Given the critical role that legal metrology plays in ensuring fairness and accuracy in trade, the eMaap portal is designed to digitize, streamline, and unify regulatory processes across all Indian states and union territories. The architectural approach not only reflects current technological standards but also allows the system to adapt to future needs and administrative changes.

Below are the key architectural strategies that define the design and structural principles of the system:

1. Technology Stack and Product Choices

The core application is being developed using **Java/J2EE** architecture on the backend, which ensures platform independence, high performance, and ease of integration with government digital ecosystems. The frontend employs **HTML5, CSS3, JavaScript**, and modern frontend frameworks such as **Angular** or **React** to deliver a responsive, dynamic, and mobile-friendly user experience.

For data persistence, relational database systems such as **PostgreSQL** or **MySQL** are used to store structured information on users, applications, licenses, and inspection records. The database schema is designed with normalization and indexing strategies to support efficient querying and high-volume transactions.

2. Modular and Reusable Design

The platform follows a **modular architecture** that decouples functionalities into independent components or services—registration, license issuance, inspections, approvals, enforcement, and dashboards. This modularity enhances maintainability, allows component reuse, and simplifies future upgrades.

The system reuses existing government services and repositories such as:

- **UIDAI** for Aadhaar-based identity verification
- **PAN and GSTN** verification via APIs
- **Digi Locker** for secure document storage and retrieval
- **e-Sign services** for digital authentication
- **State Metrology Systems** via API-based integration

By reusing these trusted and already-deployed components, the system reduces development time, minimizes redundancy, and leverages verified data sources.

3. API-Driven Architecture and Extensibility

One of the cornerstones of the eMaap architecture is its **API-first design**, which allows seamless integration with state-level portals, national ID verification services, tax databases, and other third-party systems. APIs are exposed and consumed over secure HTTPS channels, with request-response standards (typically REST with JSON) that ensure consistent communication across modules and organizations.

This strategy enables:

- Real-time data validation and synchronization
- Cross-jurisdictional updates and approvals
- Future integration with mobile apps and third-party portals
- Easy extension of features (e.g., QR code-based certificate validation)

The API-driven approach also positions eMaap for **microservices-based decomposition** in the future, should the system scale further in complexity and size.

4. User Interface and Interaction Paradigm

The user interface is role-based and context-aware. The architecture supports **Role-Based Access Control (RBAC)** to ensure that each user sees only the modules and data relevant to their responsibilities—be it applicants, inspectors, administrative officers, or controllers.

The UI is designed to:

- Be **multilingual** to support pan-India accessibility
- Support **real-time feedback and validation**
- Offer **mobile responsiveness** for field officers accessing the platform via tablets or smartphones
- Include **tooltips and helper texts** to assist users in completing forms and understanding regulatory terms

Dashboards are personalized to present meaningful analytics, pending tasks, alerts, and compliance metrics based on the user's role.

5. Security, Authentication, and Error Management

Security is integral to the system's architecture, given its role in handling legal, commercial, and personally identifiable data. Key mechanisms include:

- **Two-Factor Authentication (2FA)** using OTPs for mobile and email
- **Encryption of data in transit and at rest**
- **Authentication via government-trusted identity providers (UIDAI, PAN, GSTN)**
- **Session management and timeout policies** to prevent unauthorized access
- **Audit trails** capturing every user activity and transaction for compliance and forensic tracking

Error detection and recovery strategies are embedded across workflows. Examples include:

- OTP failures triggering retry mechanisms
- Document upload errors with size/type validations and fallback messages
- System notifications and alerts for incomplete submissions or incorrect inputs
- Graceful handling of network outages with retry logic and user alerts

6. Data Management and External Databases

The system uses a **centralized relational database** schema hosted on secure infrastructure (potentially NIC cloud or MeitY data centers). Each transaction—from registration to license issuance—is stored with audit metadata including timestamps, user ID, and application status.

Integration with **external databases** ensures data integrity and reduces manual verification effort:

- **Aadhaar** for identity matching
- **PAN database (CBDT)** for tax identity verification
- **GSTN APIs** for validating business credentials
- **State LM databases** for cross-verification and updates

Data retention policies and data access governance models are implemented in line with **IT Act, 2000** and MeitY's data governance guidelines.

7. Concurrency, Synchronization, and Workload Management

The eMaap system is designed to handle high concurrency across multiple states, users, and services. Mechanisms to ensure this include:

- **Connection pooling and thread-safe operations** in backend services
- **Token-based session management**
- **Queue-based workflows** (e.g., pending inspections or license reviews) to ensure orderly execution and load distribution
- **Database-level locking and transaction isolation** to prevent race conditions or data corruption

All actions—from login to application review—are timestamped and stored with atomicity to ensure accurate reporting and auditing.

8. Communication and Notification Mechanisms

Communication between system components and users occurs through:

- **Email and SMS gateways** for OTPs, reminders, alerts
- **Push notifications** on dashboards for internal users
- **System logs and MIS dashboards** for activity tracking and system health

A notification layer is built as a separate module to ensure independent management of delivery schedules, failure retries, and user preferences.

9. Resource and Memory Management

Uploaded files (PDF, JPEG, PNG) are stored in a **cloud-based directory structure**, organized by user ID and document type. This helps in managing access control and retrieval efficiently.

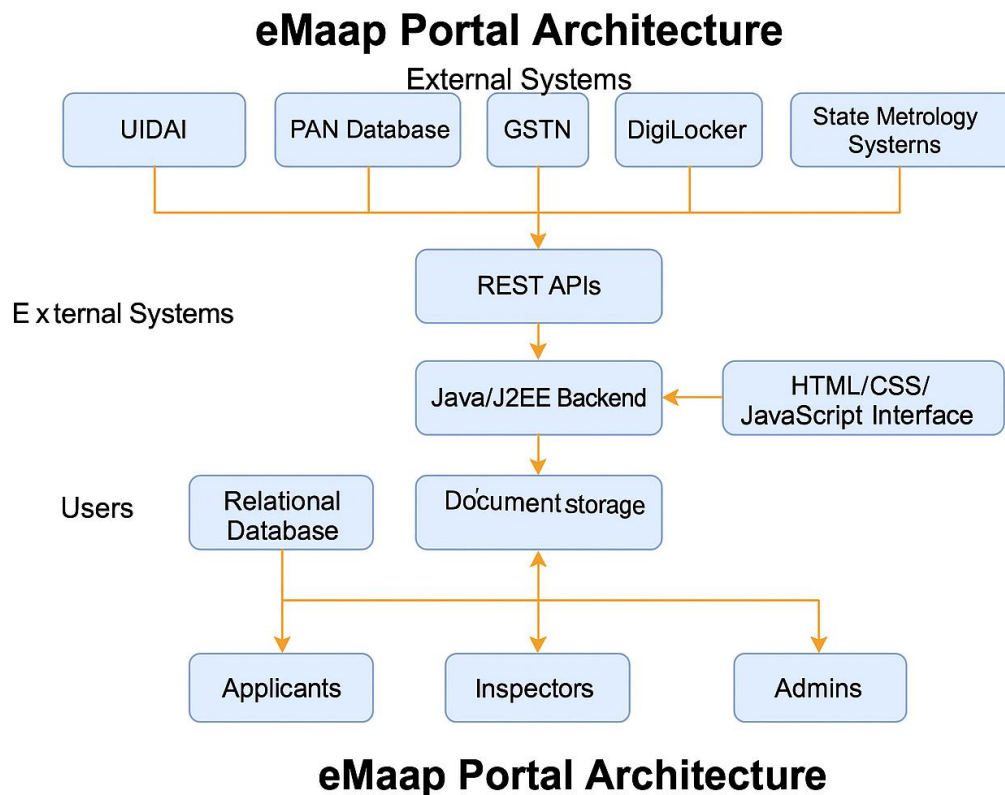
Memory-intensive operations, such as bulk data uploads, license generation, and certificate rendering, are offloaded using **background job processors** to avoid blocking the main UI thread.

10. Futureproofing and Innovation Readiness

The architecture is future proof by design and supports:

- **AI/ML integration** for predictive analytics and fraud detection in license applications
- **Blockchain support** for immutable and verifiable certificates
- **Geospatial APIs** for real-time tracking of field inspections
- **Mobile app extension** for offline inspection data entry with sync capabilities

The system aligns with **Digital India** principles and supports scalability both horizontally (more states, users) and vertically (more services, deeper analytics).



3.4 Performance Engineering

- The performance engineering process for the eMaap platform ensures the system delivers a reliable, responsive, and scalable experience across India's Centre and State-level operations. The primary objective is to validate that the portal supports a high volume of concurrent users, real-time workflows, and large-scale document processing without latency or failures. The scope of this process includes frontend responsiveness, backend workflow execution, API integrations (e.g., UIDAI, GSTIN, PAN), document uploads, and certificate generation.
- Performance testing will be implemented at various stages of the SDLC, beginning with requirement analysis, continuing through design, and actively conducted during testing and deployment. Core practices include load testing (to simulate 10,000+ users), stress testing (to evaluate system limits), and scalability testing (to ensure proper autoscaling). Key tools like JMeter, Locust, Prometheus, and Grafana will be used to execute these evaluations.
- Critical system metrics will be monitored, including response times (target ≤ 2 seconds), API latencies (≤ 500 ms), database query times, certificate rendering speed, and system uptime ($\geq 99.9\%$). Asynchronous queuing and caching (e.g., Redis, CDN) will be used to optimize performance under high traffic. Failures such as network outages, high CPU/memory usage, or external API downtimes will be mitigated using circuit breakers, fallback caching, and alert systems.

- This process aligns with MeitY's digital governance guidelines and is essential for delivering a seamless and transparent public service platform in line with the objectives of the Digital India initiative.

Category	Details
Objective	Ensure eMaap performance under varying loads, real-time use, and outages
Scope	Covers frontend, backend, APIs, workflows, DB, inspection, enforcement
Peak Load Target	10,000+ concurrent users
Key Tests	Load, Stress, Scalability, Endurance Testing
Response Time Target	≤ 2 seconds
API Latency	≤ 500 ms (Aadhaar, PAN, GSTIN, State APIs)
Certificate Gen. Time	≤ 7 seconds
Uptime Requirement	≥ 99.9%
Database Performance	≤ 300 ms for 90% of queries
Caching Strategy	Redis for sessions; CDN for static content
Queueing Mechanism	RabbitMQ / Kafka for async jobs (logs, PDFs)
Monitoring Tools	Prometheus, Grafana, ELK Stack
Testing Tools	JMeter, Locust, Postman/Newman, Apache Benchmark
Error Handling	Circuit breakers, retries, fallback cache
Compliance Frameworks	MeitY e-Gov standards, Digital India guidelines
Logging & Alerting	Role-based alerting; logs retained per compliance & SLA policies

4. System Architecture and Architecture Design

The eMaap portal is designed as a modular, API-integrated, multi-tier, cloud-ready digital governance system. It serves as a centralized platform for managing Legal Metrology services across the Centre and all 36 States/UTs, ensuring regulatory compliance, seamless integration, and scalable deployment.

Architectural Highlights:

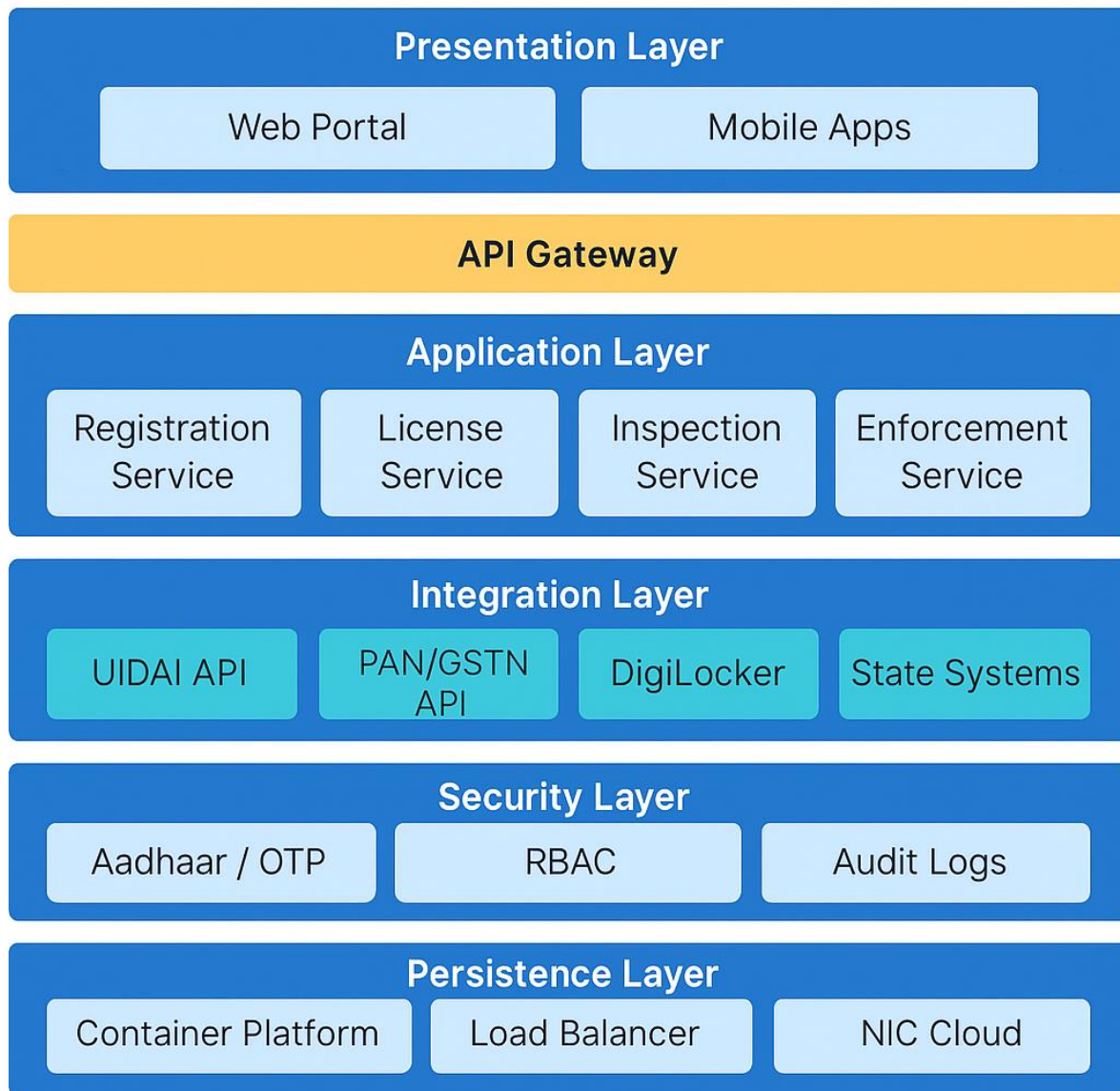
- **Modular Microservices Architecture** – Functional domains (e.g., licensing, registration, inspection, enforcement, dashboard) are encapsulated as independent services.
- **API Gateway Layer** – Facilitates communication between the frontend and backend services and supports integration with third-party/state systems.
- **Centralized Database Layer** – Holds all transactional and master data, with access governed via RBAC and encryption.
- **Cloud-Native Infrastructure** – Supports horizontal scaling, containerization (e.g., Docker), and orchestration (e.g., Kubernetes).

- **Security Layer** – Implements Aadhaar-based eKYC, PAN/GST validation, OTP, eSign, HTTPS, RBAC, and audit logs.
- **Notification Layer** – Integrated with email/SMS gateways for alerts, reminders, and acknowledgments.
- **Mobile App Support** – Parallel Android/iOS apps for inspectors and stakeholders (field inspections, status checks, uploads).

Architecture Design (Components and Interactions):

Layer	Components
Presentation Layer	Angular/React-based web portal and mobile apps; multilingual UI
Application Layer	Business logic for modules like registration, license, inspection, enforcement
Integration Layer	REST APIs with UIDAI (Aadhaar), PAN, GSTN, State LM Portals, Digi Locker, etc.
Security Layer	Role-based access control (RBAC), SSL, encryption, token-based session control
Persistence Layer	Central database (PostgreSQL / Oracle), document store (PDFs, scans)
Notification Layer	SMS gateway, Email engine, Webhooks
Reporting Layer	MIS dashboards, compliance reports, inspection logs, certificate status
Infrastructure Layer	NIC Cloud (Meghaj), Containerized environment (Docker/K8s), Load balancer

Below is the layered architecture diagram representing the eMaap system:



Data Flow & Integration Points:

1. **User Registers** on the portal → OTP sent via SMS/Email → Aadhaar/PAN/GSTN validated → User ID created.
2. **Application Submitted** (e.g., License, Model Approval) → Document validation → Payment Gateway.
3. **Workflow Engine** routes it to correct hierarchy (LMO → ZO → AC → Controller).
4. **Inspection Assignment** → Mobile-enabled field inspection → Geo-tag & upload inspection report.

5. **Certificate Generation** → eSign → Blockchain-backed record → QR code generated.
6. **Real-Time API Integration** → Status reflected in Central & State LM dashboards.

Third-Party Integrations:

System	Purpose
UIDAI	Aadhaar authentication for user verification
PAN/GSTN (CBDT/GST Network)	Tax identity verification of business entities
State LM Portals (36 APIs)	Bi-directional sync of registration/licensing data
Digi Locker	Secure document retrieval and storage
e-Sign	Digital signature of certificates and applications
Geo-tagging API	Pinpoint location of inspection or business premises
SMS/Email Gateway	Status and workflow communication to stakeholders

Deployment Model:

- **Cloud-Hosted:** Deployed on NIC's MeghRaj platform or State Data Centers.
- **High Availability:** Load-balanced across multiple nodes with failover.
- **Scalability:** Auto-scale containers during peak license/renewal periods.
- **Monitoring:** Prometheus and Grafana dashboards for infrastructure and application KPIs.
- **Backup:** Daily incremental and weekly full database/document backups.

Security Architecture:

- HTTPS across all endpoints
- OAuth 2.0 / JWT token-based session handling
- AES-encrypted document storage
- Audit trail with immutable logs
- Time-stamped approval chains

4.1 Logical View

The Logical View of the eMaap (Electronic Metrology Administration and Automation Portal) system represents the functional decomposition of the application into logically grouped components that deliver services aligned with the Legal Metrology Act. It provides a blueprint for the functional modules and their interactions, ensuring clarity for stakeholders and guiding the development and integration processes.

Core Functional Modules:

Module Name	Description
User Registration & Authentication	Handles new user onboarding with role-specific forms, OTP-based email/mobile verification, and login setup. Supports Admin, Inspector, Stakeholder roles.
Role-Based Access Control (RBAC)	Dynamically controls access to UI and backend features based on user type (Admin, Inspector, Stakeholder, Legal Officer).
License Management System	Manages applications for issuance, renewal, amendment, and duplication of licenses for Manufacturers, Dealers, and Repairers.
Document Verification Engine	Validates and securely stores uploaded documents (e.g., PAN, GSTIN, Trade Licenses) with checks on format, size, and content.
Approval Workflow Engine	Routes applications through hierarchical roles (e.g., Inspector → Assistant Controller → Controller), with state-specific logic.
Inspection Scheduling & Tracking	Facilitates assignment of inspection officers, inspection report submissions, and geo-tagged evidence uploads.
Certificate Management	Issues verifiable digital certificates post-approval. Certificates carry blockchain or QR-enabled verification mechanisms.
Package Commodities Registration	Allows manufacturers, packers, or importers to apply for and receive approval for pre-packaged commodities.
Model Approval System	Handles end-to-end workflow from model submission to RRSL forwarding and certification for weights/measuring devices.
Importer Certificate Workflow	Allows businesses to register as importers of weights and measures, with approval processes involving state and central officers.
Directorate Nomination	Allows companies to nominate legal representatives/directors under LM compliance norms.
Enforcement & Offense Tracking	Enables LM Inspectors to record field inspection findings, seizures, notices, penalties, and prosecution stages.
Notifications & Alerts System	Sends system-generated messages via email/SMS to users and officials throughout the lifecycle of application processing.
Audit Trail & History Management	Captures timestamped logs of all user and system activities for traceability and compliance with IT Act norms.
Analytics & Reporting Dashboards	AI-driven dashboards display service metrics, inspection coverage, pending queries, and compliance reports.
API Integration Layer	Interfaces with UIDAI, PAN, GSTN, state LM systems, Digi Locker, etc., for real-time validations and data sharing.

Logic View - Block Diagram:

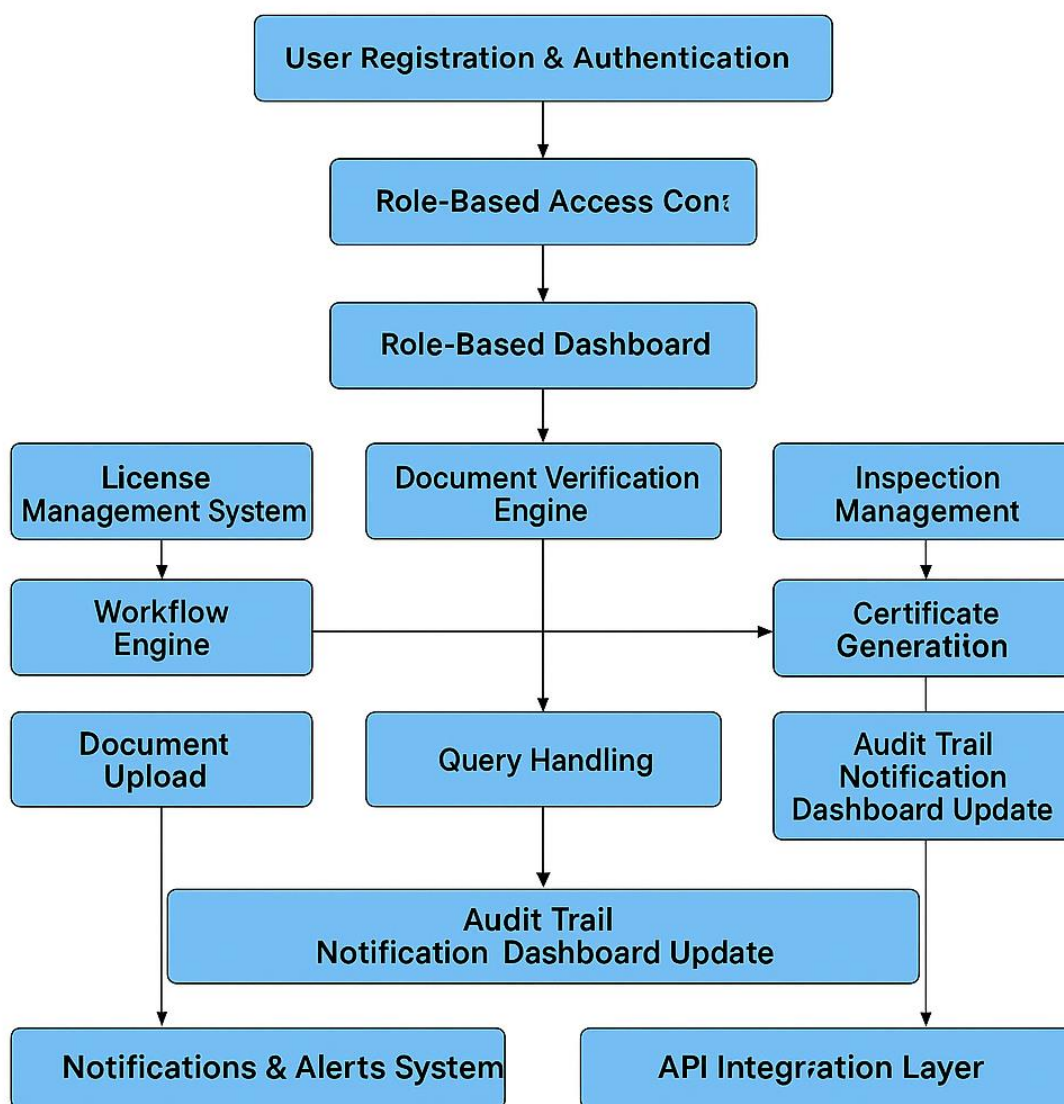


Figure: Logical Diagram of eMaap system showing component interaction and workflow.

4.2 Hardware Architecture

The hardware architecture for the eMaap (Electronic Metrology Administration and Automation Portal) system is designed as a secure, scalable, cloud-hosted infrastructure that supports legal metrology workflows across Centre and 36 States/UTs. The architecture follows a multi-tier model consisting of access, application, business logic, data processing, monitoring, and disaster recovery layers.

- **Client Access Layer**

Includes web browsers, mobile apps, and kiosks. Users interact with the portal using secure HTTPS connections. Role-based access is enforced post-login.

- **Application Delivery Layer (DMZ)**

Comprises Load Balancer, Web Servers, Reverse Proxy, and Web Application Firewall (WAF) to manage traffic distribution, performance, and protection from common cyber threats.

- **Application & Middleware Layer**

Contains App Servers, API Gateway, Notification Engine, and Audit Logger. Manages session state, form logic, user workflows, and real-time communication.

- **Business Logic & Integration Layer**

Hosts the Legal Metrology Services engine, document management system, external integrations with UIDAI, PAN, GSTIN, Digi Locker, and state portals.

- **Data Storage & Processing Layer**

Includes relational databases, secure file storage, caching via Redis, and backup vaults. Handles persistence and retrieval of structured and unstructured data.

- **Monitoring & Security Infrastructure**

Uses SIEM tools, IDS/IPS, log analyzers, and centralized dashboards for monitoring, alerting, and forensic analysis.

- **Disaster Recovery & High Availability Zone**

A mirrored environment in a different data center to ensure service continuity in case of primary system failure. Includes automatic failover databases and geo-redundant storage.

Deployment Specification:

Environment	Purpose	Configuration
Development	Internal Testing	2vCPU, 8GB RAM, 100GB Storage
UAT/Staging	Pre – Production Validation	4vCPU, 16GB RAM, 250GB storage
Production	Live portal Usage across India	8-16vCPU, 64-128GB RAM, 1+TB SSD
Disaster Recover	Failure Readiness	Mirrors Production Setup

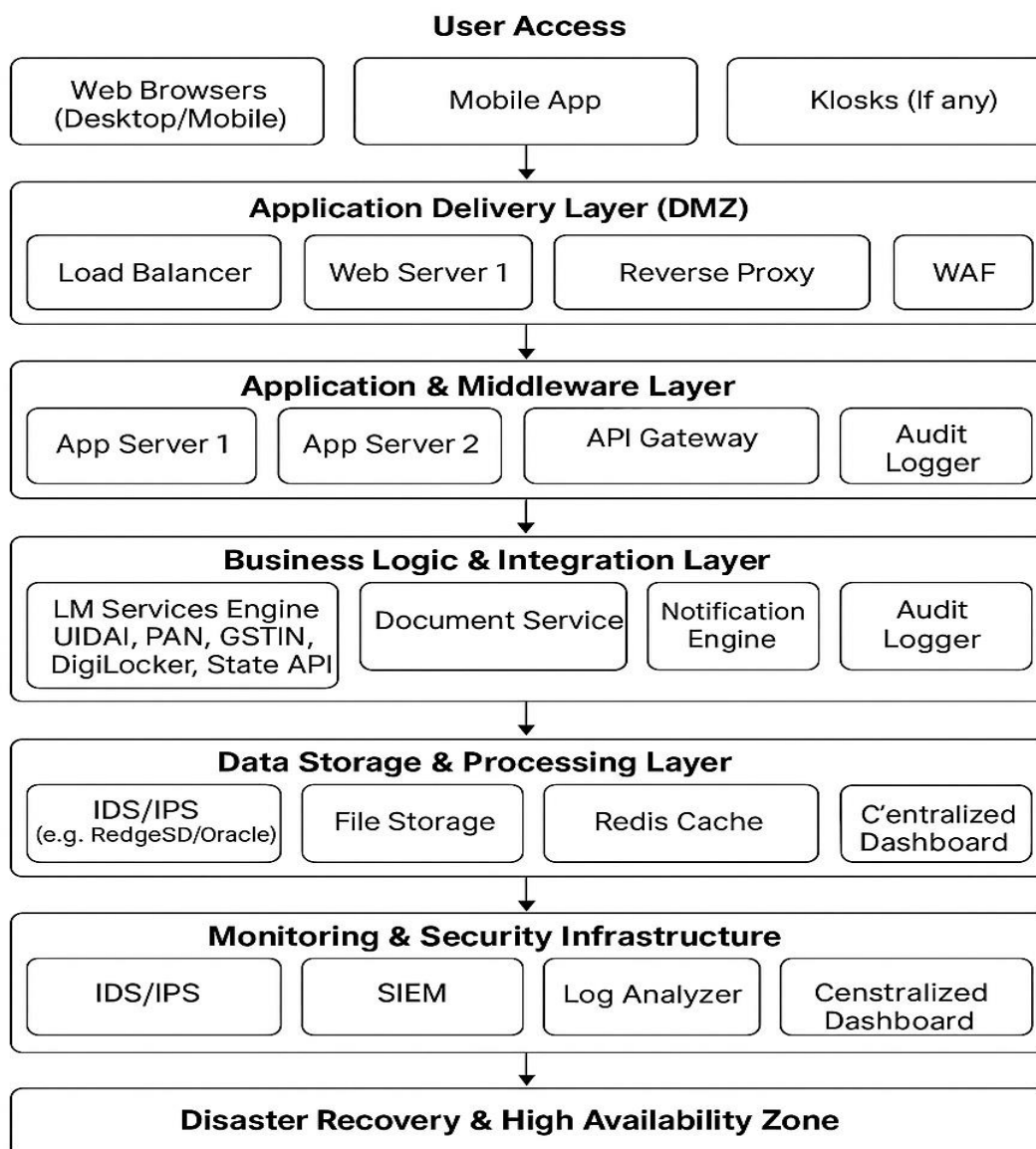


Figure: Hardware architecture diagram of the eMaap systems.

4.2.1 Security Hardware Architecture

The security hardware architecture is designed to provide **robust protection, threat detection, and user validation mechanisms**. It ensures **data confidentiality, integrity, availability, and accountability** at all layers using secure and scalable components. The system complies with national data protection standards and is capable of mitigating modern cyber threats.

Hardware Components & Configuration:

Component	Description
Web Application Firewall (WAF)	Protects against OWASP Top 10 threats at the edge (e.g., XSS, SQLi). Placed at the DMZ layer.
Reverse Proxy	Hides backend infrastructure, controls traffic flow, and caches static content.
Load Balancer	Distributes incoming traffic across multiple servers; helps with DDoS mitigation.
IDS/IPS Appliances	Intrusion Detection and Prevention Systems deployed at DMZ and Core Layers to monitor and block malicious activity.
Hardware Security Module (HSM)	Dedicated hardware for secure key management and cryptographic operations.
SIEM Appliance	Security Information and Event Management device to centralize logging, alerting, and threat analysis.
Log Analyzer Appliance	Analyzes system/application logs in real time for anomalies and security events.

➤ **Authentication for User Identity Validation (Pre-login)**

Mechanism:

- **Multi-Factor Authentication (MFA)** using OTP (SMS/email) and password-based login.
- Integration with **UIDAI (Aadhaar)** or **PAN** databases via APIs.
- Secure **TLS/SSL** sessions with HTTPS (TLS 1.2 or above).

Hardware Role:

- Authentication requests pass through **WAF** and **Reverse Proxy** to prevent credential stuffing/brute-force.
- **HSM** stores cryptographic keys used for verifying digital signatures or biometric tokens.
- **SIEM** tracks login attempts and flags suspicious access patterns.

➤ **Authentication for Functional Activity (Post-login)**

Mechanism:

- **Role-Based Access Control (RBAC)** and **Attribute-Based Access Control (ABAC)** via centralized identity engine.
- **OAuth 2.0 / JWT tokens** issued by the API Gateway, validated at every service invocation.

Hardware Role:

- API Gateway embedded with token validation logic.
- **Redis Cache** or secure memory to temporarily store session tokens.
- **Audit Logger** component logs every endpoint/user interaction.

➤ **Encryption Protocols**

Category	Protocol/Standard	Hardware Support
Data in Transit	TLS 1.2/1.3 for all APIs and portals	Reverse Proxy / Load Balancer offload TLS
Data at Rest	AES-256 encryption for file & database	HSM / Database Engine Encryption Module
Credentials & Tokens	SHA-256 hashing + salting for passwords	HSM for secure hashing operations
API Calls	JWT tokens, digitally signed with RSA keys	HSM / API Gateway with crypto module

➤ **Logging & Auditing Design**

Design Overview:

- Audit Logger (Application Layer) logs all user actions, config changes, and data accesses.
- Centralized Log Analyzer processes logs for anomalies and alerts.
- Logs are immutable and retained as per regulatory policy (e.g., 1–3 years).
- Integrated with SIEM for real-time alerts, threat correlation, and forensic investigations.

Log Type	Source	Retention	Security Control
Authentication Logs	Web/App Servers, Identity Engine	3 years	Encrypted at rest
Access Logs	API Gateway, App Servers	1 year	Token-based segregation
System Events	IDS/IPS, Load Balancer, WAF	6 months	Signature-based rules
Audit Trails	Business Services Layer	3 years	Immutable, timestamped

➤ **Security Threat Protection Coverage**

Threat Type	Mitigation Hardware/Layer
DDoS Attacks	Load Balancer, Reverse Proxy
SQL Injection / XSS	WAF with threat signatures and behavioral rules

Man-in-the-Middle	TLS 1.3 via HSM-based certificates
Credential Theft	MFA, SIEM alerting, rate limiting
Session Hijacking	Secure JWT, session timeout, token revocation
Insider Threats	Log Analyzer + SIEM + RBAC

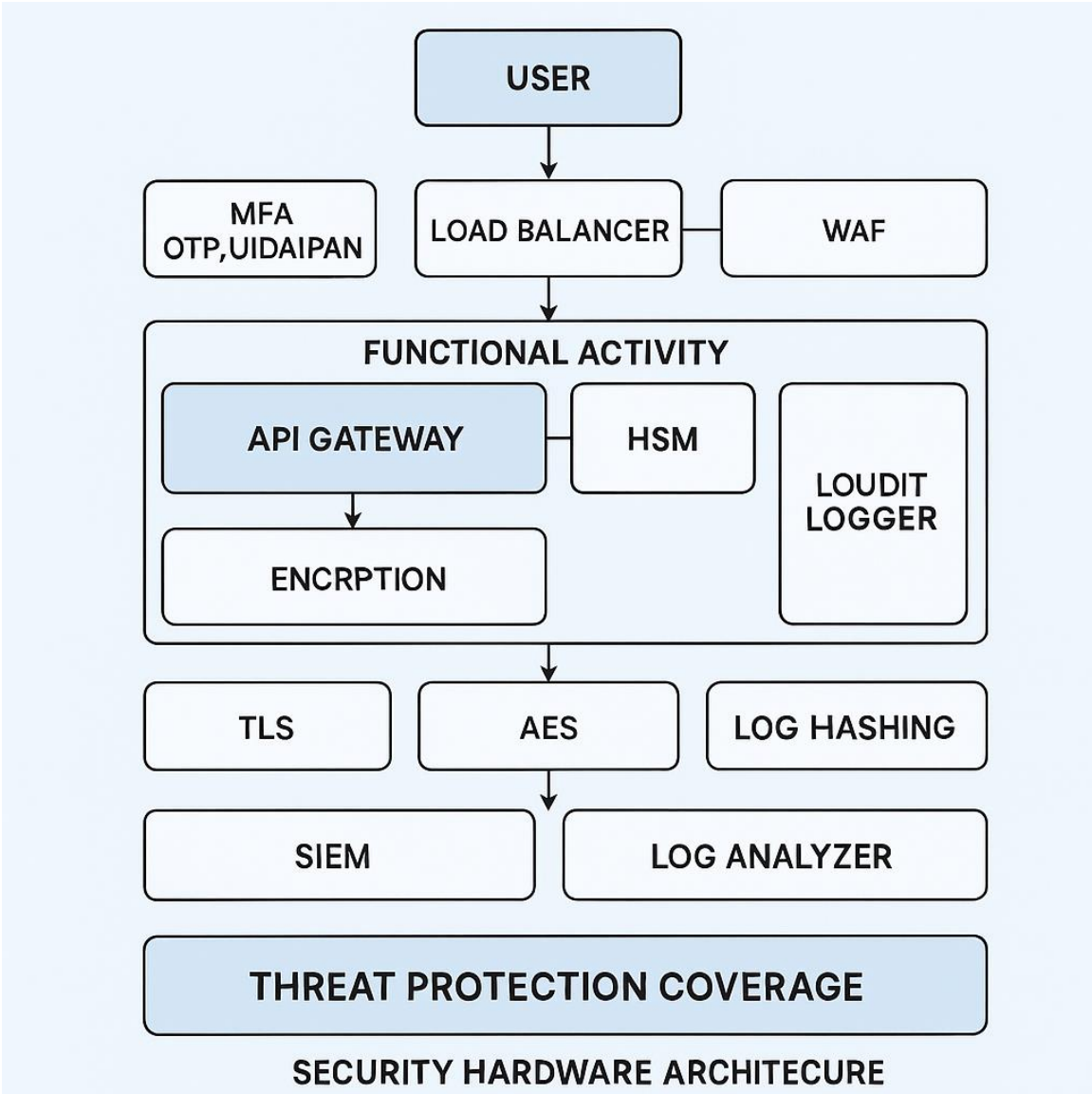


Figure: Security hardware architecture Diagram of the eMaap system.

4.2.2 Reliable Hardware Architecture

It ensures continuous availability, fault tolerance, and scalable performance across all components. The design must eliminate Single Points of Failure (SPOF) and use redundancy, clustering, and failover mechanisms.

Key Hardware Components & Configuration for Reliability:

Layer	Hardware Component	Reliability Features
Access Layer	Web Browsers / Mobile Apps / Kiosks	Redundant client access methods
Delivery Layer (DMZ)	Load Balancer, Web Servers (1 & 2), Reverse Proxy, WAF	Load Balancer uses Active-Active setup, Web servers in cluster behind proxy
Application & Middleware Layer	App Server 1 & 2, API Gateway	App Servers configured in cluster; Stateless architecture for easy scaling
Business Logic Layer	Notification Engine, Audit Logger, Document Service	Modular services with failover containers (e.g. Docker/Podman)
Data Layer	File Storage, Redis Cache, Relational DB (PostgreSQL/Oracle), Backup Vault	Master-slave DB replication, Redis cluster, backup vault with version control
Monitoring Layer	SIEM, Log Analyzer	Real-time stream processing; HA design
Security Layer	IDS/IPS, HSM	Dual appliances with mirroring
Failover & HA	DR Site, Clustering Mechanisms	DR with async replication, cluster managers (e.g., Pacemaker, Kubernetes)

Single Points of Failure (SPOFs) & Mitigation:

SPOF Area	Mitigation Strategy
Single Web Server	Two Web Servers behind Load Balancer
Single App Server	App server clustering (Active-Active/Passive)
Database Failure	Replication + Failover + Scheduled Backups
API Gateway Crash	Redundant gateway services in separate zones
Redis/Cache	Redis Sentinel or Cluster
WAF Failure	Two WAF instances configured in failover pair
Log Analyzer Failure	Dual Log processors (Elasticsearch/Fluentd setup)

Clustering & High Availability Design:

Technologies Used:

- **HAProxy/Nginx:** Load Balancing
- **Redis Sentinel/Cluster**
- **Database Replication:** Streaming replication or Oracle RAC
- **Orchestrator:** Kubernetes or Docker Swarm
- **Failover Tooling:** Pacemaker + Corosync (for system resources)

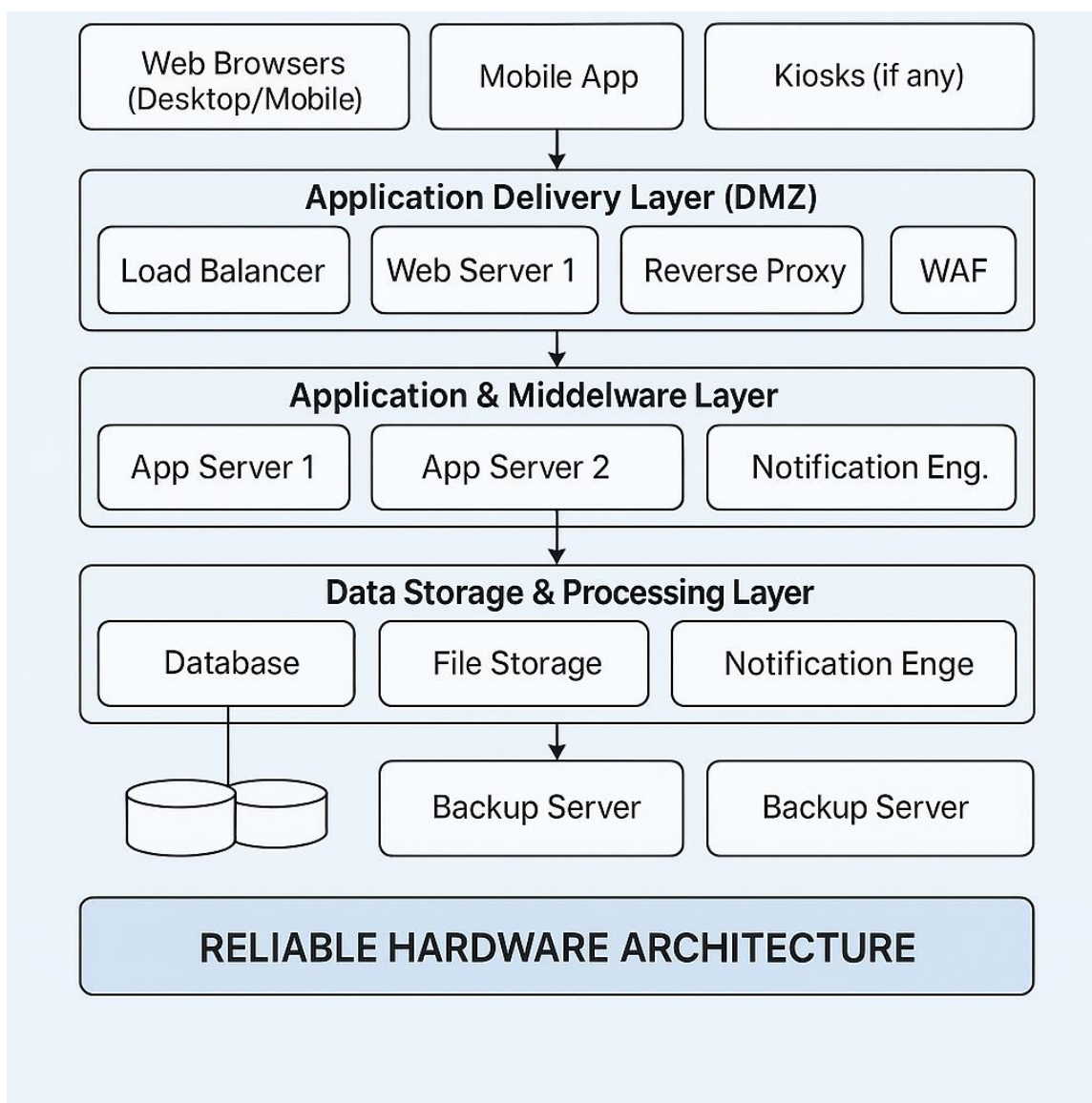


Figure: Reliable Hardware architecture Diagram of the eMaap system

4.3 Software Architecture

The eMaap Portal (Electronic Metrology Administration and Automation Platform) is designed as a modular, centralized, web-based platform that integrates legal metrology services across Centre and States. It follows a multi-tier architecture to support scalability, security, high availability, and interoperability with external systems.

The architecture adheres to MeitY's e-Governance standards, including:

- Service-Oriented Architecture (SOA)
- Role-Based Access Control (RBAC)

- API-driven integrations
- Mobile and web responsiveness
- Cloud-native deployment model

Key Architectural Components:

Layer / Component	Description
Presentation Layer	Web portal and mobile-responsive UI for users (manufacturers, dealers, officers, etc.)
Application Layer	Workflow engines, role-based dashboards, license modules, form engines
Service Layer (API Gateway)	REST APIs for Aadhaar, PAN, GSTIN, Digi Locker, eSign, and state integrations
Business Logic Layer	Modular logic for licensing, model approval, package commodity, inspection, etc.
Database Layer	Centralized RDBMS with audit trail, logs, and state-wise data segregation
Authentication Layer	OTP, Aadhaar validation, 2FA mechanisms (Email & SMS OTP), encrypted sessions
Notification Layer	SMS/Email gateway for system alerts, approvals, and status updates
Document Management System	Secure cloud-based storage with user ID-tagged folders
Dashboard and MIS Layer	AI-powered visualizations and forecasting tools for policy makers and officers
Security & Compliance Layer	Encryption, access control, audit logs, time-bound tokens, DR and backup mechanisms

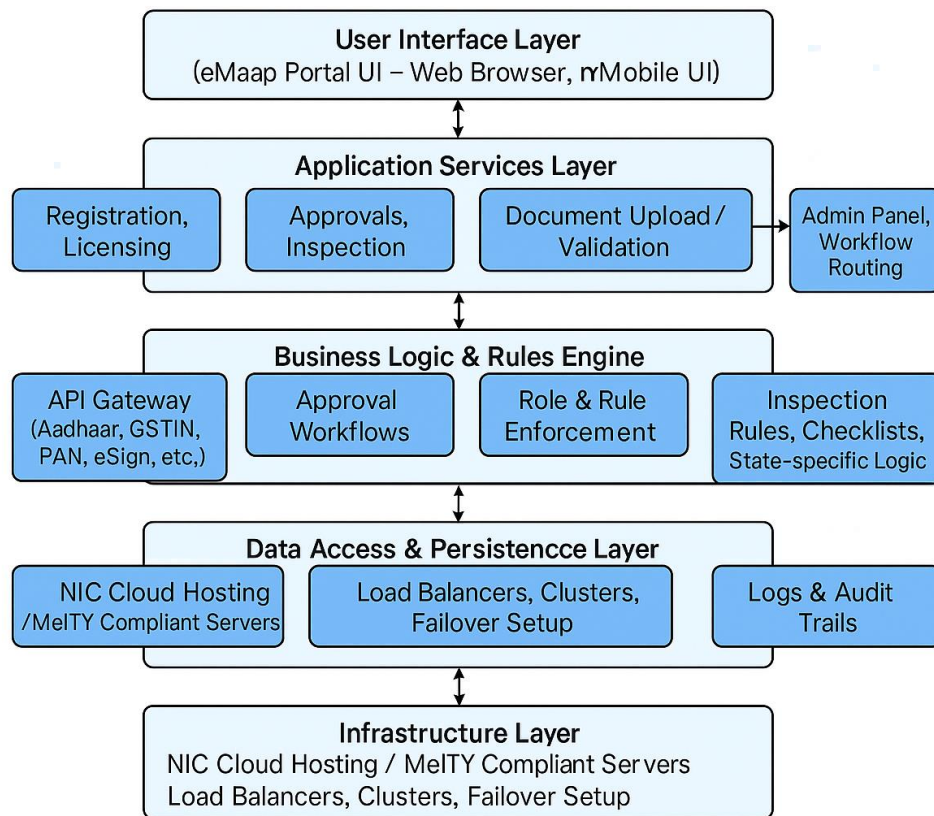


Figure: Software architecture Diagram of the eMaap system.

- **User Interface Layer**
 - Built using modern frontend frameworks (Angular/React).
 - Accessible through mobile and desktop browsers.
 - Multilingual support and dynamic form rendering.
- **Application Services Layer**
 - Handles modules like registration, licensing, renewal, inspections, grievance redressal, and certificate generation.
 - Includes dashboards for stakeholders, inspectors, and legal officers.
- **Integration Layer**
 - **Real-time APIs to:**
 - PAN Verification (CBDT)
 - GSTIN Validation (GSTN)
 - Aadhaar Authentication (UIDAI)
 - Digi Locker & eSign APIs
 - State-specific metrology systems
- **Business Logic Layer**

- Implements approval workflows (Assistant Controller → Divisional Controller → Controller).
 - Enables condition-based routing (e.g., by license type).
 - Includes role-based form rendering and query management.
- **Data Layer**
- Central relational database for transactional data.
 - Document repository using secure cloud object storage.
 - Audit logs for user actions and system events.
- **Infrastructure Layer**
- Deployed on NIC Cloud or equivalent Government Data Centre (GDC).
 - Clustered environment with load balancing and failover.
 - Encrypted communication channels, security compliance per IT Act.

Reliability, Security, and Availability Considerations:

Area	Mechanism
High Availability	Load-balanced application servers, replicated DBs, cloud hosting
Disaster Recovery	Daily backup snapshots, failover to secondary site
Single Point of Failure	Avoided via horizontal scaling and failover-ready infrastructure
Security	Role-based access, OTP validation, encrypted data, HTTPS, audit trails
Audit & Logs	Action-based audit trails for accountability and RTI compliance
Compliance	Adheres to IT Act 2000, Legal Metrology Act 2009, and UIDAI standards

4.3.1 Security Software Architecture

The eMaap portal employs a multi-layered security approach aligned with Zero Trust Principles and MeitY's e-Governance cybersecurity guidelines. It secures all components including the application, API layer, identity services, and hosting environment.

Key Security Components:

Component	Security Mechanism
User Authentication	2FA using OTP (SMS & Email), Aadhaar eKYC, password policies
RBAC	Role-based access control per user type (Admin, Inspector, Legal, Stakeholder)
Session Management	JWT-based time-bound tokens, secure cookie handling
Data Encryption	AES-256 encryption at rest, TLS 1.3 in transit
API Security	OAuth 2.0, API Gateway, IP whitelisting, rate limiting
Document Security	Encrypted cloud storage, folder-level access control
Input Validation	Server-side validation and sanitization to block SQLi/XSS
Audit Trail	Immutable logging of user actions, accessible under RTI
Infrastructure Security	NIC Cloud hosting, IDS/IPS, antivirus, regular patching
Compliance	Compliant with IT Act 2000, Aadhaar Act, UIDAI norms, Legal Metrology Act

➤ **Identity & Access Management (IAM)**

Authentication is enforced using OTPs (SMS & Email), optional Aadhaar integration, and activation links. Authorization is governed through fine-grained RBAC policies, ensuring users access only permitted modules.

➤ **Network & Hosting Security**

The platform is hosted on NIC Cloud infrastructure, secured using SSL/TLS encryption, firewalls, intrusion detection systems, and access restrictions. Daily backups and disaster recovery plans are in place.

➤ **Security Monitoring & Logging**

All system events are logged with timestamps, user IDs, and context. Monitoring tools track login attempts, file anomalies, and suspicious activity. Logs are retained for 7 years.

➤ **Security Architecture Diagram**

The following diagram visually represents the layered security design of the eMaap platform.

Security Software Architecture

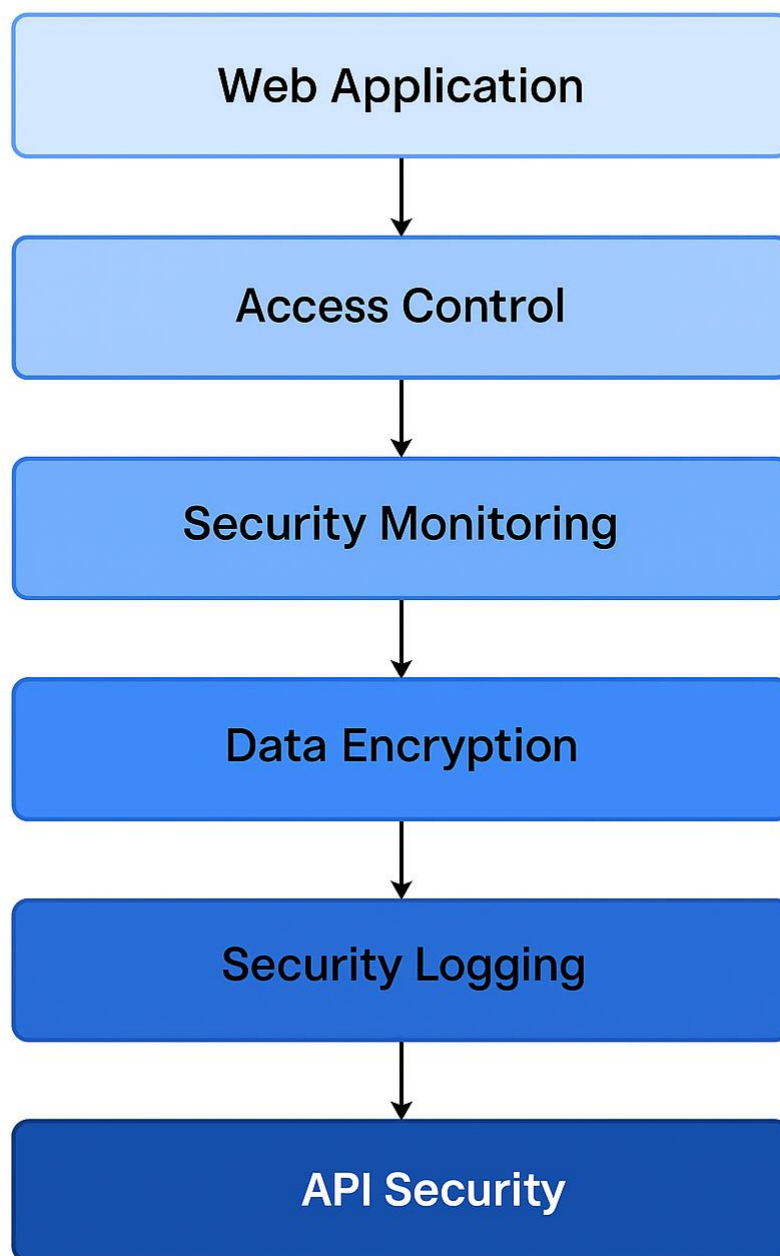


Figure: Security software Architecture Diagram of the eMaap system

4.3.2 Performance Software Architecture

The **eMaap platform** is engineered to support real-time services across Centre and 36 States/UTs. To meet this objective, the architecture incorporates performance-focused design principles such as:

- Modular and service-oriented structure
- Scalable and stateless components
- Efficient resource utilization via asynchronous processing
- Cloud-native, elastic infrastructure

Software Components Supporting Performance:

Component	Configuration & Performance Strategy
Web UI (Angular/React)	Lazy loading modules, browser caching, CDN-based static delivery
Application Server (Node.js/Java)	Stateless, horizontally scalable containers deployed on NIC Cloud
Workflow Engine	Asynchronous processing with retry queues to handle load spikes
API Gateway	Implements throttling, rate limiting, token caching, and routing logic
Database (PostgreSQL/MySQL)	Partitioned data by state/UT, read/write separation, indexing of key fields
Document Store (Object Storage)	CDN-enabled document delivery; background virus scanning
Redis Cache	Session tokens, frequently used config and lookup values cached
Load Balancer	Layer 7 load balancer for distributing HTTP traffic and failover
Auto-scaling	Based on CPU/memory usage thresholds; horizontal scaling of pods/services
Monitoring Tools	Prometheus/Grafana (or equivalent NIC stack) for live metrics and alerts

Reliability and High Availability Features:

Feature	Description
Load Balancing	Distributes traffic across multiple instances to avoid overload
Session Management	Stateless services + Redis ensures sessions are lightweight and recoverable
Failover Configuration	Application clusters are deployed in active-active zones (within NIC Cloud)
Auto-Healing Containers	Crash-loop detection and auto-restart using container orchestration (K8s)

Disaster Recovery	Full data and config snapshots taken daily and replicated to DR sites
Queue-Based Processing	Background workers and job queues for delayed/long tasks to avoid bottlenecks
Asynchronous Notifications	Emails/SMS sent through message queues to avoid slowing down main workflows

Optimizations for Performance:

- Database Query Optimization: Indexing on UID, license number, and mobile numbers.
- Response Time Targets:
 - Registration: < 2 seconds
 - License Issuance: < 5 seconds (post approval)
 - Document Upload/Preview: < 3 seconds
- Page Load Optimization: Lazy loading, modular JS bundles, and CSS minification.
- Concurrent Users Supported: Baseline support for ~5,000 concurrent users, scalable to ~50,000 under surge.

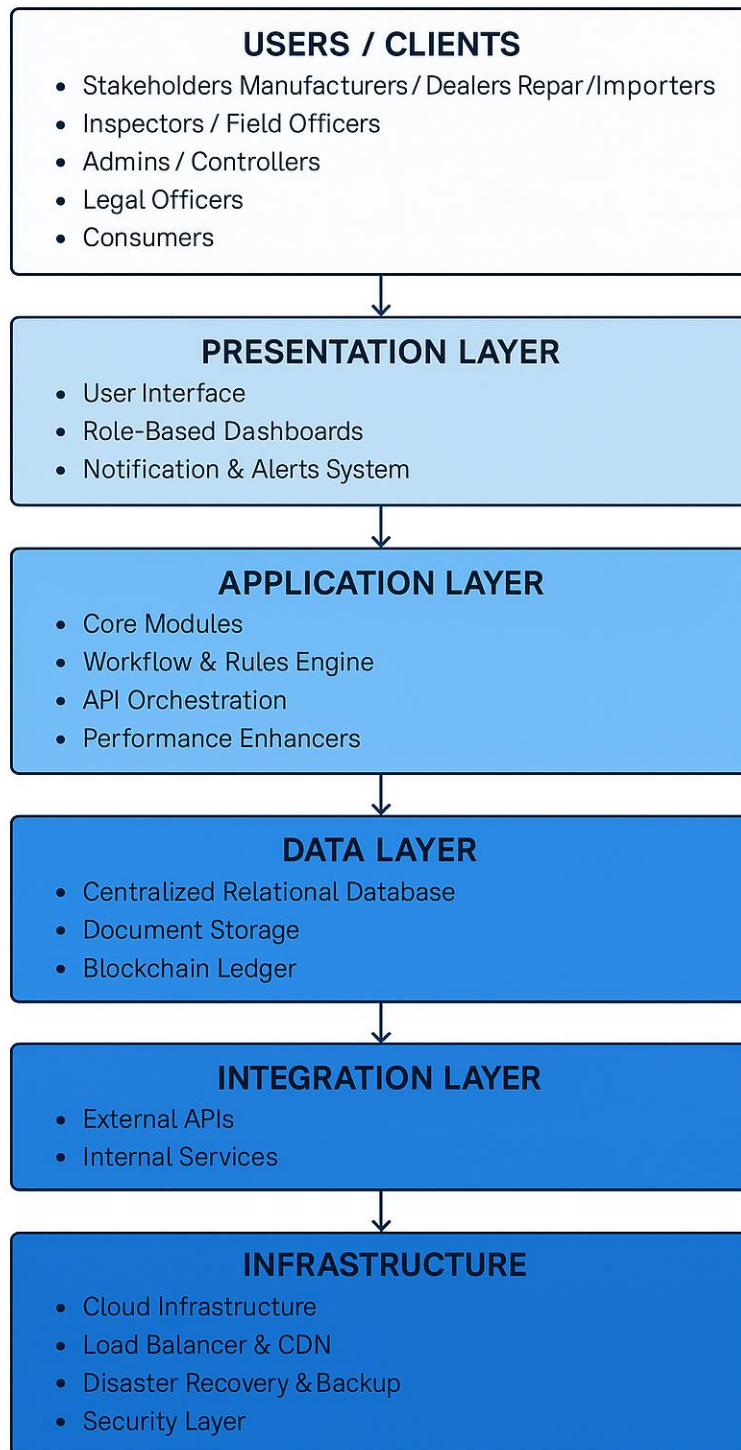


Figure: Performance Software Architecture Diagram of the eMaap software

4.4 Information Architecture

The eMaap Portal stores structured and unstructured information to support end-to-end Legal Metrology compliance processes. This includes:

- **Beneficiary & Stakeholder Information**
 - Full name, company/firm name
 - Contact details: address, phone number, email ID
 - Identity numbers: Aadhaar (optional), PAN, GSTIN, Trade License Number
 - Role and business category (e.g., manufacturer, repairer)
 - Type: Personally Identifiable Information (PII) and Individually Identifiable Information (III).
- **License & Certificate Data**
 - Types of licenses issued (manufacturer, dealer, importer, etc.)
 - License number and unique IDs
 - Validity period, renewal details, fee payment status
 - Status flags: active, suspended, cancelled
 - Type: Business operational data (linked to individuals or entities).
- **Documents & Supporting Proof**
 - Uploaded PDFs, JPEGs: ID proofs, trade licenses, compliance certificates
 - e-Signatures
 - System-generated forms, declarations, and approvals
 - Blockchain-backed immutable certificates
 - Type: PII where ID proofs are included; Business documents.
- **Verification & Inspection Data**
 - Inspector assignment data: names, designations, geo-location of inspections
 - Inspection reports: date, time, findings, non-compliance observations
 - Geo-tagging data (latitude, longitude)
 - Type: PII for inspectors; Sensitive business premises data.
- **Complaint & Grievance Data**
 - Nature of grievance
 - Stakeholder profile
 - Timeline and audit trail of complaint resolution
 - Type: May include PII of complainants and offenders.
- **System Metadata & Logs**
 - Timestamps of user actions
 - Role-based access logs

- Notifications and alert logs (SMS, email)
- Audit trails for compliance and legal traceability
- Type: Indirect PII, for accountability and evidence in legal disputes.

Sensitive / Regulated Data:

Data Element	Sensitivity	Notes
Aadhaar Number	PII, III	Regulated under Aadhaar Act; must use UIDAI-compliant APIs.
PAN / GSTIN	PII / Business ID	Must follow CBDT/GSTN guidelines for validation & storage.
Mobile Number & Email ID	PII	Critical for OTP & notifications; must be encrypted.
Uploaded ID proofs	PII	Government IDs must be secured.
Geo-location Data	Sensitive	Tracks physical inspections; should use secure APIs.
Blockchain Certificates	Sensitive	Must be immutable and traceable for non-repudiation.

Compliance & Privacy Considerations:

- Encryption at rest and in transit for PII, especially Aadhaar, PAN, and mobile numbers.
- Role-Based Access Control (RBAC) ensures only authorized officers can view or process sensitive data.
- Explicit consent required for Aadhaar usage as per UIDAI norms.
- Audit Trail: Every user action is logged for legal compliance.
- Data Minimization: Only necessary data collected for statutory purposes.

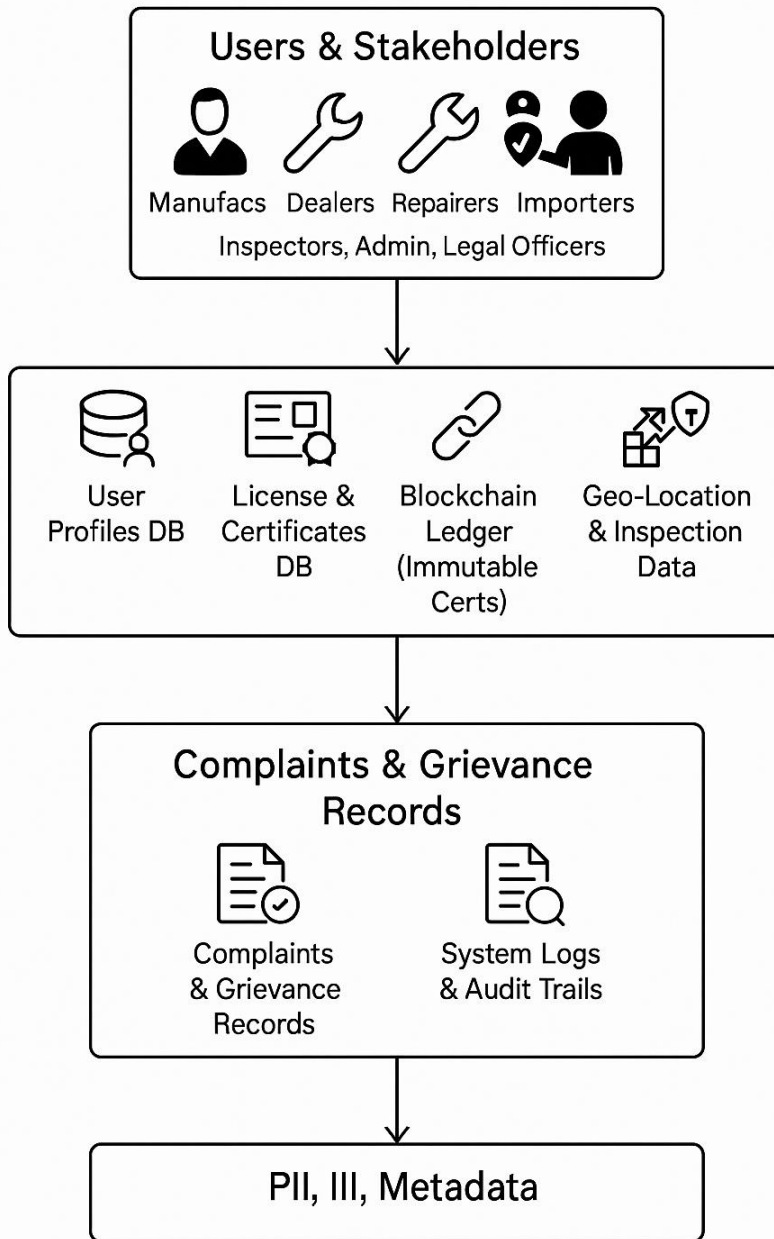


Figure: Information Architecture Diagram for eMaap Platform

4.4.1. Data

The **eMaap Platform** collects and manages a broad spectrum of data.
Below is a clear structure:

User & Stakeholder Identity Data:

Data Type	Format	Supplied By	Note
Name, Address	Manual input	User/Stakeholder (via portal)	Text fields; validated in real-time
Mobile Number, Email	Manual input + OTP	User/Stakeholder	OTP verified; stored encrypted
Aadhaar Number	Electronic (via API)	UIDAI	Real-time API fetch/verify
PAN, GSTIN	Manual input + API	User, Verified by CBDT/GSTN API	Format-validated; verified electronically
Business License/Trade Certificate	Scanned document (PDF/JPEG)	User/Stakeholder	Uploaded via portal; format check max size

Application & Licensing Data:

Data Type	Format	Supplied By	Notes
Application Forms	Electronic	User/Stakeholder	Dynamic forms; validated
License Category, Validity	Manual input	User+Admin	Dropdown & Date field
Fee Payment	Electronic	Payment Gateway	Payment status + receipt stored

Verification & Inspection Data:

Data Type	Format	Supplied By	Notes
Inspector Assignment	Manual/Electronic	Admin Controller	Via backend UI
Inspection Report	Electronic	Inspector (mobile app)	Location (geo-tag) + photos
Geo-Location	Electronic	GPS API	Auto-captured during inspection

Complaints & Grievance Data:

Data Type	Format	Supplied By	Notes
Complaint Details	Manual input	User/Stakeholder	Text, categories selected
Response Log	Electronic	Admin/Legal Officers	Comments, status updates

Supporting Documents:

Data Type	Format	Supplied By	Notes
Identity Proofs	PDF/JPEG	User/Stakeholder	File upload; stored in secure cloud
Compliance Certificates	PDF	Generated by System	Auto-created & blockchain backed

System Metadata & Logs:

Data Type	Format	Supplied By	Notes
Login & Actions	Electronic	System	Audit trail; timestamped
Notifications	Electronic	System	Email/SMS logs stored

4.4.2 Manual Inputs

The **eMaap Platform** collects and manages a broad spectrum of manual inputs.

Below is a clear structure:

1. **Users enter data:** Direct input into portal fields (text, dropdowns, uploads).
2. **System validation:**
 - o Format check (e.g., PAN, GSTIN format)
 - o Mandatory fields
 - o OTP verification for mobile/email
3. **Admin/Inspector review:** For workflows needing approvals (e.g., license issuance, complaints resolution).
4. **Once verified:**
 - o Data is committed to the **Master Database**.
 - o Supporting documents are stored in secure cloud folders linked by user ID.

- Immutable certificates are pushed to the Blockchain ledger for tamper-proof status.

Example:

- A manufacturer applies for a license → fills form → uploads Trade License (PDF) → Admin reviews → Inspector inspects premises → status updated → approved license issued → certificate saved to master DB and Blockchain ledger.

4.4.3 Master Files

The **Master Files** in the system act as the *Single Source of Truth* for all modules. These include:

Master File	Description
User Profiles DB	Stores verified identities: Name, Address, Contact, Aadhaar, PAN, GSTIN, User Roles
Licenses & Certificates DB	License type, category, validity, status (active, expired, suspended), fees paid, issue history
Documents Repository	All uploaded documents linked with unique user/application IDs; stored with version control
Blockchain Ledger	Immutable copies of issued certificates, audit flags for revocation
Inspections Master	Inspector assignments, reports, geo-location evidence, photos
Complaints & Grievance DB	Complaint details, status, resolution logs
System Logs & Audit Trails	Every action: who did what and when, with timestamps and role-based tagging

4.5 Internal Communication Architecture

The eMaap platform's internal communications architecture is designed to provide a secure, scalable, and efficient communication framework between its distributed components and services, which span central and state legal metrology departments. This architecture leverages a hybrid **Ethernet-based star topology** within the data center and a **secured WAN architecture** to support inter-department communication and integration with external services (e.g., Aadhaar, PAN, GSTIN, State APIs).

Type of Communication Architecture:

- **Ethernet-based Star Topology (LAN):**
All core system components (web servers, application servers, database servers, file storage) are interconnected through high-speed Ethernet switches, forming a star topology within the National Data Center managed by NIC.
- **WAN Communication (via VPN/MPLS):**
For interactions between central and state-level departments, secure MPLS VPN tunnels or NIC's SWAN (Statewide Area Network) are used to create private, encrypted communication channels.

Components and Interconnections:

Component	Description	Communication Protocols Used
Web Servers	Hosts the frontend portal for user interaction (registration, login, dashboards, etc.)	HTTPS, REST APIs over TLS
Application Servers	Processes business logic, workflows, API orchestration, and validation routines	REST APIs, RPC, WebSocket (internal)
Database Servers	Stores structured data (user info, application records, audit logs, etc.)	JDBC/ODBC over secure internal LAN
Document Storage Server	Cloud-based or NAS for storing uploaded documents (PDF/JPEG/PNG)	NFS/SMB or HTTPS with authentication
Firewall and IDS/IPS	Filters incoming/outgoing traffic and prevents unauthorized access	Layered firewall (L3/L7), Snort/Suricata
Load Balancer	Distributes load across multiple web and application servers for performance and availability	HTTP(S), SSL Termination
API Gateway	Manages traffic between internal services and external integrations (Aadhaar, PAN, etc.)	REST/JSON, OAuth 2.0, JWT Tokens
Email/SMS Gateway	Sends notifications and OTPs	SMTP (TLS), SMPP over VPN
State Server Gateways	Connects to State LM Portals using defined APIs	Secure API over SWAN/MPLS with IP whitelisting

Communication Layers:

- **Presentation Layer**
 - User interactions over HTTPS via responsive UI.

- Mobile and browser-based access.
- **Service Layer**
 - Stateless microservices communicate via REST APIs.
 - Token-based authentication using OAuth 2.0.
- **Data Layer**
 - Centralized RDBMS hosted on NIC infrastructure.
 - Redundant storage and periodic backups.

Equipment and Infrastructure:

Equipment/Software	Role
Routers (Layer 3)	Route traffic between LAN segments and connect to MPLS/VPN.
Switches (Layer 2/3)	Facilitate internal communication across servers.
Firewall Appliances	Enforce traffic rules, NAT, and VPN configurations.
Reverse Proxy (e.g., Nginx)	Protects backend services and handles SSL termination.
Virtual Load Balancer	Balances user traffic between redundant servers.
VPN Gateway	Secures remote or inter-state access to the central portal.
HSM (Hardware Security Module)	Stores cryptographic keys securely for certificate/e-sign.

Port and Protocol Usage:

Port	Protocol	Usage
443	HTTPS	Secure web access and API communication
22	SSH	Secure server administration
25/587	SMTP	Email notifications
80	HTTP	Redirects to HTTPS
3306/5432	MySQL/PostgreSQL	Database communication between app and DB
8080	HTTP API	Internal microservice communication
Custom	Web sockets	Real-time notification within dashboards

Security Measures in Communication:

- TLS 1.2+ Encryption for all data in transit.
- Firewalls and Access Control Lists (ACLs) to block unauthorized access.
- API Whitelisting: Only approved state systems can access APIs.
- Audit Trail: All communication logs (API calls, notifications, admin actions) are recorded.
- Two-Factor Authentication (2FA) for admin and inspector logins.
- IP Whitelisting & VPN for sensitive services (e.g., admin dashboards).

Integration with External Systems:

The system is integrated with several external services using secure APIs:

- UIDAI (Aadhaar)
- GSTN
- PAN Verification (CBDT)
- Digi Locker/eSign APIs
- State Legal Metrology Portals via SWAN

All API communications are authenticated via OAuth tokens, JWT, or mutual TLS certificates to ensure security and non-repudiation.

Internal Communications Architecture Diagram – eMaap Platform:

The following diagram illustrates the internal communications architecture of the eMaap system, showing how system and subsystem components are interconnected through LAN and WAN networks.

Internal Communications Architecture

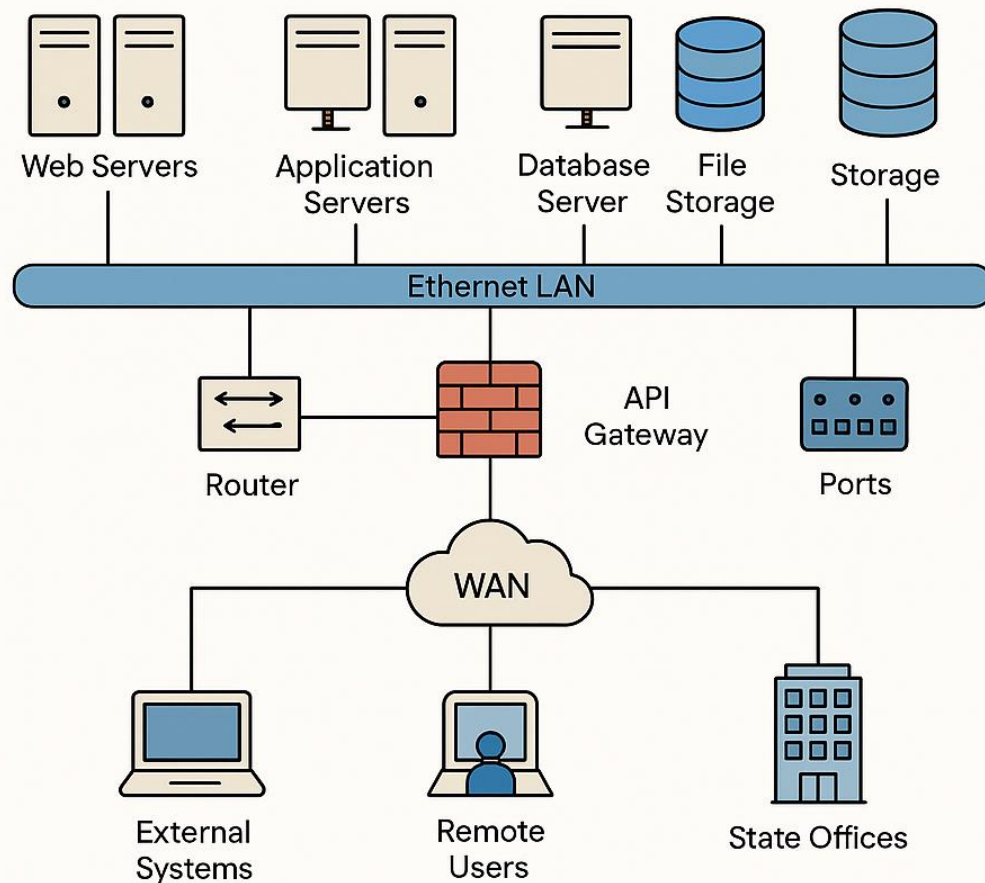


Figure: - Internal Communication Architecture Diagram of eMaap system.

4.6 Security Architecture

The purpose of the security architecture is to protect the eMaap platform from internal and external threats while ensuring the confidentiality, integrity, and availability of its services and data. This includes:

- Enforcing secure access for users and administrators
- Safeguarding personal and business information
- Preventing unauthorized system access or data manipulation

Ensuring legal compliance with Indian IT and data protection laws

Architectural Model:

The security architecture is implemented using a **defense-in-depth model** spanning the following layers:

➤ **Network Security Layer**

- **Firewalls (Layer 3/7):** Control ingress/egress traffic to/from application servers
- **DMZ (Demilitarized Zone):** Isolates the public-facing web servers from backend systems
- **VPN/MPLS (NICNet / SWAN):** Secure tunneling between Central & State nodes
- **Intrusion Detection/Prevention Systems (IDS/IPS):** Real-time network monitoring
- **Port Management:** Only necessary ports (443, 22, 587) are exposed

➤ **Application Security Layer**

- **Authentication via OAuth 2.0 & JWT Tokens:** Every user interaction and API request is authenticated
- **2FA/OTP:** Dual-channel OTPs for critical logins (Admin, Inspector roles)
- **Role-Based Access Control (RBAC):** Enforced at both UI and API levels
- **Session Management:** Token-based session expiry and renewal mechanisms

➤ **Data Security Layer**

- **Encryption In Transit:** All data is transmitted via HTTPS using TLS 1.2+
- **Encryption At Rest:** AES-256 encryption for uploaded documents, logs, and sensitive records
- **Data Masking:** Sensitive fields (e.g., Aadhaar, PAN) are masked for non-admin users

➤ **Document & Digital Identity Controls**

- **eSign Integration:** All certificates/licenses digitally signed via NIC's eSign
- **Digi Locker Integration:** Enables secure document storage and retrieval
- **QR Code on Licenses:** Allows offline validation of approved documents
- **Checksum Validation:** Hashing (SHA-256) of uploaded files to prevent tampering

➤ **Monitoring & Audit Layer**

- **Audit Trail Logging:** All user actions are timestamped and linked to IP and role
- **SIEM (Security Info & Event Management):** Real-time anomaly detection and correlation
- **Log Retention:** All logs retained for **minimum 2 years** for RTI or legal audit compliance
- **Notification Alerts:** Unauthorized login attempts, or API misuse generate admin alerts

Tools, Protocols & Equipment:

Component	Description
Firewall	Fortinet / Palo Alto for L3/L7 packet filtering
Web Server	Apache/Nginx with SSL offloading
VPN Gateway	NICNet/MPLS for secure inter-state communication
Database	PostgreSQL/MySQL with TDE and row-level security
API Gateway	Kong or NIC-Gateway with token introspection & rate limits
SIEM Tool	ELK Stack / QRadar for central log monitoring
eSign	NIC/MeitY Gateway-compliant integration
Document Store	NIC Cloud with object-based ACLs

Compliance & References:

Framework/Act	Description
IT Act 2000 (with 2008 Amendments)	Governs digital transactions & data protection
MeitY e-Governance Security Guidelines	Covers API, authentication, and data handling
CERT-In Guidelines	National guidelines for incident handling
UIDAI Regulations	Covers Aadhaar data verification usage
RTI Act, 2005	Data auditability and disclosure responsibility
OWASP Top 10	Form and API protection incorporated

Integrity & Failover Controls:

- Backup & Disaster Recovery: Daily encrypted backups with RPO < 24 hrs., hosted at secondary NIC DC
- High Availability: Load balancer (HAProxy/NIC infra) for redundant failover
- Data Validation Rules: Implemented at form, API, and DB level
- Secure Session Handling: Session tokens expire after inactivity (20 mins default)

Future Enhancements:

- AI/ML-based behavioral threat detection
- Blockchain hashing for certificate tracking
- Biometric login for Inspectors via field mobile app

Security Architecture Diagram:

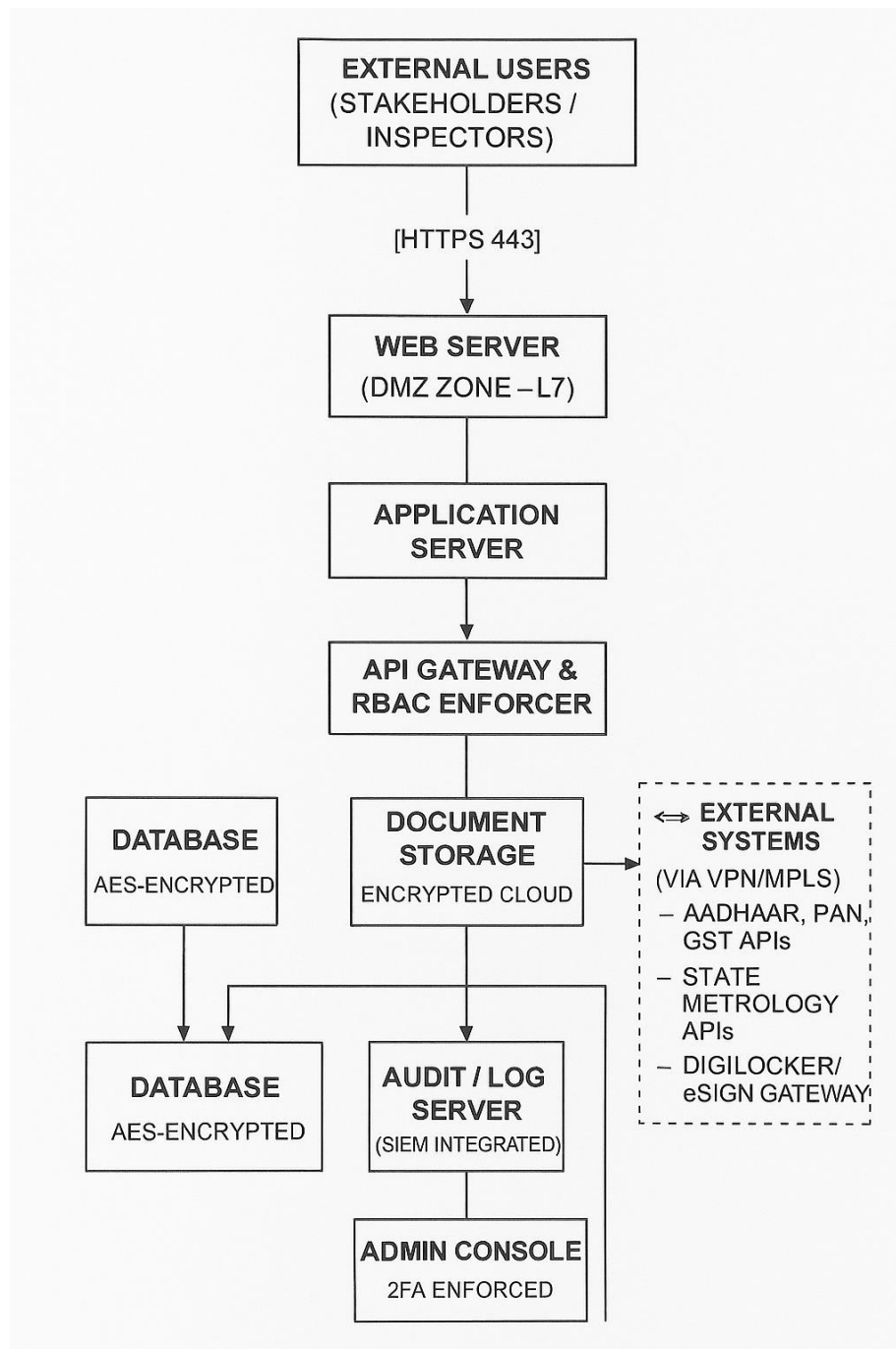


Figure: Security Architecture Diagram of eMaap system.

4.7 Performance

The performance architecture of the eMaap system is designed to ensure high availability, scalability, responsiveness, and fault tolerance to handle the digital governance workload for legal metrology services across all 36 States and Union Territories of India.

Performance Goals:

- **Response Time:** All critical user actions (e.g., form submission, document uploads, dashboard rendering) should complete within **3–5 seconds** under normal load.
- **Concurrent Users:** Support for **5,000+ concurrent users**, with auto-scaling infrastructure provisions.
- **System Uptime:** **≥ 99.9% availability** through clustering and failover mechanisms.
- **Throughput:** The platform must handle **200+ transactions per second (TPS)** during peak usage.
- **Latency:** Backend processing and workflow transitions must not exceed **2 seconds** for 95% of transactions.

Performance-Enabling Features from FRS:

The following features from the FRS directly contribute to system performance:

- **Role-Based Dashboards:** Personalized views based on user type reduce unnecessary processing and data loading.
- **Save as Draft:** Minimizes backend load by allowing incremental data storage.
- **OTP Verification with Timeout:** Optimized token lifecycles to reduce unnecessary retries and resource consumption.
- **Cloud-Based Document Handling:** Files are stored in structured, secure directories to improve access speed and scalability.
- **Real-Time Notifications:** Asynchronous processing (e.g., email/SMS alerts) using integrated gateway services avoids blocking user workflows.

Infrastructure Configuration:

- **Load Balancer:** Front-end requests are routed via a load balancer to evenly distribute the load across multiple servers.
- **Auto-Scaling:** Backend servers (application and database tiers) are horizontally scalable based on CPU, memory, and traffic thresholds.
- **Asynchronous Processing Queues:** Heavy-lifting tasks like PDF generation, certificate issuance, and API responses are handled asynchronously.
- **Caching Layer:** Frequently accessed data (e.g., master dropdown values, license types) are cached to reduce DB load.

Monitoring and Optimization:

- **Real-Time Monitoring:** Performance metrics (CPU, memory, response time, request queues) are monitored using tools like Prometheus/Grafana.
- **Performance Testing:** Load and stress tests are performed quarterly with tools like Apache JMeter or Locust.
- **Logging & Audit Trails:** Structured logs and audit trails are maintained to trace latency sources and system bottlenecks.

Performance Validation Approach:

- **System Integration Testing (SIT)** and **User Acceptance Testing (UAT)** cycles will include performance benchmarks.
- **Service Level Agreement (SLA)** compliance for each transaction is logged and reviewed regularly.

5 System Design

5.1 Functional Requirement

Module: 1	
Registration Process on eMaap Portal: Manages new user registration for Inspectors, Stakeholders, and Admins. Includes OTP verification, role-based forms, document uploads, admin review, and dashboard setup	
Functional Requirement	Description
New Registration (#FR001)	Provides a dynamic form for new users to register based on role (Inspector/Stakeholder/Admin).
Role-Based Registration (#FR002)	Displays different fields and validations based on the user role selected during registration.
OTP Verification (#FR003)	Sends OTPs to both mobile and email for verification. Requires success on both to proceed.
Document Upload Handling (#FR004)	Allows users to upload documents in PDF, JPG, or PNG, with validations on type and size.
Submission & System Notification (#FR005)	After form submission, sends notifications and logs data with reference IDs.
Admin Review Panel (#FR006)	Admin reviews user details, approves/rejects applications, and triggers credential generation.

User Login and Dashboard Access (#FR007)	Manages login, password reset, and role-based dashboard redirection.
Module: 2	
Licenses for Manufacturer / Dealer / Repairer (Centre & States): Manages license lifecycle for stakeholders including new license, renewal, amendments, and inspection with approvals.	
Functional Requirement	Description
Login and Dashboard Access (#FR001)	Personalized dashboard showing relevant licensing actions post-login.
Form Filling & Selection (#FR002)	Dynamic form rendering based on license type; includes save as draft.
Attachment & Document Upload (#FR003)	Allows users to upload license-specific documents based on selected license type.
Fee Calculation & Payment (#FR004)	Auto-calculates fees based on category; supports multiple payment modes.
Submission & Acknowledgement (#FR005)	Acknowledges application post-payment and queues it for inspection.
Inspection Assignment & Execution (#FR006–FR007)	Assigns inspectors, allows field visits, inspection report upload.
Query Management & Resubmission (#FR008–FR010)	Allows raising/resolving queries post-inspection for clarification.
Multi-Level Review (#FR011–FR014)	Routed to Assistant Controller, Divisional Controller, or Controller for tiered review.
License Issuance (#FR015)	System issues final license if all reviews are passed successfully.
Module:3	
Package Commodities Registration: Enables online registration and approval for manufacturers/packers/importers of packaged commodities.	
Functional Requirement	Description
Request Submission (#FR001)	Applicant submits initial request with necessary documents.
Notifications & Logging (#FR002)	Sends system-generated alerts; logs key steps.
Application Review & Feedback (#FR003–FR004)	Admin reviews and may request resubmission or reject.
Certificate Generation (#FR005)	Approved cases result in downloadable certificate.

Module:4	
Verification & Re-verification of Weights & Measures: Handles new verification and periodic re-verification of weighing/measuring instruments.	
Functional Requirement	Description
Application Submission (#FR001–FR002)	Form submission and upload of verification details.
Document Upload & Submission (#FR003–FR004)	Upload of supporting documents and confirmation.
Nodal Officer Review (#FR005–FR006)	Reviewed by District/Lab Nodal Officers for final validation.
Module:5	
Model Approval System (Centre): Manages approval of weight/measure models through RRSL and Director reviews.	
Functional Requirement	Description
Submission & Routing (#FR001–FR003)	Routes applications to LM Director and RRSL.
Final Review & Certification (#FR004–FR005)	Final Director approval and certificate generation.
Post-Certification Handling (#FR006)	Adds approved model to database for public reference.
Module:6	
Certificate of Importers (Centre): Provides a structured workflow for issuing importer certificates.	
Functional Requirement	Description
User Actions & Upload (#FR001–FR002)	Application initiation with required data/documents.
Director/Controller Review (#FR003–FR004)	Reviewed at central and state levels.
Certificate Issuance (#FR005)	Certificate issued and stored digitally.
Module:7	
Nomination of Directors of Companies: For companies to nominate directors under Legal Metrology rules.	
Functional Requirement	Description
Application, Document Upload & Review (#FR001–FR003)	Form filling, document upload, and inspection.
Query Resolution & Approval (#FR004–FR005)	Applicant addresses raised queries and awaits final approval.
Post-processing (#FR006)	Generates audit trails and maintains nomination records.

Module:8	
End-to-End Processing & Enforcement Tracking: Manages backend processing workflows and enforcement actions.	
Functional Requirement	Description
UI, Workflow & Review Engine (#FR001–FR003)	Unified UI with dynamic processing workflow and hierarchical review.
Query, Certification & Monitoring (#FR004–FR006)	Covers queries, certificate delivery, and system monitoring.
Module:9	
Real-Time Enforcement Tracking: Tracks real-time inspection, seizure, and legal processes.	
Functional Requirement	Description
Authentication & Case Initiation (#FR001–FR005)	Users initiate and record enforcement cases.
Offense Management & Judicial Steps (#FR006–FR010)	Tracks offenses, compounding, and legal proceedings.
Record, Reporting & Compliance (#FR011–FR016)	Generates reports, sends alerts, and ensures audit and compliance logs.

5.2 System Use Cases

Below is the generalized case model of System Use cases containing elements:

Module 1: Registration process on eMaap portal

Use Case 1: Stakeholder (Offender/Trader) Registration

Field Name	Description
Use Case ID	UC_REG_001
Use Case Name	Stakeholder (Offender/Trader) Registration
Created By (Date)	eMaap Functional Team (16-June-2025)
Last Update (Date)	16-June-2025
Primary Actor	Business Entity Representative / Trader
Secondary Actor	Admin (Reviewer), OTP Service, Email Gateway
Trigger Activating Use Case	User clicks on “Register New User” → selects “Stakeholder” from the role dropdown
Inter Linkages	OTP Verification, Document Upload, Admin Review Panel, Role-Based Dashboard
Normal Event Flow	1. User opens Stakeholder registration form 2. Fills in Business Name, Trade License, GSTIN, etc.

	3. Verifies mobile via OTP 4. Uploads Trade License/ID Proof 5. Submits form 6. Admin reviews and approves 7. Credentials are shared
Alternate Event Flow	• OTP fails → Retry up to 3 times • Missing document → Rejection or resubmission • Invalid business details
Frequency of Use	Once per stakeholder during account setup
Assumptions	Business has required documents, valid mobile/email
Additional Note	Business Category is selected via dropdown; dashboard includes notice and case tracking functionality

Use Case 2: Admin (Super / Regional Admin) Registration

Field Name	Description
Use Case ID	UC_REG_002
Use Case Name	Admin (Super/Regional) Registration
Created By (Date)	eMaap Functional Team (16-June-2025)
Last Update (Date)	16-June-2025
Primary Actor	Admin User (Applicant)
Secondary Actor	Approving Authority, OTP Service, Email Gateway
Trigger Activating Use Case	Admin applicant selects "Admin" as role during registration
Inter Linkages	OTP Verification, Document Upload, Nomination Letter Review, Role Assignment
Normal Event Flow	1. Applicant fills admin registration form with employee ID, authorization letter, etc. 2. Mobile/email OTP 3. Submit nomination 4. Admin approval 5. Credentials assigned
Alternate Event Flow	• Nomination not valid → rejection • OTP fails • Missing employee code or incorrect role selected
Frequency of Use	As needed for new admin onboarding
Assumptions	Role is approved internally before registration; proper authorization letter present
Additional Note (if any)	Access includes Master Data and User Management modules

Use Case 3: Legal Officer Registration

Field Name	Description
Use Case ID	UC_REG_003
Use Case Name	Legal Officer Registration
Created By (Date)	eMaap Functional Team (16-June-2025)
Last Update (Date)	16-June-2025
Primary Actor	Legal Officer / Judicial Liaison
Secondary Actor	Admin, OTP Service, Approving Authority
Trigger Activating Use Case	User selects "Legal Officer" role during registration
Inter Linkages	OTP Verification, Document Upload, Admin Review Panel
Normal Event Flow	1. Officer fills registration form 2. Enters employee ID and uploads authorization letter 3. OTP verification 4. Admin approves request 5. Account activated
Alternate Event Flow	• Invalid authorization • OTP expired • Rejected due to incomplete form
Frequency of Use	Low (only during onboarding of new Legal Officers)

Module 2: Licenses for Manufacturer/ Dealer / Repairer (Centre & States)

Field Name	Description
Use Case ID	UC_eMaap_LIC_004
Use Case Name	License Application, Inspection & Issuance
Reference	Based on FRS Numbers: FR001 to FR015
Created By (Date)	eMaap Functional team (16-June-2025)
Last Updated (Date)	16-June-2025
Primary Actor	Applicant (Manufacturer / Dealer / Repairer)
Secondary Actor	Inspector, Assistant Controller, Divisional Assistant Controller, Controller
Trigger Activating Use Case	User login to eMaap portal and chooses to apply for a new license or related service
Inter Linkages	Dashboard > Form Filling > Document Upload > Payment > Inspection > Review > Issuance
Normal Event Flow	1. Applicant logs into portal 2. Selects license type (New / Amendment / Renewal / Duplicate)

	3. System displays dynamic form based on role (Manufacturer/Dealer/Repairer) 4. Applicant fills and saves form (drafts supported) 5. Applicant uploads required documents 6. System validates document types/sizes 7. Fee auto-calculated and user pays (via UPI, Net banking, etc.) 8. Application Reference Number (ARN) generated 9. Inspector assigned based on jurisdiction 10. Inspector inspects premises, uploads report 11. Status marked as Verified / Pending / Non-compliant 12. If query arises, system notifies applicant 13. Applicant addresses and resubmits 14. Inspector reviews response and resolves 15. Application forwarded to appropriate Controller (based on type) 16. Controller reviews and approves or raises query 17. Final license document auto generated and stored in dashboard
Alternate Event Flow	1. If incomplete or incorrect document: system raises error before submission 2. If query is raised post-inspection: applicant receives notification and has 7 calendar days to respond 3. If query not resolved in two cycles, application may be rejected or reinitiated 4. In case of manual override required by Controller, physical verification can be initiated
Frequency of Use	Daily, depending on new applications and renewals across 36 states + Centre
Assumptions	• Applicant has access to required documents • Internet access and OTP/email verification functional • Jurisdiction mapping of inspectors and controllers is maintained in the backend
Additional Note	• Auto-trail maintained throughout for audit • Email/SMS updates sent to applicant after every major step • License is stored in applicant's dashboard under "My Licenses" section

Module 3: Package commodities Registration

Field Name	Description
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Use Case ID	eMaap_PC_005
Use Case Name	Registration / Amendment of Packaged Commodities
Reference	FRS Numbers: FRS_001 to FRS_005
Created By (Date)	eMaap Functional team (16-June-2025)
Last Update (Date)	16 June 2025
Primary Actor	Business User (Manufacturer, Packer, Importer)
Secondary Actor	Controller / Director, Legal Metrology (Centre or State)
Trigger Activating Use Case	Business user initiates application for registration or amendment of packaged commodities.
Inter-linkages	1. User Authentication System 2. Dashboard and Notification Engine 3. Certificate Generation Module 4. Document Management System 5. Audit Trail and Logging
Normal Event Flow	1. User logs in via eMaap portal 2. Navigates to “Packaged Commodity Registration/Amendment” module 3. Fills application form with details 4. Uploads necessary documents 5. Submits application (eMaap_RS_005) 6. System logs and notifies Controller/Director 7. Controller reviews application 8. If approved, certificate is auto generated 9. Certificate is available for download in user dashboard
Alternate Event Flow	1. Application is Rejected or requires Update 2. Status is updated to “Rejected” or “Update Required” 3. User receives comments, resubmits updated application 4. Application is re-reviewed for final decision
Frequency of Use	As required per business need – typically during: 1. New product registration 2. Packaging changes 3. Regulatory updates 4. Amendments or renewals
Assumptions	1. User is a valid registered business entity with login credentials 2. Controller/Director has appropriate permissions for review 3. Portal has uptime and document upload functionality 4. Regulatory rules are configured in the system
Additional Note	1. The QR code on the certificate provides a

	mechanism for field officers and consumers to verify the authenticity of the registration. 2. System maintains full audit logs for all approvals, rejections, and modifications.
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Module 4: Verification & Re-verification of weights and Measures

Field Name	Description
Use Case ID	eMaap_VR_006
Use Case Name	Verification & Re-verification of Weights & Measures
Reference	FRS Numbers: FR001 to FR006S
Created By (Date)	eMaap functional team (16 June 2025)
Last Update (Date)	16 June 2025
Primary Actor	Applicant (Repairer / Manufacturer / Dealer)
Secondary Actor	District Nodal Officer, Lab Nodal Officer, Legal Metrology Officer (LMO)
Trigger Activating Use Case	Applicant initiates a request for verification or re-verification of a weight or measure instrument.
Inter-linkages	1. User Authentication Module 2. Dashboard and Notification Engine 3. Document Management System 4. Payment Gateway 5. Certificate Generation Module 6. Audit Log and Rejection Tracking
Normal Event Flow	1. Applicant logs in and selects 'Verification/Reverification' 2. Fills instrument details, uploads documents, and selects jurisdiction 3. Documents are uploaded and validated 4. System validates form and submits application 5. District Nodal Officer reviews and assigns Lab Nodal Officer 6. Lab Officer processes application, schedules verification, performs inspection 7. Upon conformity, certificate is generated and shared with applicant
Alternate Event Flow	1. If query is raised, applicant is notified and must update/resubmit 2. If instrument is non-compliant, rejection certificate issued 3. Applicant may correct and reapply; system maintains traceability to previous rejected application

Frequency of Use	As per statutory requirements – typically: 1. Fresh verification (prior to use) 2. Re-verification (annually or biennially based on instrument type) 3. Post repair or modification
Assumptions	1. Applicants are registered and verified on the eMaap Portal 2. Legal Metrology infrastructure and Labs are geo-tagged and functional 3. Officers are trained in digital verification flow 4. Payment integration and receipt generation systems are live
Additional Note	1. System logs every action by officer and applicant for audit 2. QR-enabled certificates facilitate field verification 3. Rejected applications are version-controlled and traceable

Module 5: Modal approval system

Field Name	Description
Use Case ID	eMaap_MA_007
Use Case Name	Model Approval System – Centre
Reference	FRS Number: FR001 to FR006
Created By (Date)	eMaap functional team (16 June 2025)
Last Update (Date)	16 June 2025
Primary Actor	Applicant (Manufacturer/Importer submitting model approval request)
Secondary Actor	Director, Legal Metrology (Centre); Regional Reference Standard Laboratory (RRSL)
Trigger Activating Use Case	Applicant logs in to the eMaap portal and submits a new model approval request
Inter-linkages	1. Dashboard Module 2. RRSL Interface Module 3. Certificate Management Module 4. Audit Trail 5. Notification Engine
Normal Event Flow	1. Applicant logs in and navigates to “Model Approval” 2. Fills out model details (e.g., instrument type, category, purpose) 3. Uploads technical documents 4. Submits the application 5. Director LM receives and reviews

	- Forwards to RRSL for evaluation - RRSL performs tests and submits report - Director LM takes final action (Approve/Reject/Send for update) - If approved, certificate is auto generated and available to applicant
Alternate Event Flow	- If documents are incomplete or invalid, application is returned to applicant with comments - Applicant updates and resubmits - If RRSL report is negative, application is rejected
Frequency of Use	As and when a new model is to be introduced in the market (per manufacturer or importer)
Assumptions	- RRSL is digitally integrated with the eMaap system - Users have access to full technical specs - Certificates are digitally signed and stored
Additional Note	- Audit logs are maintained for every user action - Model once approved can be used for further manufacturer or import licenses - The certificate contains QR code for verification

Module 6: Certificate of Importers

Field Name	Description
Use Case ID	eMaap_CIWM_008
Use Case Name	Certificate of Importers for Weights and Measures
Reference	FRS Number: FR001 to FR005
Created By (Date)	eMaap functional team, (16-06-2025)
Last Update (Date)	16-06-2025
Primary Actor	Business User / Importer
Secondary Actor	Director (Legal Metrology), State Controller (LM)
Trigger Activating Use Case	Submission of Importer Certificate Application on eMaap Portal
Inter Linkages	Dashboard module, Document Upload module, Notification module, Certificate generation module
Normal Event Flow	- User logs in to the portal. - Navigates to Importer Certificate module. - Fills application form and uploads required documents. - Submits the application. - Director reviews and forwards to Controller. - Controller verifies and returns remarks. - Director approves and generates certificate. - Certificate available in user dashboard.

Alternate Event Flow	1. If incomplete, Director returns for update. 2. User re-submits with corrections. 3. If Controller raises issue, Director informs user. 4. User may reapply after corrections.
Frequency of Use	As per business need, typically once per importer annually or when changes occur.
Assumptions	1. User has valid login credentials. 2. Internet connectivity is stable. 3. Uploaded documents meet size and format criteria.
Additional Notes	1. Certificate is digitally signed. 2. Notifications are sent via SMS/email. 3. All actions are audit logged for transparency.

Module 7: Nomination of Directors of companies

Field Name	Description
Use Case ID	eMaap_NDC_009
Use Case Name	Nomination of Directorates of companies
Reference	FRS Number: FR001 to FR006
Created By (Date)	eMaap functional team, (16-06-2025)
Last Updated (Date)	16-06-2025
Primary Actor	Applicant / Business User
Secondary Actor	Legal Metrology Controller, System Admin
Trigger (Activating Use Case)	Business logs in to nominate a new director
Inter Linkages	User Registration, Document Upload Module, Notification System
Normal Event Flow	Typical sequence of steps: 1. Applicant logs in 2. Fills nomination form 3. Uploads documents 4. Submits to Controller 5. Controller reviews 6. Application approved or query raised.
Alternate Event Flow	Steps taken when issues arise: 1. Controller raises query 2. Applicant modifies & resubmits form 3. Re-inspection happens 4. Approval or rejection based on corrections.
Assumptions	Preconditions or expectations: 1. Applicant is a registered user 2. Director details and documents are valid 3. Legal Metrology Controller has access to verify documents.

Additional Notes	Any special notes, such as: 1. Acknowledgement is digitally available 2. Nomination is subject to Legal Metrology Division's jurisdiction. 3. Notifications are system-generated via portal dashboard/email/SMS.
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Module 8: End to End processing enforcement

Field Name	Description
Use Case ID	eMaap_EEP_010
Use Case Name	End to End Processing
Reference	FRS Number: FR001 to FR006
Created By (Date)	eMaap functional team, (16-06-2025)
Last Updated (Date)	16-06-2025
Primary Actor	eMaap Backend System / LMD Backend Operator.
Secondary Actor	State API Systems, Controllers, Admins.
Trigger (Activating Use Case)	Registration, license, verification via eMaap portal.
Inter Linkages	User Dashboard, State Approval Workflows, API Integration Layer, Notification System.
Normal Event Flow	Typical sequence of backend actions: 1. Request submission by user 2. Routed to appropriate state/center backend 3. Approval workflow initiated 4. API call made 5. Status updated and displayed.
Alternate Event Flow	If a state's API is unresponsive or fails: 1. Manual intervention logged 2. Notification sent to backend admin 3. Retry or escalate depending on configuration.
Frequency of Use	Frequently used as all user-initiated services pass through backend processing.
Assumptions	All state APIs are functional and accessible. Approval hierarchy is properly mapped for each state and Centre.
Additional Notes	Module supports centralized monitoring. All logs are stored for audit purposes. Integrated notification for real-time status updates.

Module 9: Real-time enforcement tracking

Field Name	Description
------------	-------------

Use Case ID	eMaap_TTE_011
Use Case Name	Time Tracking Enforcement
Reference	FRS Number: FR001 to FR016
Created By (Date)	eMaap functional team, (16-06-2025)
Last Updated (Date)	16-06-2025
Primary Actor	Inspector, Assistant Controller
Secondary Actor	Business Representative, Witness, Controller, Judicial Authority, eMaap Notification System.
Trigger (Activating Use Case)	Starts when an inspector begins an inspection or logs an enforcement activity via the portal.
Inter Linkages	Connected with: 1. Geo-tagging Service 2. Notification Module 3. Case History Database 4. Compounding Fee Payment System 5. Court Case Filing Module.
Normal Event Flow	1. Inspector conducts inspection with witness 2. Deficiency check 3. If compliant, case closed 4. If not, seizure and offence logging 5. Check repeat offence 6. For first-time, notices issued (up to 3) 7. Compounding fee requested 8. On payment, issue receipt and close case 9. If refused, escalate to court and update records.
Alternate Event Flow	1. If Geo-tag fails – fallback to manual entry 2. If compounding not accepted – judicial sanction required 3. If court judgment delays – system marks case as pending and notifies admin.
Frequency of Use	Frequent – used by all enforcement officers during routine and complaint-based inspections.
Assumptions	1. Officers have real-time access to eMaap 2. Business representatives are reachable via email/SMS 3. Geo-tagging service is operational Legal framework supports digital compounding.
Additional Notes	Supports transparency, accountability, and progressive enforcement Appeal logging by businesses is integrated Data available for audit and reporting purposes.

Module 10: Dashboard Module

Field Name	Description
Use Case ID	eMaap_DM_012
Use Case Name	Role-Based Dashboard and Reporting
Created By (Date)	eMaap functional team, (16-06-2025)
Last Updated (Date)	16-06-2025
Primary Actor	Applicant, Inspector, Zonal Officer, Department Head
Secondary Actor	System Scheduler, Admin, Report Viewer, Notification System.
Trigger (Activating Use Case)	Initiated when a user logs in to the portal and accesses the dashboard module.
Inter Linkages	Connected to: 1. MIS Reporting Service 2. Data Aggregation Layer 3. Scheduling & Export Module 4. User Role Management.
Normal Event Flow	1. User logs into eMaap portal 2. Navigates to dashboard 3. Selects report parameters (role, date, region, status, etc.) 4. System retrieves and visualizes data 5. User views/downloads or schedules reports.
Alternate Event Flow	1. If data source unavailable – show alert with retry option 2. If report exceeds size limit – user advised to refine filters 3. If export fails – allow retry or report issue to admin.
Frequency of Use	Frequent – used daily by officers, weekly/monthly by department heads for performance and revenue monitoring.
Assumptions	1. Data is synchronized in real-time 2. Users have assigned roles and access rights 3. System can handle large data volume and simultaneous requests.
Additional Notes	Supports drill-down and graphical reports Exportable in PDF/Excel/CSV Custom MIS reports and auto-scheduling supported.

5.3 Sequence Diagram

User Registration and Activation Sequence Diagram

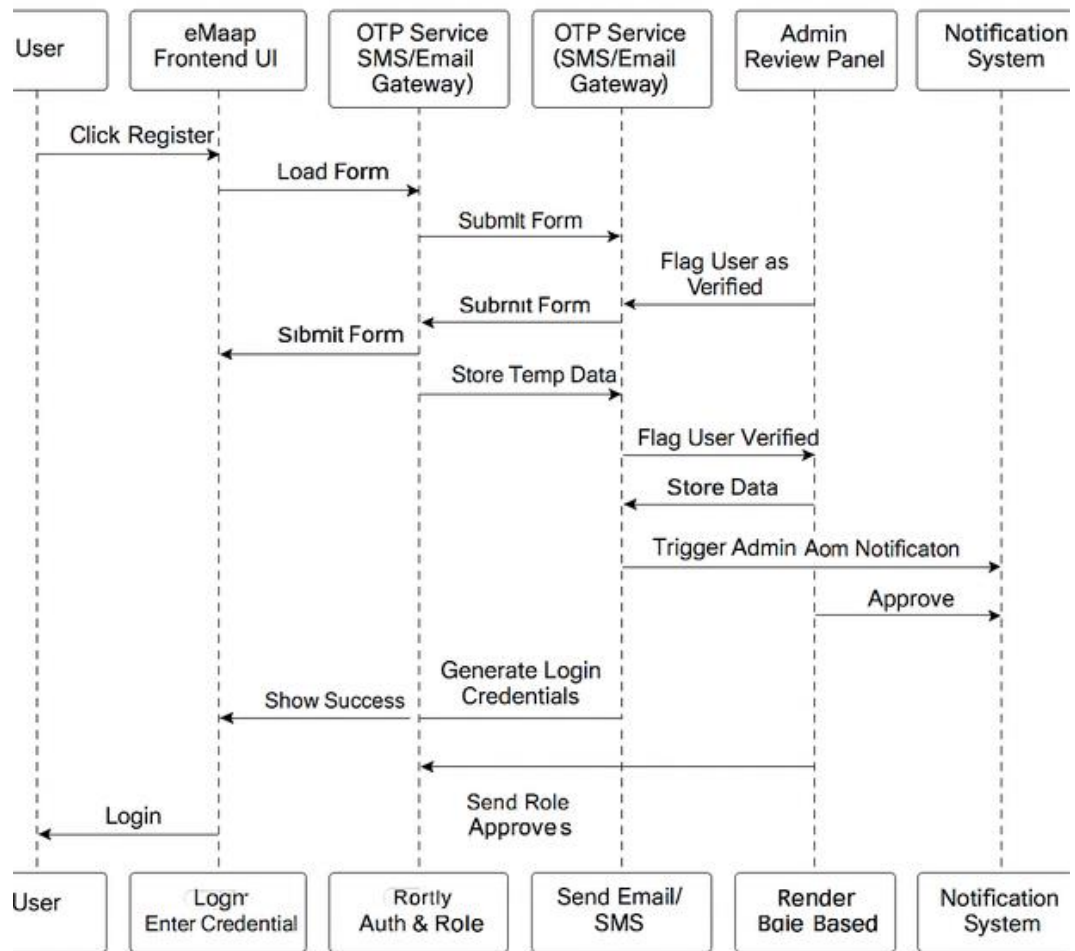
Actors & Objects (Lifelines):

- User (Stakeholder / Inspector / Admin)
- eMaap Frontend UI
- eMaap Backend Services
- OTP Service (SMS/Email Gateway)
- Admin Review Panel
- Database
- Notification System

Key Information Exchanges:

1. **Registration Initiation:** Role-based form rendering based on user type.
2. **OTP Verification:** Dual verification via email and SMS with retry logic and time-bound validation.
3. **Form Submission:** Data validation, document upload, and temp storage.
4. **Admin Review:** Approval or rejection with audit trail, reason logging, and credential provisioning.
5. **Notification Services:** Real-time alerts to user and admin via email/SMS.
6. **Login & Role-Based Routing:** First-time login triggers password reset and routes to specific dashboard.

Sequence Diagram – User Registration & Activation (eMaap)



Sequence Diagram – User Registration & Activation (eMaap)

5.4 Database Design

The eMaap system supports modules like Registration, Licensing, Verification, Package Commodity Approval, Enforcement Tracking, and more. This database design captures all key entities and relationships, aligned with those modules.

Logical Data Model (Key Tables)

Table Name	Description
Users	Stores all platform users including stakeholders, inspectors, and admins.
Roles	Defines roles assigned to each user (Inspector, Admin, Stakeholder, etc.)
Registrations	Tracks user registration attempts and statuses.

Licenses	Stores license applications and issue license details.
Documents	Stores metadata of uploaded documents.
Otp_Verification	OTP details for mobile/email verification.
Inspections	Tracks inspections for license approval or enforcement.
Payments	Stores payment details for services.
Queries	Stores system-generated and user-resolved queries.
Notifications	Tracks email/SMS notifications.
Audit_logs	Logs all critical system/user events.
Geotags	Geo-coordinates for premises or inspections.

Data Dictionary (Sample)

Field Name	Table	Type	Length	Description	Source	Validation Rules	CRUD
User_id	users	UUID	36	Unique user identifier	System	Auto-generated UUID	C/R/U/D
Full_name	users	VARCHAR	100	Full name of the user	User	Not null	C/R/U
Email	users	VARCHAR	100	Email address	User	RFC 5322-compliant, unique	C/R/U
Mobile	users	VARCHAR	10	Mobile number	User	Starts with 6/7/8/9, 10 digits	C/R/U
Role_id	users	FK (UUID)	36	User's role	System	FK to roles	C/R/U
License_type	licenses	VARCHAR	30	Type of license	User	Must match dropdown values	C/R/U/D
Document_type	documents	VARCHAR	50	Type of uploaded document	System	Dropdown based on license type	C/R/U/D
Geo_latitude	geotags	DECIMAL(9,6)	-	Latitude of the premises	System	Must be a valid geo-coordinate	C/R/U/D
Geo longitude	geotags	DECIMAL(9,6)	-	Longitude of the premises	System	Must be a valid geo-coordinate	C/R/U/D

CRUD Maintenance Summary

Table Name	Create	Read	Update	Delete
Users	✓	✓	✓	✓
Roles	✓	✓	✓	✗ (soft delete)
Licenses	✓	✓	✓	✓
Documents	✓	✓	✓	✓
Otp_Verification	✓	✓	✓	✓
Inspections	✓	✓	✓	✓
Payments	✓	✓	✓	✗
Queries	✓	✓	✓	✓
Notifications	✓	✓	✗	✗
Audit_logs	✓	✓	✗	✗
Geotags	✓	✓	✓	✓

5.4.1 Data Objects and Resultant Data Structures

This section describes the data structures passed between functional components (e.g., UI, backend services, DB), along with their associated functions, input/output parameters, and usage in workflows. These structures support modules such as registration, licensing, inspections, verification, notifications, payments, and geo-tagging.

Data Objects and Structures

User Registration Object: -

Used to register a new stakeholder, inspector, or admin.

Field	Type	Description
Full_name	string	User's full name
email	string	RFC 5322-compliant unique email
Mobile	string	10-digit Indian mobile
Role_type	enum	Inspector / Stakeholder / Admin

Employee_id	string	Required for inspectors/admins (optional)
Business_id	string	Required for stakeholders (optional)
Otp_verified	boolean	Flag for OTP status
Address	string	Physical address
Created_at	datetime	System-generated timestamp

Document Upload Object:

Used to upload and validate supporting documents.

Field	Type	Description
User_id	UUID	Reference to `users` table
Document_type	string	E.g., PAN, GST, Trade License
File_name	string	Filename with extension
File_type	enum	PDF / JPEG / PNG
File_size	integer	Max: 5MB
Upload_status	boolean	Success or failure
Storage_path	string	Cloud location

OTP Verification Object:

Manages dual OTP (email + mobile) verification.

Field	Type	Description
User_id	UUID	Foreign key
Otp_type	enum	Email / Mobile
Otp_code	string	6-digit numeric or 6–8 alphanumeric
Expiry_time	datetime	OTP expiry (5 minutes)
Retry_count	integer	Max 3 attempts

License Application Object:

Captures application data for New/Renewal/Amendment of licenses.

Field	Type	Description
License_id	UUID	Primary key
User_id	UUID	Applicant
license_type	enum	Manufacturer / Dealer / Repairer
status	enum	Submitted / Under Review /

		Approved
Submission_date	datetime	Timestamp
Payment_id	UUID	Reference to payment object
Document_ids	array of UUID	Linked documents

Inspection Report Object:

Stores inspection metadata and outcome.

Field	Type	Description
Inspection_id	UUID	Primary key
License_id	UUID	License under inspection
Inspector_id	UUID	User ID of inspector
Scheduled_date	datetime	Date assigned
Inspection_status	enum	Scheduled / Completed / Queried
Remarks	string	Inspection notes
Geo_tag_id	UUID	GPS location of premises

Geo Tag Object:

Captures geo-location data of user premises or inspections.

Field	Type	Description
Geo_tag_id	UUID	Unique ID
Latitude	decimal (9,6)	Geo coordinate
Longitude	decimal (9,6)	Geo coordinate
Captured_by	UUID	User ID
context	string	Inspection / Registration
timestamp	datetime	When location was recorded

Query Object:

Stores system-generated queries to applicant for correction.

Field	Type	Description
Query_id	UUID	Primary key
License_id	UUID	Application associated
Created_by	UUID	Admin/Inspector
Query_text	text	Description of issue
status	enum	Open / Resolved
Resolved_at	datetime	Timestamp of resolution

Notification Object:

Sends notifications to users at every critical stage.

Field	Type	Description
Notification_id	UUID	Primary key
User_id	UUID	Recipient
type	enum	SMS / Email
message	text	Notification content
status	enum	Sent / Failed
Delivery_time	datetime	Actual delivery timestamp

Payment Object:

Tracks fees paid via UPI/Credit Card/Net Banking.

Field	Type	Description
Payment_id	UUID	Primary key
User_id	UUID	Payer
License_id	UUID	Associated license
Amount	decimal	Fee calculated
Mode	enum	UPI / Card / Net Banking
Status	enum	Initiated / Successful / Failed
Txn_ref_no	string	Payment gateway reference
Paid_at	datetime	Timestamp

Functions and Functional Parameters

Function Name	Description	Input	Output
Register User	Initiates user registration and stores data	User Registration Object	user_id
Verify OTP	Validates email/mobile OTP	OTP Verification Object	boolean
Upload Document	Validates and stores uploaded files	Document Upload Object	upload_status
Submit License Application	Persists license application with docs	License Application Object	license_id
Assign Inspector	Assigns inspector to license	license_id, inspector_id	assignment_status
Record Inspection	Stores inspection report and geo-tag	Inspection Report Object	inspection_id

Capture Geo Tag	Saves GPS coordinates	Geo Tag Object	geo_tag_id
Raise Query	Raises query to applicant	Query Object	query_id
Send Notification	Sends email/SMS	Notification Object	delivery_status
Process Payment	Confirms payment and links license	Payment Object	payment_status

5.4.2 File and Database Structures

The eMaap portal is a centralized, cloud-based system designed to manage end-to-end Legal Metrology operations across Centre and States. Its data architecture follows a modular, multi-tenant design aligned with statutory workflows. The data model will ensure integrity, security, traceability, and auditability of registrations, licenses, inspections, verification records, and enforcement actions.

Logical Data Model (LDM)

The Logical Data Model identifies and normalizes the main entities and their relationships. Core entities include:

Entity	Key Attributes	Relationships
User	User_ID (PK), Role_ID (FK), Name, Mobile, Email, Aadhaar/PAN, Status	One-to-Many: Roles, Applications
Role	Role_ID (PK), Role_Name, Permissions	One-to-Many: Users
Application	Application_ID (PK), User_ID (FK), License_Type, Status, Submission_Date	One-to-One: Payment; One-to-Many: Documents, Inspections
Document	Document_ID (PK), Application_ID (FK), File_Path, Type, Upload_Date	One-to-One: Application
Payment	Payment_ID (PK), Application_ID (FK), Amount, Status, Transaction_Ref	One-to-One: Application
Inspection	Inspection_ID (PK), Application_ID (FK), Inspector_ID (FK), Date, Outcome, Geo_Location	One-to-One: Application
Notification_Log	Notification_ID (PK), User_ID (FK), Message, Channel, Status,	One-to-Many: User

	Timestamp	
Audit_Trail	Log_ID (PK), User_ID (FK), Action, Timestamp, IP_Address	One-to-Many: User
Master_Data	Various lookup tables: States, LGD Codes, License_Types, Fee_Structured	Referenced by multiple modules

Key LDM features:

- Each core business process (registration, licensing, verification, inspection) maps to a distinct entity.
- Referential integrity is enforced through foreign keys.
- Sensitive attributes (Aadhaar, PAN) are encrypted.
- Audit Trail captures all CRUD operations

Physical Data Model (PDM)

The PDM translates the LDM into actual physical structures for the chosen DBMS: **PostgreSQL** (or other enterprise-grade RDBMS).

DBMS Chosen:

- **Type:** Relational DBMS with ACID compliance.
- **Deployment:** Cloud-hosted (e.g., NIC Cloud or MeitY empaneled CSP).
- **Clustering:** Master-slave replication for high availability.
- **Backup:** Daily full backups and hourly incremental backups.

Table Structures:

Tables will be created for each entity, with:

- Proper indexing on PKs and FKs.
- Partitioning where appropriate (e.g., by State/UT for multi-tenant scaling).
- Views for role-based data access.

File Storage:

Documents uploaded by users (PDF, JPEG, PNG) are **not stored in the DBMS** but in **secure cloud object storage**, referenced by file paths in the Document table.

File Storage Structure:

```
/secure_bucket/
├── User_ID/
```



```

└─ Application_ID/
   └─ document_type/
      └─ filename.pdf

```

- Access control lists (ACLs) ensure that only authorized roles (Inspectors/Admins) can view or download documents.
- Files are encrypted at rest and in transit.

Location of Data Structures

Component	Location
Database Tables & Views	Cloud-hosted RDBMS (Primary Data Center, NIC Cloud)
Files/Documents	Object storage (e.g., S3-compatible) within India, under government cloud policy
Configuration & Master Data	Separate schema, managed centrally with version control
Backup Storage	Secondary DR site with periodic sync

Data Manipulation & Access

Operation	Mechanism
CRUD	RESTful APIs with RBAC and audit trails
Integration	Secure APIs for UIDAI, PAN, GSTIN, State LM Portals
Bulk Uploads/Imports	Admin interfaces with validation and rollback on failure
Reports & Dashboards	Materialized views + caching for high-frequency queries
Encryption	PII fields in tables encrypted with industry-standard algorithms
Monitoring	DB logs, file access logs, and system logs consolidated for SIEM analysis

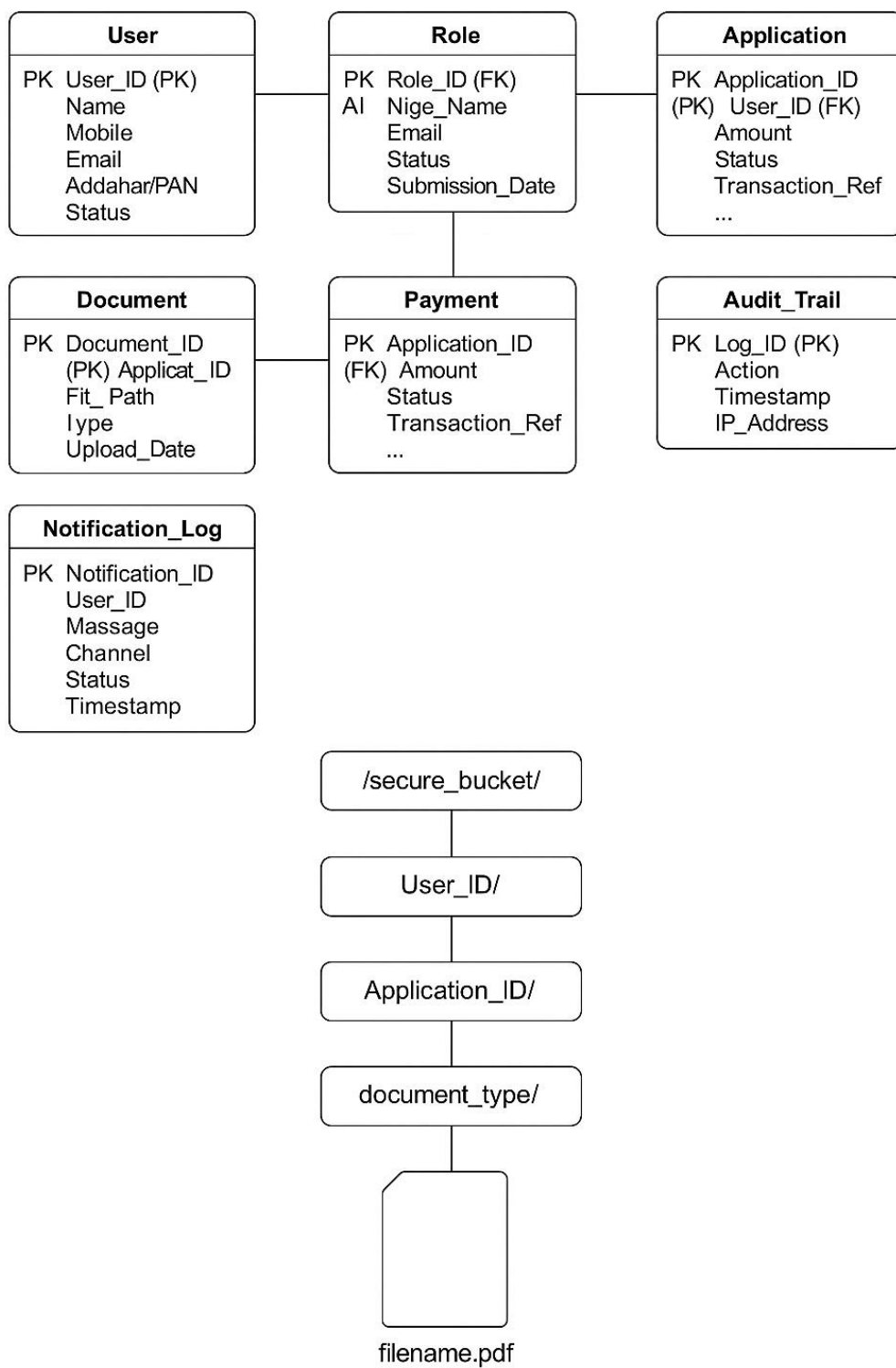
DBMS-specific features

- **PostgreSQL JSONB:** For storing unstructured form data snapshots.
- **Triggers & Stored Procedures:** For automatic audit log generation.
- **Partitioning:** By financial year or State for large license datasets.
- **Full-Text Search:** For inspecting uploaded documents metadata.
- **Geo-Spatial Extensions:** For storing geo-tagged inspection locations.

Entity-Relationship Diagram (ERD) and File Structure Flowchart

The complete Entity-Relationship Diagram (ERD) for the eMaap Legal Metrology system with the secure file storage flowchart. It shows all entities, attributes, primary and foreign keys, relationships, and the organized file storage hierarchy from /secure_bucket/ to document files.

File and Database Structures



This structure ensures secure, transparent, and scalable data handling, aligning with regulatory and governance requirements for Legal Metrology.

5.5 Data Conversion

Data conversion in the eMaap system is necessary to transition legacy or paper-based records from existing Legal Metrology (LM) systems (at both State and Central levels) into the new unified digital platform. This ensures business continuity, regulatory consistency, and historical data accessibility for stakeholders such as manufacturers, dealers, inspectors, and government administrators.

Scope of Data Conversion

The conversion encompasses the following types of legacy data:

- **Licensing Records** (Manufacturers, Dealers, Repairers)
- **Registration Data** (Packers, Importers)
- **Inspection and Verification Logs**
- **Case History and Enforcement Data**
- **Historical Certificate Information**
- **Stakeholder Information** (Contact, PAN, GST, Aadhaar)
- **Application Status & Document Trails**

Sources of Data

- State Legal Metrology Systems (digitized data from individual states)
- Manual paper-based records (digitized and validated)
- Excel-based databases or Access DBs used by district offices
- Archived PDFs, scanned licenses, and inspection documents
- Data collected via offline inspections and legacy mobile apps

Conversion Approach

➤ **Data Mapping**

A detailed **Data Mapping Document** (DMD) is to be created. This will:

- Map each field from the source schema to the target schema in eMaap.
- Define transformation rules (e.g., converting free text to dropdown-based enums).
- Include validation rules and format changes (e.g., date format normalization, document ID structures).

➤ **Data Cleansing**

Legacy data will be subjected to:

- De-duplication
- Field standardization (e.g., state codes aligned with LGD codes)

- Format conversions (e.g., PDF scanning to text-tagged records using OCR)
- Removal of corrupt/incomplete records (with audit logs)

➤ **Import Templates**

Custom **Excel/CSV import templates** will be prepared for:

- Licensees
- Stakeholders
- Inspection logs
- Uploaded legacy documents (with metadata)

Note : These will be shared with state departments for manual input or assisted data entry.

Documented Artifacts

The following documents are to be created and stored for auditing and rollback purposes:

Document	Description	Storage Location
Data Mapping Document (DMD)	Defines source-to-target field relationships and transformations	/conversion_docs/mapping/
Data Cleansing Log	Track of removed, corrected, or modified data entries	/conversion_logs/cleansing/
Validation Reports	Reports showing conversion errors, format mismatches	/Conversion_reports/validation/
State-wise Import Sheets	Excel templates used to collect and input state data	/Conversion_templates/states/
Pre-conversion Backups	Raw dumps from source systems	/Conversion_backups/pre_migration/
Post-conversion Verification Report	Output comparison against expected records	/Conversion_reports/post_validation/

System and Storage

All converted data is to be stored in the **centralized eMaap data repository** under modular folders:

- /Central_repo/licenses/
- /Central_repo/inspections/
- /Central_repo/stakeholders/
- /Central_repo/documents/

Cloud object storage (e.g., NIC cloud or S3-compatible) will be used for:

- Historical documents (PDF, JPEG)
- Scanned forms
- Signed certificates

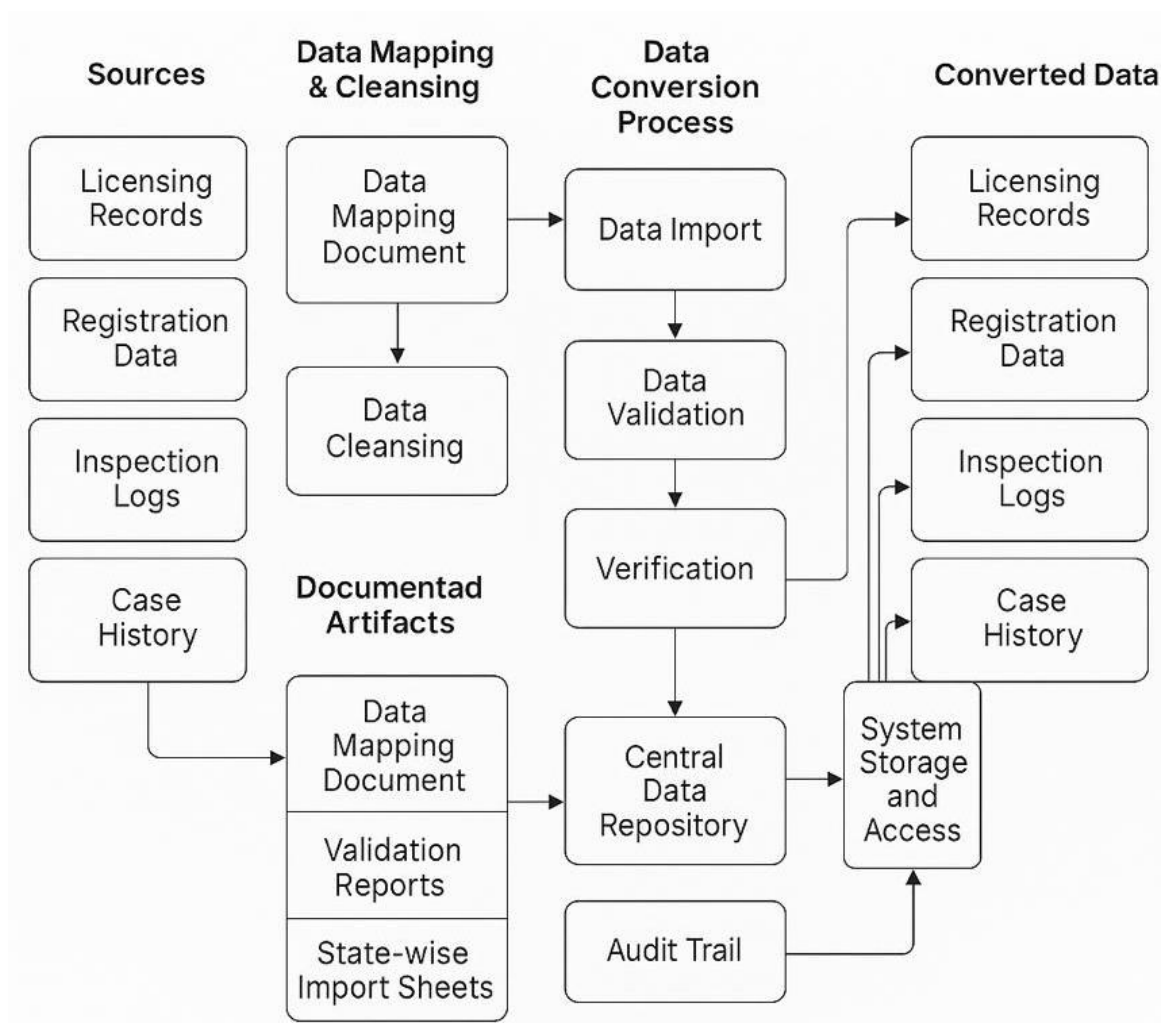
Access Control and Audit

- Only authorized Admins and Data Migration Teams will have access to conversion folders.
- Every data import operation is logged with:
 - Timestamp
 - User ID
 - Number of records processed
 - Success/Failure count
- Audit trail maintained as per NIC/MeitY e-Governance Guidelines.

Post-Conversion Activities

- Manual validation by district/state nodal officers.
- Automated cross-verification using random sampling.
- Feedback loop for states to raise issues within 30 days.
- Freeze old systems or make them read-only post full validation.

Data Conversion Flowchart



5.6 User-Machine Readable Interface

This section outlines the user classes interacting with the eMaap system and describes their typical roles, interaction modes, skill levels, and access expectations. The eMaap platform has been designed to support a range of user profiles—each with customized dashboards, workflows, and authorization levels. Interaction with the system is predominantly through web-based and mobile interfaces, with access governed via role-based authentication and secure session management.

Stakeholders (Manufacturers / Dealers / Repairers / Importers / Packers)

- **Responsibilities:** Submit applications for licenses, upload documents, respond to inspection queries, track application status, and download certificates.
- **Skill Level:** Basic to moderate digital literacy.

- **Work Activities:** Form submissions, document uploads, fee payments, viewing dashboards, and responding to system notifications.
- **Interface Interaction:** Web portal or mobile app via guided forms and dashboard widgets.
- **Estimated Users:** 1,00,000+ entities pan-India.
- **Concurrent Users:** ~5,000.
- **External Users:** All users in this class are external.

Inspectors / Field Officers

- **Responsibilities:** Conduct inspections, upload reports, flag non-compliance, raise queries, geo-tag premises.
- **Skill Level:** Moderate.
- **Work Activities:** Use mobile-enabled inspection forms, location tagging, uploading images and documents.
- **Interface Interaction:** Mobile app and web portal dashboards.
- **Estimated Users:** 5,000–7,000.
- **Concurrent Users:** ~1,000.
- **External Users:** Government field personnel, considered semi-internal.

Admins / Controllers / Assistant Controllers

- **Responsibilities:** Review applications, assign inspectors, approve/reject licenses, manage workflow queues.
- **Skill Level:** High.
- **Work Activities:** Case evaluations, system notifications, report generation.
- **Interface Interaction:** Advanced role-specific dashboards and master data modules.
- **Estimated Users:** 1,000.
- **Concurrent Users:** 200–300.
- **External Users:** No (Internal government users).

Legal Officers / Enforcement Officials

- **Responsibilities:** Register offenses, track violations, manage judicial proceedings, compound fees.
- **Skill Level:** Moderate to High.
- **Work Activities:** Offense profiling, notice generation, documentation uploads.
- **Interface Interaction:** Enforcement module dashboard with document handling features.
- **Estimated Users:** 500.
- **Concurrent Users:** 50–100.
- **External Users:** No (Internal to enforcement/legal divisions).

System Operators / NIC IT Support

- **Responsibilities:** Maintain application uptime, manage data backups, system health, and configurations.
- **Skill Level:** Expert.
- **Work Activities:** Configuration, log review, API management.
- **Interface Interaction:** Admin and backend configuration panels.
- **Estimated Users:** 50–100.
- **Concurrent Users:** 10–20.
- **External Users:** No (Internal technical support personnel).

Trainers / Support Personnel

- **Responsibilities:** Guide users in onboarding, conduct training sessions, provide operational help.
- **Skill Level:** Moderate.
- **Work Activities:** Helpdesk management, query resolution, training program management.
- **Interface Interaction:** Helpdesk dashboards, knowledge base systems.
- **Estimated Users:** 100–200.
- **Concurrent Users:** ~50.
- **External Users:** Semi-external (contracted or NIC-aligned personnel).

5.6.1 Inputs

Input Media & Data flow Mapping

Input Medium	Purpose	Mapped Module
Web Forms (portal)	Registration, license applications, document upload	Registration, License, Verification modules
Mobile App Forms	Field inspection reports, geotagging, evidence upload	Inspection & Enforcement modules
Scanned Forms (if any)	Offline capture of enforcement or court documents	Enforcement & Legal Filing modules
APIs (external integrations)	PAN/GSTIN/Aadhaar validation, license verifications	State/Center department APIs

Data Element Definition & Edit Criteria

Field	Data Type	Length	Mandatory	Validations / Value Range
Full Name	Alphanumeric	3–100	Yes	No special characters, must contain space
Mobile Number	Numeric	10	Yes	Starts with 6-9; OTP validation
Email Address	Alphanumeric	5–100	Yes	RFC 5322 format; OTP verification
PAN	Alphanumeric	10	Conditional	5 letters + 4 digits + 1 letters
GSTIN	Alphanumeric	15	Conditional	Format: State Code + PAN + Suffix
Aadhaar	Numeric	12	Optional	Must pass UIDAI mask checks
File Upload (ID/License)	PDF/JPEG/PNG	≤ 5MB	Yes	Format & size restriction; virus scan
Password	Alphanumeric + Special Char	8–16	Yes	Includes uppercase, special char, number

Data Entry Controls

Client-side validations using JavaScript for real-time checks.

- Server-side validation on form submission.
- Duplication checks (email, PAN, mobile).
- File size and format enforced using MIME-type checking.
- Input sanitization to prevent XSS/SQL injection.
- Logs and audit trails for all user submissions.

Messages Associated with User Inputs

Scenario	System Message
Invalid mobile/email format	Please enter a valid 10-digit mobile / correct email.
Duplicate registration	User already exists with the provided credentials.

Invalid OTP	Incorrect OTP. You have X attempts remaining.
Missing mandatory fields	Please complete all required fields before submission.
File exceeds size limit	File exceeds 5MB limit. Upload a smaller document.
Server validation failed	Unexpected error. Please try again later.

Printed form Inputs (if applicable)

Printed forms may be scanned and uploaded in specific modules such as enforcement or court processing:

- Accepted formats: Scanned JPEG or PDF
- Input path: Uploaded via enforcement or legal module dashboard
- Use: For registering seizures, offenses, or compounding actions

Access Restriction & Security Consideration

Role-Based Access Control (RBAC) with restricted permissions

- OTP-based user authentication
- Secure session tokens with encryption and timeout
- Logging and audit trails for user actions
- Virus scanning and secure cloud storage for uploaded files

Translation – Based system Metadata

Transaction Name	Transaction Code	Description
New Registration	TXN_REG_001	Registers a new user (Inspector/Stakeholder/Admin)
License Application	TXN_LIC_001	Initiates a license request for manufacturer/dealer etc.
Verification Submission	TXN_VER_001	Submits application for inspection
Offense Registration	TXN_ENF_001	Logs offense in enforcement module

Certificate Generation	TXN_CERT_001	Triggers approval-based certificate generation
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Privacy statement in compliance

The eMaap platform integrates a Privacy Statement in accordance with the IT Act, 2000. It is displayed:

- Before registration and data submission
- On profile and document upload screens
- Via footer links across the portal

5.6.2 Outputs

This section describes the system outputs of the eMaap platform in terms of user-facing reports, dashboards, and interactive displays. It highlights how each output is designed, the intended purpose, associated user roles, and any relevant distribution or access controls. The system supports data output in the form of on-screen displays, downloadable reports, query results, and graphical user interfaces (GUIs).

Identification of codes and names for reports and data display screens

Output Code	Report/Screen Name
OUT_DASH_001	Stakeholder Dashboard
OUT_REP_001	License Certificate PDF
OUT_GUI_002	Inspection Officer Task List
OUT_GUI_003	Admin Approval Panel
OUT_REP_002	Offense and Enforcement Summary Report
OUT_GUI_004	Legal Case Status Viewer
OUT_GUI_005	Analytics and Compliance Dashboard

Description of reports and screen context

- Stakeholder Dashboard: Displays application status, submitted documents, pending queries, and downloadable certificates.
- License Certificate PDF: Digitally signed certificate with license number, validity, applicant name, and QR code.
- Inspection Officer Task List: Lists assigned inspections, location maps, date/time slots, and inspection notes section.
- Admin Approval Panel: Showcases registration requests, form data, document previews, and decision buttons.

- Enforcement Summary Report: Tabular report including seizure details, offense category, offender identity, and legal action taken.
- Legal Case Status Viewer: Summarizes ongoing case files with filing date, status updates, next hearing date.
- Analytics Dashboard: Graphs and charts on registration volume, compliance rates, inspection frequency, and turnaround time.

Description of purpose and primary users

Each output screen/report is designed to serve specific users within the system:

- Stakeholders (e.g., manufacturers, dealers) use dashboards to track their application/licensing lifecycle.
- Inspectors utilize GUI-based task lists to manage inspections and submit compliance reports.
- Admin and Controllers rely on approval panels and summary tables to review, approve, and forward applications.
- Legal Officers access case status screens to monitor ongoing proceedings.
- NIC & Department Heads use analytics dashboards for decision-making and policy review.

Reports Distribution Requirements

License Certificates are auto generated as downloadable PDFs and also emailed to verified applicants.

- Enforcement and legal reports are restricted to internal users and viewable only from authorized accounts.
- Dashboards and GUIs are available after user login based on role-based access.
- Monthly MIS summaries may be exported and distributed to DoCA and State-level nodal officers as part of compliance reporting.

Access Restriction and security Consideration

- Role-Based Access Control (RBAC) ensures that each user sees only relevant screens/reports.
- Reports containing sensitive data (e.g., offender details, legal actions) are encrypted and accessible only by legal/admin roles.
- License and inspection outputs carry digital signatures to prevent tampering.
- Session timeouts and secure tokens are enforced during output generation and viewing.
- All output interactions are logged for auditing purposes.

5.7 User Interface Design

This section provides an overview of the user interface (UI) design principles, key components, and structure of the eMaap system. The design adopts a user-centric approach to ensure ease of use, clarity, accessibility, and mobile responsiveness. It supports diverse user roles such as stakeholders, inspectors, administrators, and legal officers, each with a customized UI experience through dynamic dashboards and module-specific interactions.

UI Design Element overview

The eMaap portal includes the following major UI components:

- **Dashboard Layout:** Customized per user role (e.g., stakeholder, inspector, admin).
- **Form Wizards:** Step-by-step guidance for processes like registration and license applications.
- **Tab-Based Navigation:** Seamless movement across modules such as License, Inspection, Enforcement, etc.
- **Notification Panel:** Real-time alerts and system messages related to status changes and actions.
- **Analytics & Graphical Views:** Charts and graphs for inspection tracking, application trends, compliance analytics.
- **Multilingual Support:** Interface available in English and Hindi, expandable to other Indian languages.
- **Accessibility Features:** Keyboard navigability, scalable fonts, and contrast adherence.

Key UI Screen Referenced

Screen Name	Code	Description
Registration Page	UI001	Role selection, OTP verification, ID upload
Stakeholder Dashboard	UI002	Track applications, query status, license download
Inspector Dashboard	UI003	View assigned inspections, upload reports, location tagging
Admin Approval Panel	UI004	Manage user approvals, review applications, workflow control

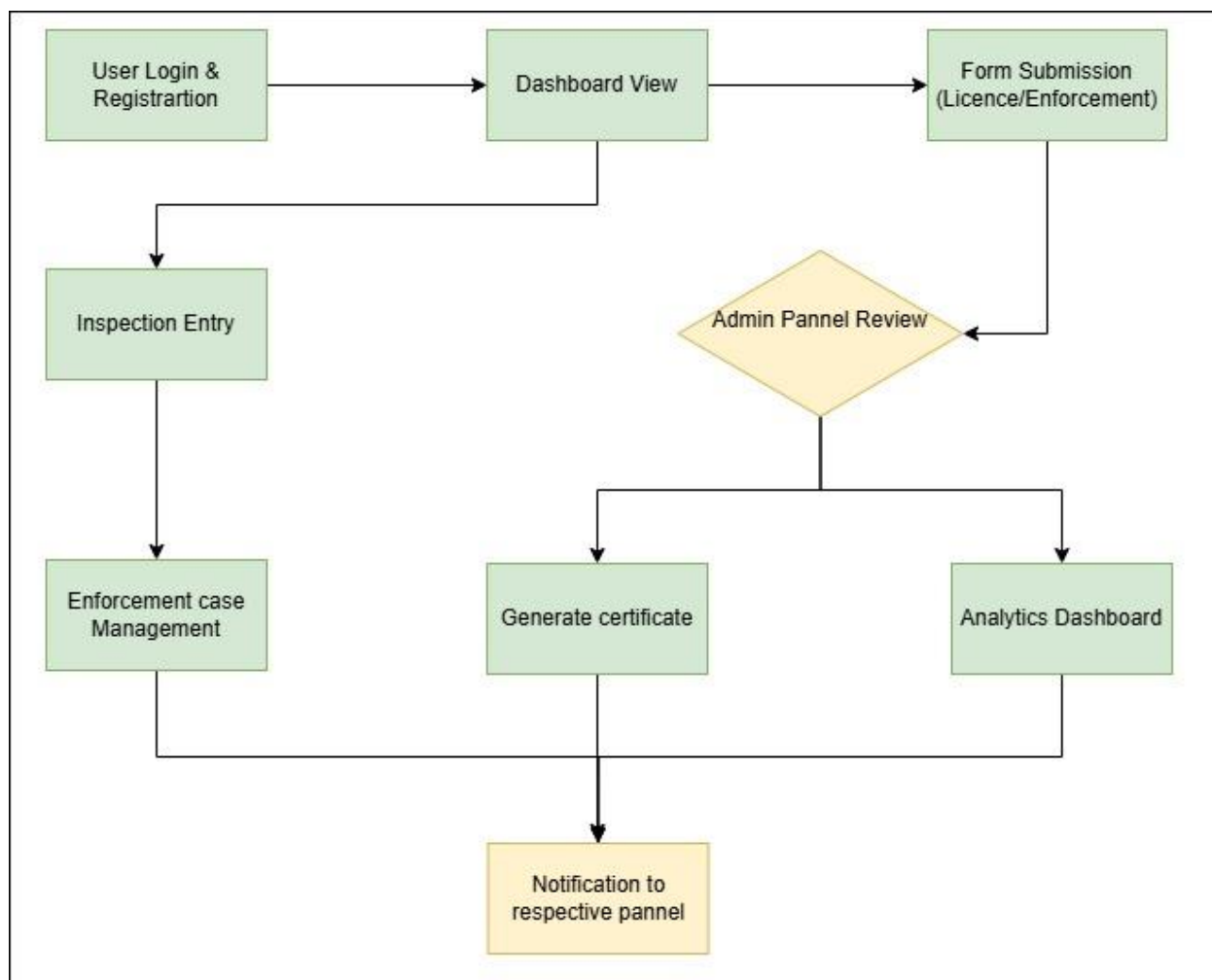
Enforcement Case Panel	UI005	Manage offense entries, seizures, legal actions
Certificate Viewer	UI006	Display and allow download of digitally signed licenses
Analytics Dashboard	UI007	Visual analytics for administrators and policy makers

Compliance & Standards

The UI is developed in accordance with the Digital India guidelines and e-Governance Standards published by MeitY. The platform adheres to:

- Government Web Design Guidelines
- Accessibility norms (WCAG 2.1-compliant)
- Responsive and device-independent layout best practices

Workflow



6. Operational Scenarios

The eMaap (Electronic Metrology Administration and Automation Portal) platform provides a unified digital infrastructure for Legal Metrology across India. This section presents in-depth operational scenarios covering how each user class interacts with the system under different circumstances. It also details how various modules interlink, the steps followed in each transaction, transaction volume, and exceptions. These scenarios are presented in a hybrid format — by major function, threaded transaction flow, and user interaction flow — to provide clarity and accommodate the complex user ecosystem of eMaap.

User Registration and Onboarding

1) User Classes Involved:

- Stakeholders (Manufacturers, Dealers, Repairers, Importers, Packers)

- Legal Metrology Inspectors
- Admin / Legal Officers

2) Scenario Overview:

New users begin by registering themselves on the eMaap portal. The registration process is role-based and dynamically renders the appropriate form depending on whether the user is a Stakeholder, Inspector, or Admin. This scenario ensures that only validated users enter the system and have access only to their permitted functionalities.

3) Step-by-Step Flow:

- User accesses the eMaap portal and clicks on “Register New User.”
 - The system displays a role selection interface — Stakeholder / Inspector / Admin.
 - Based on the role selected:
 - Stakeholders enter details such as Business Name, Trade License Number, GSTIN/PAN, address, contact info, and upload required documents.
 - Inspectors input Department Name, Employee ID, upload ID proofs and duty orders.
 - Admin/Legal Officers must provide official letters of authorization and nomination.
- 4) System performs frontend validations (mandatory fields, formats).
- 5) Captcha verification is completed.
- **Upon submission:**
 - System triggers dual OTP — 6-digit numeric OTP via SMS and 6–8 alphanumeric OTP via Email.
 - User must validate both OTPs within 5 minutes.
- 6) **On successful verification:**
- Status flags (email_verified, mobile_verified) are updated.
 - Registration data is stored.
 - Admin receives notification to review pending approvals.
- 7) Once approved by admin, user credentials are issued, and they may now log in to the portal.
- Exception Scenarios:**
- Duplicate registrations (email/mobile): Blocked.
 - OTP fails or times out: Retry enabled with a cooldown.
 - Missing documents or incorrect format: Application not submitted.

Note:

- **Frequency:** 5,000–6,000 new registrations/month
- **Peak Load:** Start of financial year, post-policy circulars

License Application, Inspection, and Issuance

1) User Classes Involved:

Stakeholders, Inspectors, Controllers, Admins

2) Scenario Overview:

Stakeholders such as manufacturers, dealers, or repairers apply for new licenses, renewals, or amendments. The process includes document upload, fee payment, inspection, and final approval.

3) Step-by-step Flow:

- Registered user logs into the eMaap portal.
- Navigates to the “Licensing” section and selects the type: New / Renewal / Amendment / Duplicate.
- Dynamic form is loaded, capturing:
 - Business identity, type of instruments, operating location, previous license details.
 - Uploads supporting documents (address proof, prior licenses, compliance certificates).
- 4) System calculates fee and redirects user to online payment gateway.
- 5) Once paid, application is marked “Under Processing.”
- 6) Jurisdiction mapping is performed; appropriate Inspector is assigned.
- 7) Inspector visits the business premises, verifies the equipment and compliance with rules, and submits a report via mobile/tablet interface with geo-tag and timestamp.
- 8) If discrepancies found, Inspector raises a query via system.
 - Applicant is notified and allowed to resubmit clarifications within 7 days.
- 9) Once cleared, application moves to Assistant Controller, then Controller for approval.
- 10) Upon approval:
 - Unique License Number is generated.
 - PDF License Certificate with QR code is issued and made available in “My Licenses.”
 - Email/SMS confirmation is sent.

Alternate Paths:

- If query unresolved in 2 cycles → Application marked “Rejected.”
- If license is to be overridden → Manual intervention enabled via Controller dashboard.

Note:

- **Frequency:** 15,000–20,000 applications/month
- **Peak Load:** April–June, October

Verification and Re-verification of weights & Measures

1) User Classes Involved:

Stakeholders, Lab Nodal Officers, Inspectors

2) Scenario Overview:

Manufacturers, dealers, or repairers submit requests to verify instruments before or after deployment, post-repair, or periodically as per legal requirement. Lab officers inspect the instrument and issue certificates.

3) Step-by-Step Flow:

- User logs in and selects “Verification / Re-verification” from dashboard.
- Selects:
 - Type: Fresh or Re-verification.
 - Instrument: Type, Make, Model, Serial Number.
 - Usage: Business type, location.
- Uploads required documents (e.g., invoice, calibration certificate).
- System auto-assigns a Lab Officer based on location/jurisdiction.
- Lab Officer reviews and either:
 - Approves for physical inspection and performs stamping,
 - Raises a query, or
 - Rejects if non-compliant
- If verified, the system generates a Verification Certificate with a QR code.
- Certificate is accessible in user dashboard.

Exception Handling:

- Application rejected → Reapply with changes.
- Instrument fails inspection → Rejection certificate issued.
- Scheduling conflict → User notified and reschedules.

Note:

- Frequency: ~25,000 requests/month
- Peak Load: Sept–Nov (festival season, trading uptick)

Packaged Commodities Registration & Amendment

1) User Classes Involved:

Business Users (Packers, Importers), Controllers, Directors

2) Scenario Overview:

This module handles registration/amendment of commodity packaging declarations under LMPC Rules, ensuring label compliance and packaging accuracy.

3) Step-by-Step Flow:

- Business user logs in → opens “Packaged Commodity Registration” module.
- Selects: New or Amendment.
- Enters: Product name, packaging type, units, net weight, volume, MRP, declarations.
- Uploads: Sample label, compliance certifications, GST/license copies.
- Application submitted → unique ID generated.
- System routes to the Controller/Director (Centre or State).
- Reviewer validates application:

- If compliant - Approval and digital Certificate issued with QR.
- If not – Application marked “Updated Required” or “Rejected”.
- User may modify and resubmit.

Note:

- Frequency: ~3,000–4,000/month
- Peak Load: Aug–Oct (pre-festival inventory rollouts)

Enforcement and Inspection

1) User Classes Involved:

Inspectors, Controllers, Admins, Offenders (Retailers/Traders)

2) Scenario Overview:

This scenario supports real-time regulatory inspections, raids, and offense handling. Geo-tagged inspections, seizure documentation, and notice generation ensure legal compliance.

3) Step-by-Step Flow:

- Inspector logs in and opens the Enforcement dashboard.
- Accesses a list of assigned inspections or initiates new ones.
- Visits the business site and performs inspection in witness presence.
- Inputs inspection results:
 - Satisfactory – Close closed
 - Deficiency found – Seizure initiated, photos / Documents.
- Offense is logged – ticket issued to violator via SMS/Email.
- Checks:
 - First-time violator → Notice 1 → Notice 2 → Notice 3.
 - Repeat violator → Case escalated to Court.
- If compounding fee accepted and paid → Case closed.
- If refused → Legal sanction sought → Court process begins.

Note:

- Frequency: ~10,000 enforcement cases/month
- Peak: FCI/transport audits

Certificate Generation and QR-Based Authentication

1) Scenario Overview: Every approved transaction (license, verification, packaged goods registration) results in certificate generation. These are digitally signed and embedded with QR codes.

2) Step-by-Step Flow:

- Application approved - triggers certificate generation.
- Certificate includes Applicant Name, Application ID, Type, Approval Date.

- QR code generated and embedded.
- PDF certificate generated, stored in user profile.
- Notification sent to user.

3) QR Code Functionality:

- Scannable in field by officers or consumers.
- Verifies authenticity, status, issue date.
- Links back to eMaap record.

Transaction Load and volume summary

Function	Avg Monthly Transactions	Peak Frequency	Peak Period
Registration	5,000	~1,000/day	April, July
Licensing	18,000	~2,500/day	April–June, Oct
Verification / Re-verification	25,000	~3,000/day	Sept–Nov
Packaged Commodity Registration	3,500	~800/day	Aug–Oct
Enforcement	10,000	~1,500/day	Festivals, audits
Certificate Generation (across all)	50,000+	Continuous	Year-round

7. Detailed Design

The eMaap (Electronic Metrology Administration and Automation Portal) is a comprehensive, centralized, and integrated platform designed to automate legal metrology services across India. This detailed design describes how the system's hardware, software, data flows, and external integrations are implemented to deliver a seamless, scalable, and compliant service aligned with the Legal Metrology Act, 2009.

This section provides:

- A modular breakdown of the eMaap architecture.
- Description of inter-module data flows and workflows.

- Integration blueprint for COTS components.
- Communication protocols for APIs and hardware coordination.
- Design considerations for performance, security, and compliance.

System Architecture

Modular Layered Architecture

The eMaap platform is designed using a multi-tiered modular architecture, ensuring separation of concerns and ease of maintenance:

- **Presentation Layer:** User Interface (UI) modules for web and mobile platforms, supporting multilingual, responsive design.
- **Application Layer:** Business logic implementation with role-based service routing, approval hierarchies, and workflow engines.
- **Integration Layer:** APIs for real-time data exchange with state systems, Aadhaar/UIDAI, PAN/CBDT, GSTN, and enforcement tracking services.
- **Data Layer:** Centralized repository for user profiles, licenses, documents, transactions, logs, and audit trails.
- **Security Layer:** End-to-end encryption, OTP verification, RBAC (Role-Based Access Control), secure session tokens, and audit logs.

COTS Components and Tools

The following Commercial Off-The-Shelf (COTS) packages and services are planned for integration:

- DigiLocker for document storage and access.
- eSign for digital authentication.
- NIC's SMS Gateway for mobile OTP and alerts.
- Email Service (Gov.in) for secure transactional communication.
- Payment Gateway (Bharat Kosh / SBI ePay) for online fee payment.
- Cloud Infra (MeghRaj or NIC Cloud) for hosting and scalability.
- Geo-location APIs for real-time inspection mapping.

Software Components and Interfaces

UI Components

- **Web Portal:** Built using Angular/React, accessible on desktops and tablets.
- **Mobile Interface:** Android-compatible app for field officers and inspectors.
- **Dashboard Views:** Customized per role (Admin, Inspector, Stakeholder, Legal Officer).

Each screen handles real-time data rendering from APIs, integrated validation, and field-specific input logic.

Business Logic

The system handles complex workflows such as:

- Multi-step approval chains for license issuance and inspection (center/state level).
- Conditional branching for escalation and resubmission based on compliance.
- Document validation using rule-based logic (e.g., PAN, GSTIN formats).

Integration Interfaces (APIs)

Interface	Purpose	Protocol	Frequency
Aadhaar UIDAI API	User ID verification	HTTPS	At registration
GSTN API	Business validation	REST/JSON	At application
PAN Verification API	Tax ID validation	REST/JSON	On form submission
State LM APIs	License sync, enforcement	REST/JSON	Near real-time
OTP Service	SMS/email alerts	NIC Gateway	Real-time
DigiLocker/eSign	Document signing	OAuth2, REST	On document action

Hardware Design and Infrastructure

Server Architecture

- Hosting: Deployed on NIC Cloud infrastructure.
- Servers: Load-balanced clusters with autoscaling enabled for production workloads.
- Storage: Secure object-based storage for documents (e.g., S3-compatible).
- Backup & Recovery: Daily incremental backup with disaster recovery zone.

Devices for Field Users

- Mobile Devices: Android smartphones/tablets with camera and GPS.
- Inspection Equipment: GPS-enabled weighing/scaling devices integrated via BLE or OTG if required.
- Connectivity: Secure 4G/5G access with fallback to offline data sync.

Workflow Integration

Registration & Authentication

- User initiates registration → OTP verification → Form submission.
- System validates entries and routes to Admin Panel for review.
- Upon approval, user credentials and dashboard access are generated.

License Application and Review

- Stakeholder submits application with documents.
- Application flows through multi-layered approval — Inspector → Assistant Controller → Controller.
- Queries raised are tracked through audit trails and routed for user resolution.
- On approval, system generates a digitally signed license accessible via user dashboard.

Real-time Enforcement Tracking

- Inspector initiates case on mobile.
- Captures geotagged evidence, uploads images.
- Workflow continues for evaluation, seizure, compounding, or legal action.
- Admin reviews, sanctions actions, or escalates to court filing.

COTS Integration Procedure

To combine and integrate multiple external and third-party tools into the single eMaap system:

1. Compatibility Review: Assess API formats (REST/SOAP), authentication models (OAuth2, Token-based).
2. Sandbox Testing: Each API tested in isolation in a staging environment.
3. Wrapper Service Creation: Custom adapters created to normalize and route data from COTS packages to internal models.
4. Data Security Checks: Apply encryption at transit and rest; log all API calls.
5. Integration Validation: Perform SIT (System Integration Testing) and UAT (User Acceptance Testing).
6. Logging & Monitoring: Establish ELK/Prometheus-based monitoring for uptime, error tracking.

Security & Compliance Design

- Authentication: Two-factor login with OTP (email/SMS).
- Authorization: RBAC for feature-based access.
- Data Protection:
 - AES-256 encryption at rest.
 - TLS 1.2+ in transit.
- Audit Trail: Immutable logs for registration, inspection, approvals.
- Compliance: IT Act 2000, Legal Metrology Act 2009, and UIDAI/GST regulations.

Testing and Deployment Strategy

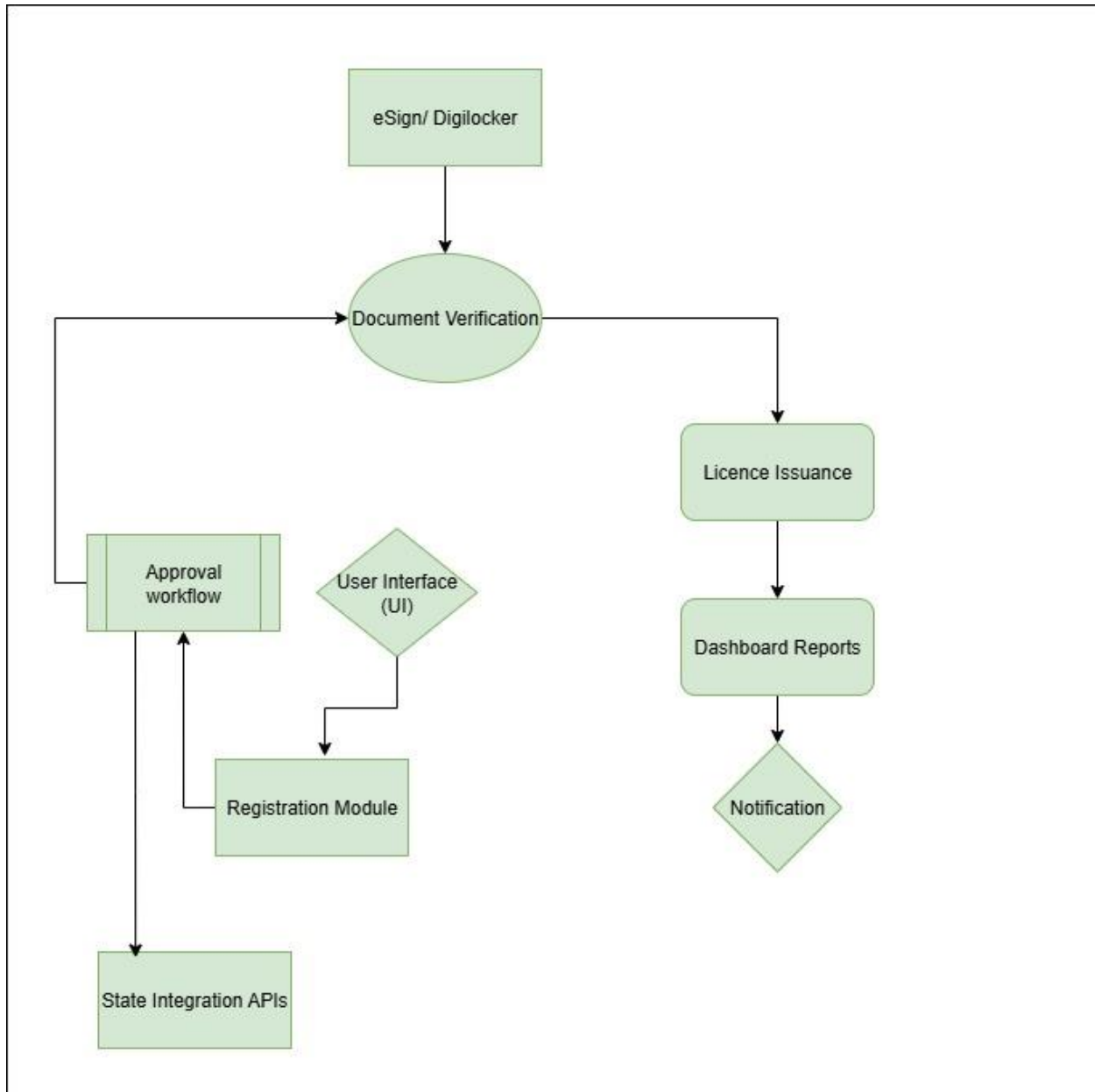
Phase	Activity	Tools Used
Unit Testing	Each module individually tested	JUnit, Jasmine
Integration Testing	Validate inter-component flow	Postman, JMeter
Regression Testing	Post-deployment check	Selenium
UAT	Real-user testing before Go-Live	Manual
Load Testing	Assess system under stress	JMeter

Deployment is done via CI/CD pipelines (Jenkins/GitHub Actions) and monitored via Grafana dashboards.

Post-Deployment Maintenance

- **Incident Management:** L1/L2 support by NICSI.
- **Patch Management:** Quarterly update cycle.
- **Scalability Plan:** Horizontal scaling enabled for peak usage.
- **Monitoring Tools:** Real-time alerts for downtime, performance issues.

Workflow of eMaap Architecture Diagram

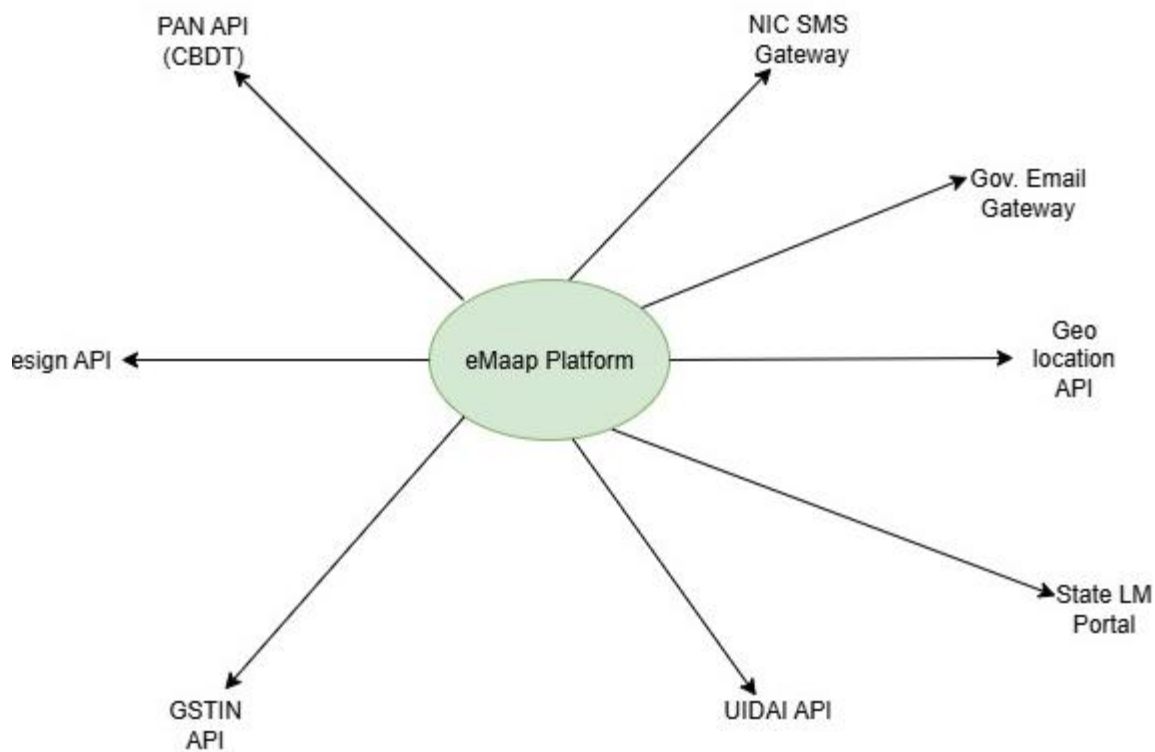


API Integration Matrix

API Name	Functionality	Protocol
UIDAI API	Aadhaar-based identity verification for users	HTTPS (REST)
GSTN API	Validate GSTIN numbers for businesses	REST/JSON

PAN API (CBDT)	PAN number validation for identity compliance	REST/JSON
State LM Portals	Exchange licensing and inspection data with state systems	REST/JSON
DigiLocker API	Fetch/store digitally signed user documents	REST + OAuth2
eSign API	Apply digital signatures to applications and certificates	REST + OAuth2
NIC SMS Gateway	OTP and alert message delivery via SMS	HTTPS
Gov Email Gateway	Send system notifications, credentials, and status updates	SMTP over TLS
Geo-Location API	Capture user/inspector GPS coordinates during activities	REST/JSON

eMaap-External API Integration Diagram



7.1 Hardware Detailed Design

The eMaap system architecture is envisioned as a cloud-hosted, scalable, secure, and performance-optimized infrastructure to support Legal Metrology functions across central and state jurisdictions. This section outlines the detailed hardware components, their specifications, roles, and interconnections required for successful implementation and procurement.

Server Infrastructure (Data Centre)

Component	Description
Type	Cloud Virtual Machines or Blade Servers (for on-premises)
Purpose	Hosts core application backend, APIs, services, and databases

COTS	Dell EMC PowerEdge R750 / HP ProLiant DL380 / AWS EC2 C7i instances
Power Input	110V–240V AC, Dual Redundant Power Supplies, 500–750 Watts
Memory Specification	128 GB ECC DDR5 RAM (minimum for production)
Storage	2 TB SSD (RAID-10), Expandable to 10 TB; separate backup volume
Processor	Intel Xeon Gold 6330 or AMD EPYC 7443P; 2.6+ GHz, 32+ cores
Interfaces	4x Ethernet Ports, IPMI / iLO Remote Management Interface
OS Support	Red Hat Enterprise Linux / Ubuntu Server 22.04 LTS

Database server (Dedicated or Virtualized)

Component	Description
Type	Physical Server or RDS (Cloud Managed Database)
Purpose	Store user data, registration records, inspection logs, audit trails
Power Input	230V AC, Dual Power Supply, 500W–650W
Memory Specification	128–256 GB RAM (based on load)
Storage	5 TB NVMe SSD (with hourly snapshots + replication)
Processor	Minimum 16-core, 2.5 GHz+ Xeon/EPYC
Database Engine	PostgreSQL 14 / MySQL 8.0 / Oracle 19c
Backup Strategy	Incremental (hourly), full (daily), with 7-day rotation

Application firewall & Load balancer

Component	Description
Type	Hardware or Cloud-native (e.g., AWS ALB + WAF / Fortinet FortiGate)
Purpose	Protects web services from attacks and distributes load
Power Input	250–400W (for hardware WAF)
Processor	Multi-core Security ASIC / x86 chip
Functionality	OWASP Top 10 Protection, Geo-blocking, SSL Inspection, Session Control

Workstations for Admin & Inspection Officers

Component	Description
COTS	Dell OptiPlex / Lenovo ThinkCentre / HP EliteDesk
Power Input	90–260V AC, 200–350W
Memory	16 GB DDR4 RAM
Storage	512 GB SSD
Processor	Intel i5 12th Gen / AMD Ryzen 5, 2.4+ GHz
OS	Windows 11 Pro / Ubuntu Desktop 22.04
UI & Display	1080p Resolution Monitor (21" or higher); minimum 1920x1080 pixels
USB Ports	Minimum 4; for smart card readers, printers

Network Equipment

Component	Description
Firewall	UTM Firewall (SonicWall, Cisco Meraki) with VPN
Router/Switch	1 Gbps Managed Layer 3 Switch with VLAN support (Cisco/HP)

Power	90–150W (POE switches)
Redundancy	Dual ISPs with failover routing, optional SD-WAN setup

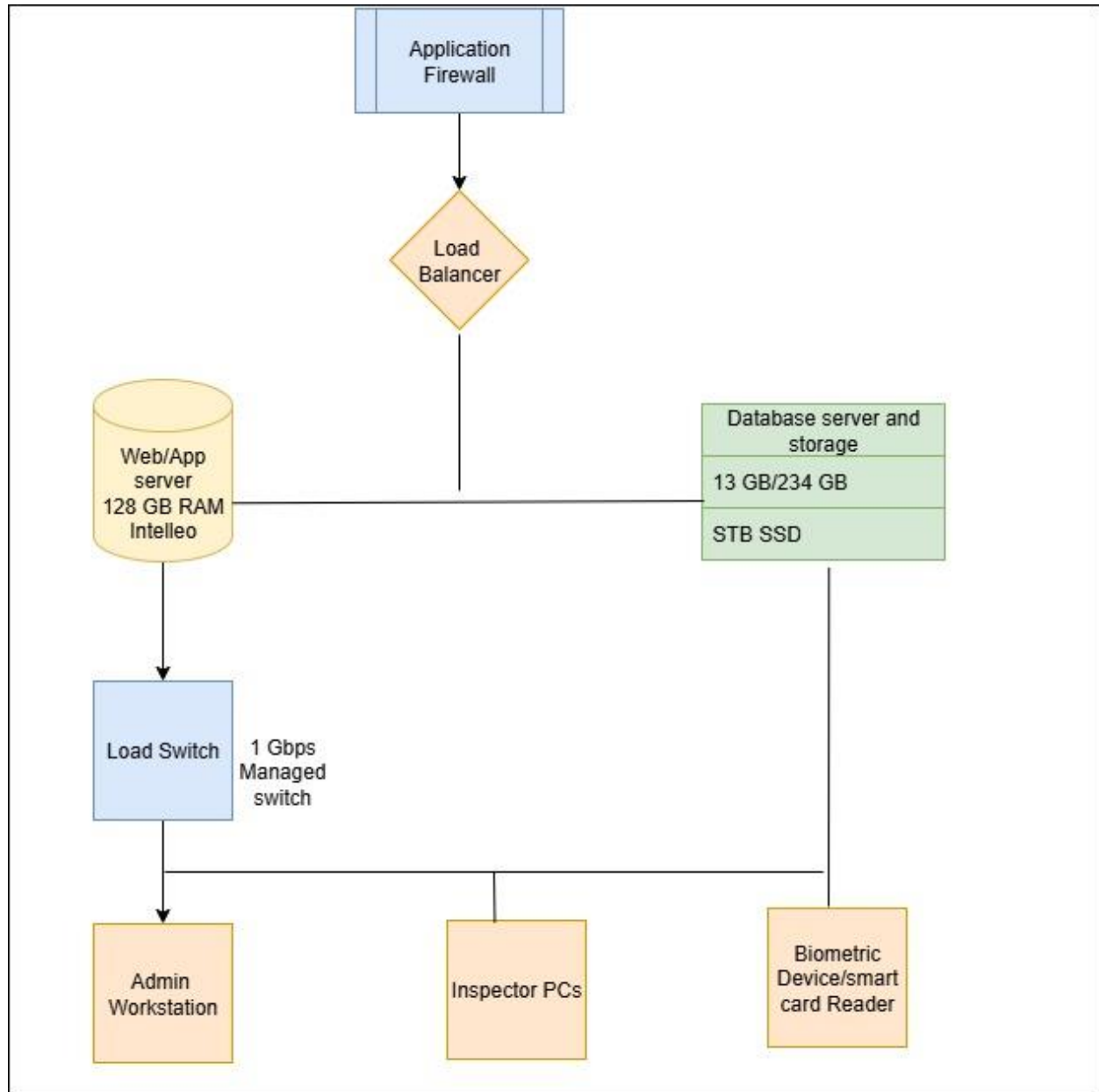
Biometric Device / Smart Card Reader (Optional, If Integrated)

Component	Description
Type	STQC Certified Biometric Scanner (e.g., Morpho 1300)
Power Input	5V USB powered
OS Support	Windows/Linux
Integration	Aadhaar e-KYC via UIDAI APIs

UPS & Power Backup System

Component	Description
Capacity	3–5 KVA Online UPS per server rack
Runtime	30–45 minutes with Auto-Switch Generator Backup
Power Input	Single-phase or 3-phase AC supply

Graphical Representative of Hardware Architecture



Bill of Material (BOM) for eMaap Hardware Systems setup

Component Name	Qty	Specification	Estimated Unit Cost (INR)	Total Cost (INR)
Web/App Server (Dell EMC / HP ProLiant)	1	128 GB RAM, 2 TB SSD, Intel Xeon	₹4,50,000	4,50,000
Database Server	1	256 GB RAM, 5 TB NVMe SSD, Intel Xeon	₹6,50,000	6,50,000
Application Firewall (Fortinet/Cisco)	1	Hardware-based WAF with DDOS protection	₹3,50,000	3,50,000
Load Balancer (AWS ALB / Hardware)	1	High Availability Load Balancer	₹2,50,000	2,50,000
Admin Workstations	1	16 GB RAM, 512 GB SSD, Intel i5	₹55,000	55,000
Inspector PCs	1	16 GB RAM, 512 GB SSD, Ubuntu Desktop	₹50,000	50,000
Biometric/Smart Card Devices	1	STQC Certified, Aadhaar eKYC ready	₹8,000	8,000
1 Gbps Managed Switches	1	24 Port Layer 3 Switch, VLAN support	₹45,000	45,000
UPS Backup System (5 KVA)	1	Online UPS, 30–45 min backup	₹90,000	90,000

7.2 Software Detailed Design

This section provides a comprehensive breakdown of the software services that will power the eMaap platform. Each software service is described with respect to its identifiers, classification, functionality, internal data structures, constraints, user interactions, and interfaces. This design is intended to guide development teams in implementing the system in a modular, scalable, and maintainable manner.

Registration Service

Attribute	Description
Service Identifier	REG_SERVICE
Classification	Application Service
Definition	Manages new user registrations including OTP verification, role-based form rendering, and admin approval.
Requirements	Handles mobile/email OTP, document upload, user role mapping, secure data handling, confirmation notifications.
Internal Data Structures	UserProfile, VerificationToken, UserRoleMapping, DocumentUpload, AuditTrail
Constraints	Email/mobile uniqueness; OTP expiry in 5 minutes; supported file types only (PDF/JPEG/PNG); maximum file size 5MB
Compositions	OTP service, Document Upload Handler, Admin Approval Workflow
Users/Interactions	Applicants (Stakeholders, Admins, Inspectors), Admin Officers
Processing	Captures form data, validates OTP, uploads docs, queues for approval

Interface/Export	REST API for registration; integration with OTP/email gateway
Report Design & Integration	Audit logs, registration confirmation, dashboard updates for admin

Licensing Service

Attribute	Description
Service Identifier	LIC_SERVICE
Classification	Business Application Service
Definition	Manages issuance, renewal, amendment, and cancellation of licenses for Manufacturers, Dealers, Repairers.
Requirements	Dynamic forms, document validation, fee calculation, payment gateway integration, multi-level approvals
Internal Data Structures	LicenseApplication, DocumentMetadata, FeeDetails, InspectionReport, WorkflowStep
Constraints	State-specific workflows; license uniqueness; one application at a time per user
Compositions	Form Renderer, Payment Module, Inspection Engine, Approval Chain
Users/Interactions	Stakeholders, Inspectors, Controllers, Admins
Processing	Collects data, validates input, routes for inspection and approval
Interface/Export	APIs for form submission, inspection update, payment acknowledgment

Report Design & Integration	License certificates (PDF), MIS dashboards, license status logs
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OTP & Authentication Service

Attribute	Description
Service Identifier	AUTH_OTP_SERVICE
Classification	Security & Validation
Definition	Verifies user identity during registration via OTPs sent to email and mobile.
Requirements	OTP generation, secure token handling, retry logic, expiration control
Internal Data Structures	OtpRecord, VerificationFlag, UserSessionToken
Constraints	6-digit numeric for SMS, 6-8 alphanumeric for email; 3 retry attempts; 5 min expiry
Compositions	SMS/Email Gateway, Session Manager
Users/Interactions	All user types during registration/login
Processing	Generates OTPs, validates input, updates verification flags
Interface/Export	Gateway APIs, UI prompts for OTP input
Report Design & Integration	Verification logs, failed attempt tracking, admin alerts on retries

Document Upload Service

Attribute	Description
Service Identifier	DOC_UPLOAD_SERVICE
Classification	File Handling Service

Definition	Handles secure uploading of documents in compliance with format and size constraints.
Requirements	File type/size checks, secure cloud storage, metadata tagging
Internal Data Structures	DocumentReference, UserFolder, FileMetadata
Constraints	Allowed types: PDF, JPG, PNG; max 5MB; file path by User_ID/DocType
Compositions	Cloud Storage API, Virus Scanner, File Validator
Users/Interactions	Stakeholders, Inspectors, Admins
Processing	Accepts upload, validates, stores securely, links to application
Interface/Export	Document upload UI, API for preview/download
Report Design & Integration	Upload logs, document checklist status per application

Dashboard and Role-Based Access

Attribute	Description
Service Identifier	DASHBOARD_SERVICE
Classification	UI/UX Application Layer
Definition	Renders personalized dashboards based on user roles and permissions.
Requirements	Role mapping, module visibility, application/task summary
Internal Data Structures	UserRole, DashboardWidget, ApplicationSummary
Constraints	Access control enforced strictly; unauthorized access blocked
Compositions	Dashboard Renderer, Widget Generator

Users/Interactions	Stakeholders, Inspectors, Admins, Legal Officers
Processing	Loads dashboard based on session role, fetches relevant data
Interface/Export	UI modules (Angular/React), Role APIs
Report Design & Integration	MIS summaries, license counts, pending actions

Workflow & Approval Engine

Attribute	Description
Service Identifier	WORKFLOW_ENGINE
Classification	Business Logic Service
Definition	Routes applications through hierarchical approval processes (state/central).
Requirements	Configurable routes, conditional logic, action history
Internal Data Structures	WorkflowRule, TaskQueue, ApprovalLog
Constraints	Rule engine must support multi-tier routing and fallback
Compositions	Task Manager, State Router, Status Updater
Users/Interactions	Controllers, Assistant Controllers, Admins
Processing	Evaluates rules, updates status, logs actions
Interface/Export	Admin interface to define/edit workflows
Report Design & Integration	Approval trails, application journey tracker

Notification & Alert System

Attribute	Description
Service Identifier	NOTIFY_SERVICE
Classification	Communication Utility

Definition	Sends alerts and notifications via SMS, email, and portal.
Requirements	Event-triggered messages, retry mechanism, failure logging
Internal Data Structures	NotificationLog, UserContact, Template
Constraints	Gateway rate limits, message formats, language compatibility
Compositions	Email Gateway, SMS Gateway, Notification Scheduler
Users/Interactions	All users depending on event (submission, approval, rejection)
Processing	Detects events, fetches templates, dispatches messages
Interface/Export	Notification templates management interface
Report Design & Integration	Notification delivery report, status logs, admin alerts

Verification & Re-verification Service

Attribute	Description
Service Identifier	VERIFY_SERVICE
Classification	Application Service
Definition	Enables users to submit verification or re-verification requests for weights and measures, with follow-up review and approval by lab and district officers.
Requirements	Form submission, document upload, scheduling inspections, lab/field nodal officer integration
Internal Data Structures	VerificationRequest, InspectionAssignment, VerificationReport, NodalOfficerReview

Constraints	Verification tied to unique instrument ID; only one active request per instrument at a time
Compositions	Inspection Scheduler, Officer Role Mapper
Users/Interactions	Business users, District Nodal Officers, Lab Nodal Officers
Processing	Request creation, routing for inspection, report generation, nodal review
Interface/Export	User forms, inspector panels, nodal dashboards
Report Design & Integration	Verification certificates, compliance reports, status tracking

Inspection & Enforcement Service

Attribute	Description
Service Identifier	ENFORCE_SERVICE
Classification	Enforcement/Compliance Service
Definition	Manages field inspections, offense registration, seizures, notices, case filing, and legal compliance processes.
Requirements	Real-time location tagging, photo evidence, notice generation, compounding fee module
Internal Data Structures	InspectionRecord, OffenseRegister, CaseFile, NoticeHistory, AuditTrail
Constraints	GPS tagging required for inspection validity; no duplicate offense entries per case ID
Compositions	Mobile App Integration, Case Management Engine

Users/Interactions	Inspectors, Offenders/Traders, Legal Officers, Controllers
Processing	Inspector initiates inspection, submits findings, triggers legal flow
Interface/Export	Mobile UI for field use, legal officer dashboard, search/report APIs
Report Design & Integration	Offense logs, compounding fee reports, legal notices (PDF)

Certificate Generation & Post Approval System

Attribute	Description
Service Identifier	CERT_SERVICE
Classification	Utility Service
Definition	Generates downloadable certificates for licenses, verification, model approvals, and nominations with embedded QR code and optional blockchain validation.
Requirements	Auto-fill templates, digital signature integration, QR code generation, storage in user dashboard
Internal Data Structures	CertificateTemplate, CertificateRecord, BlockchainLog
Constraints	Certificate can be generated only after final approval; QR must match encoded metadata
Compositions	Certificate Renderer, QR Generator, Digital Sign Service
Users/Interactions	Stakeholders, Controllers, Inspectors
Processing	Pulls approved data, populates template, generates certificate
Interface/Export	UI for download/preview, Certificate Verification API
Report Design & Integration	Issuance report, expiration tracker, reprint logs

Nomination Service (For Company Directors)

Attribute	Description
Service Identifier	NOMINATION_SERVICE
Classification	Application Service
Definition	Allows company representatives to submit nominations of responsible directors for regulatory accountability.
Requirements	Form validation, director detail capture, controller review workflow
Internal Data Structures	NominationForm, DirectorDetails, ApprovalLog
Constraints	One nomination per company; cannot proceed without valid document upload
Compositions	Nomination Form Handler, Controller Review Queue
Users/Interactions	Business users, Controllers, Legal Officers
Processing	Form submission → document check → approval/rejection
Interface/Export	Director nomination form, approval dashboard
Report Design & Integration	Nomination logs, certificate issuance, audit trail tracking

Dashboard & Analytics MIS Report

Attribute	Description
Service Identifier	MIS_SERVICE
Classification	Data Service
Definition	Provides role-based dashboards, visual analytics, trends, performance metrics, and compliance statistics.

Requirements	Data aggregation, AI-based forecasting, export (PDF/Excel), role-based metrics display
Internal Data Structures	ApplicationStats, ServiceTrends, UserActivityLog, GeoAnalytics
Constraints	Time-based aggregation; role-level access to sensitive data only
Compositions	Dashboard Widgets, Analytics Engine, Data Export Handler
Users/Interactions	Central/State Admins, Controllers, NIC/MeitY Officers
Processing	Real-time data pull from transaction logs, display in charts/tables
Interface/Export	MIS dashboard, report export API, heatmaps
Report Design & Integration	Daily/weekly/monthly service usage, compliance scorecards, user engagement charts

7.3 Security Detailed Design

This section defines the security architecture and control mechanisms necessary to ensure the confidentiality, integrity, and availability of the eMaap platform. Based on the sensitivity and criticality of legal metrology operations and citizen/business data, the system is classified under a “High Assurance” security level, as per MeitY and NIC security directives.

Authentication & Authorization

Feature	Design Specification
Authentication	Multi-Factor Authentication (MFA) using mobile OTP and email OTP during onboarding and login. Integrates with Aadhaar for identity verification (UIDAI-compliant).
Protocol	OAuth2.0 with support for refresh tokens and role-based scopes
Session Management	Secure session tokens with inactivity timeout and renewal thresholds
Authorization	Role-Based Access Control (RBAC) based on the user profile: Inspector, Stakeholder, Admin, Legal Officer, Controller
Password Policy	Minimum 8 characters, including upper/lowercase, number, and special character; periodic expiration and no reuse for 5 cycles
Account Lockout	After 5 failed login attempts, account temporarily disabled for 15 minutes

Logging & Auditing

Feature	Design Specification
Logging Engine	Centralized log management using ELK Stack (Elasticsearch, Logstash, Kibana)
Log Events Captured	Login attempts, failed authentications, role escalations, application submissions, approval actions, file uploads/downloads, audit trail

Retention Period	Minimum 5 years (as per IT Act and Digital Locker compliance)
Tamper Protection	Hashing of logs using SHA-256; logs stored in append-only format on secure file systems
Audit Trails	Every user activity mapped to session ID, IP address, role, and timestamp; queryable by role for compliance reviews
Access Control	Only Admin and Audit Officers have visibility into audit logs via dashboard interface

Encryption

Feature	Design Specification
Data at Rest	AES-256 encryption for all user and application data in RDBMS
Data in Transit	TLS 1.3 enforced on all HTTPS endpoints, with strong cipher suites (ECDHE-RSA-AES256-GCM)
File Uploads	All uploaded documents (PDF/JPEG/PNG) encrypted before being saved to the cloud
Database-Level	Column-level encryption for sensitive fields (Aadhaar, PAN, email, mobile number)
Key Management	NIC-managed HSM (Hardware Security Module) with regular rotation of keys and audit logs

Network Port Usages

Component	Ports Open	Justification
Web Server (HTTPS)	443 (TCP)	Secure access for all external and internal users

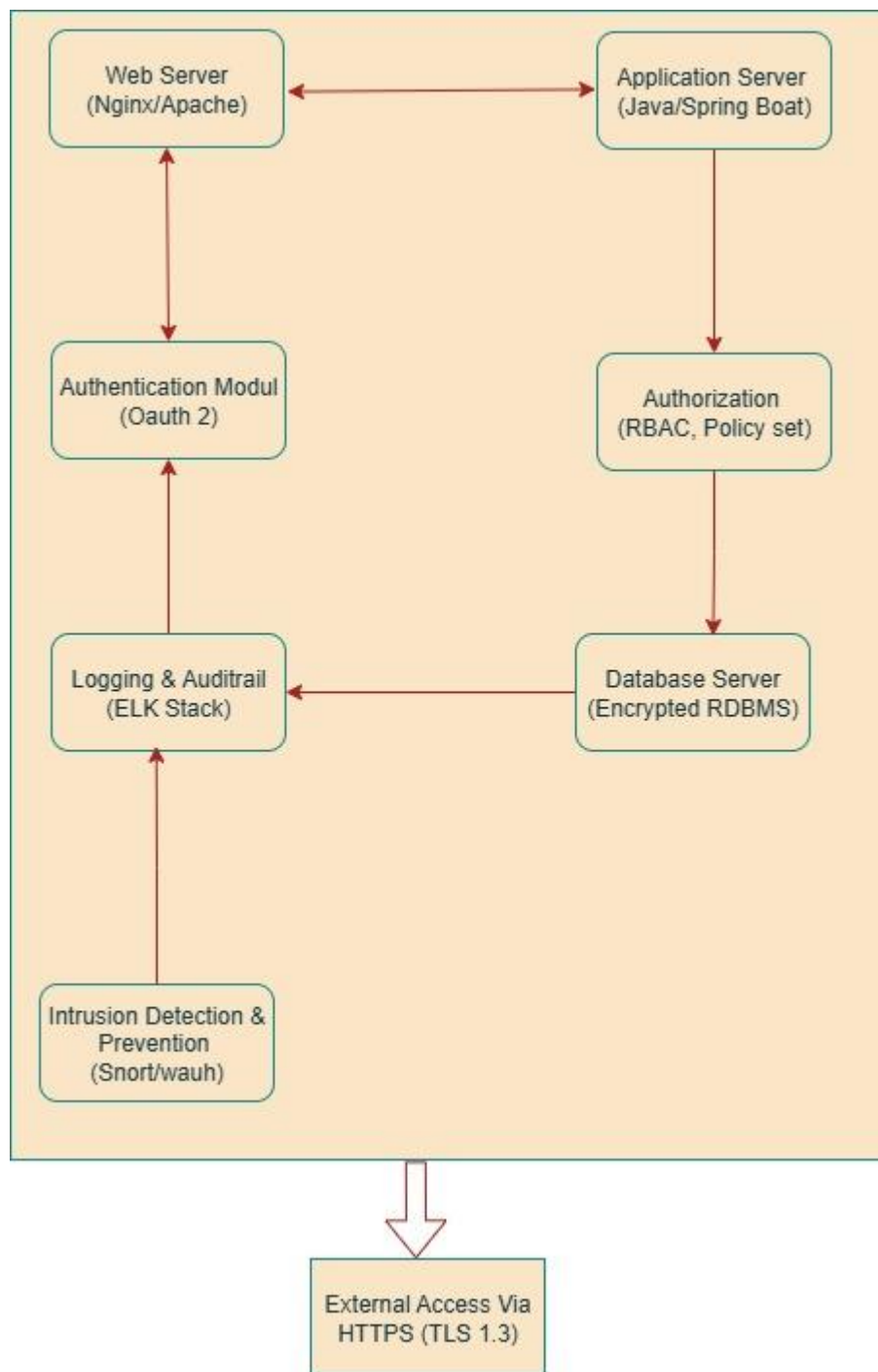
Admin Dashboard	8443 (TCP)	Admin panel secured via VPN tunnel and TLS
Application Server	8080 (Internal)	Backend services exposed internally only
Logging Agent	9200/5044 (TCP)	ELK Stack communication (Elasticsearch/Logstash)
Database Server	5432 (PostgreSQL)	Encrypted DB communication with backend services only
IDS/Monitoring	1514 (UDP/TCP)	Wazuh/Snort agent log forwarding
Blocked Ports	20-21, 23, 25, 110, 3389	Disabled to prevent FTP, Telnet, SMTP abuse or RDP exposure

Intrusion Detection & Prevention

Component	Specification
Tool	Snort + Wazuh IDS/IPS
Monitoring Points	Web server traffic, API endpoints, authentication service, document upload module
Threat Detection	SQL injection, XSS, brute force, port scanning, directory traversal
Alerts	Real-time alerts to system admin with severity levels; auto-escalation to NIC-SOC for high threats

Prevention Measures	Auto block offending IPs; CAPTCHA & lockout enforcement on multiple failed actions
Integration	Feeds alerts into SIEM; generates periodic security incident reports for audit review

eMaap Security Boundary Architecture



7.4 Performance Detailed Design

This section outlines how the eMaap system is engineered to meet defined performance and availability goals across high-load regulatory operations, statewide user base, and real-time enforcement workflows. The design ensures the system is scalable, resilient, and optimized for continuous high performance under national-level usage.

Capacity and Volume Requirements/Estimates

Parameter	Estimate
Concurrent Users	Up to 50,000 concurrent users nationwide
Registered Entities	Approx. 5 million records (stakeholders, inspectors, admins)
Document Volume	Over 20 million uploaded PDFs/images in first 3 years
Daily Transactions	2-3 million transactions (logins, registrations, verifications, submissions)
Data Growth Rate	20-30 GB/month initially; scaling up with adoption
Peak Load Periods	State/central application deadlines, inspection seasons, licensing drives

Performance Expectations

Metric	Target
Page Load Time	< 2 seconds for major dashboard views
OTP/Verification	< 5 seconds (round trip)
Document Upload	< 10 seconds for 5MB
License Generation	< 5 seconds per issuance
System Uptime	≥ 99.9% monthly availability (non-maintenance)
Throughput	≥ 500 transactions/second (scalable under load balancer)
API Latency (Internal)	< 250 ms

Performance Design to Meet Capacity Requirements

Design Element	Implementation
Horizontal Scaling	Load balancer with Nginx and Spring Boot microservices allows scaling web/app tiers dynamically
Auto-scaling Groups	Automatically add/remove server nodes based on CPU/RAM thresholds
Distributed Caching	Redis/Memcached layer accelerates OTPs, dashboards, role lookups, and UI rendering

Parallel Processing	Asynchronous job queues (RabbitMQ/Kafka) for document processing and notifications
DB Partitioning	Logical partitioning of state-wise user/application data ensures query speed and I/O isolation
API Rate Limiting	Prevents overloads via throttling under extreme usage

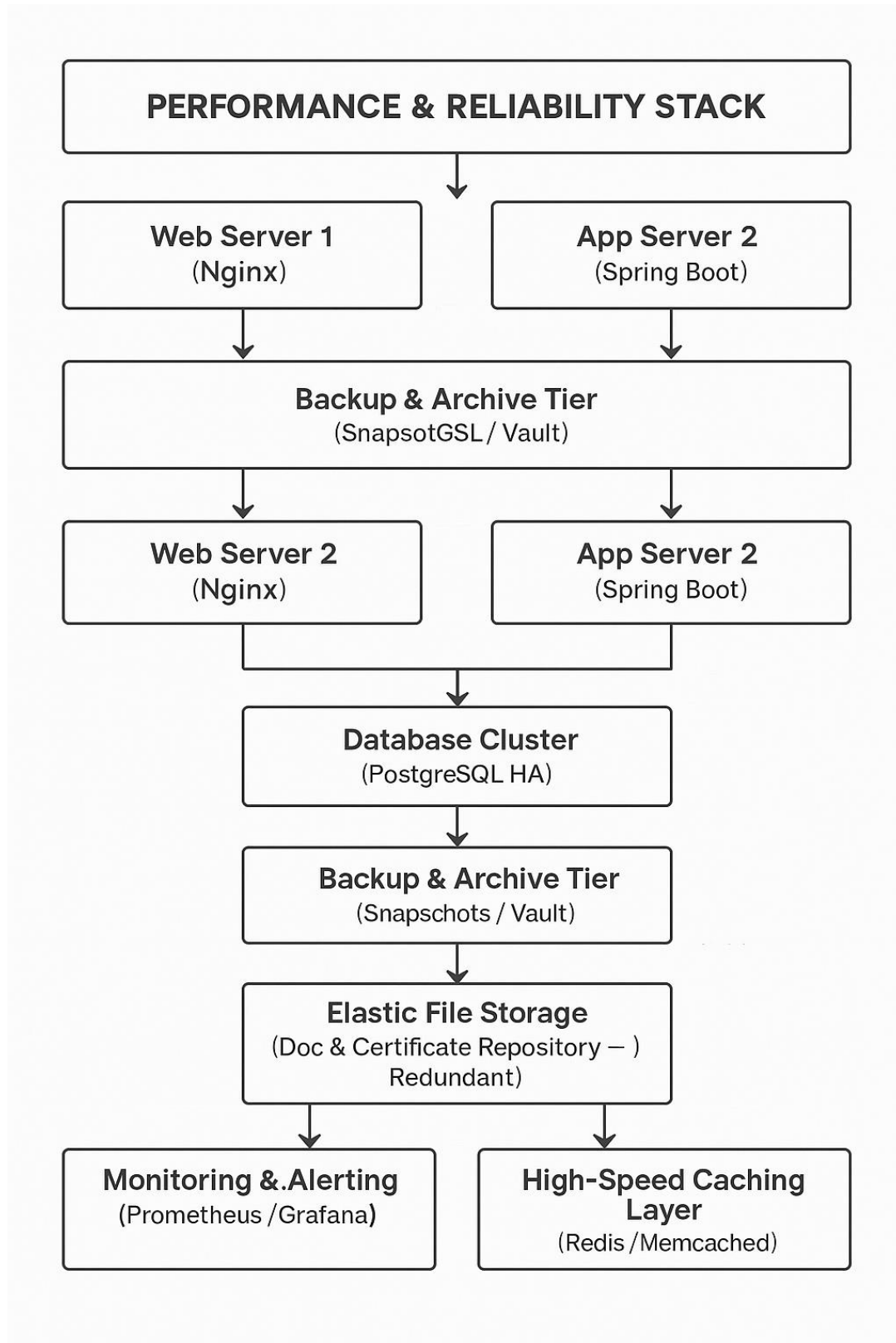
Reliability Design to Meet Availability Requirements

Reliability Feature	Implementation
High Availability (HA)	2N redundancy at each web/app tier; PostgreSQL configured with HA clusters and replication
Failover Systems	Automatic failover for DB via standby replicas; load balancer reroutes failed node traffic
Zero Downtime Deployment	Blue-green deployment strategy for version upgrades without user disruption
Monitoring & Auto-Healing	Prometheus with Grafana alerts + self-healing instances through scripts or orchestration tools
Redundant File Storage	Dual-zone storage for document/certificate repositories
Disaster Recovery (DR)	Periodic snapshots to secondary data center (RPO ≤ 4 hours, RTO ≤ 2 hours)

Backup recovery and Archive Diagram

Function	Details
Database Backup	Daily incremental + weekly full backups using logical dumps and snapshots; 30-day retention
File Archive	Long-term archival of certificates/documents into cold storage (6-month rotation)
Versioning	Every document upload is version-controlled; previous revisions retrievable for audit
Recovery Time Objective (RTO)	≤ 2 hours
Recovery Point Objective (RPO)	≤ 4 hours
Backup Validation	Automatic restore testing in sandbox environments bi-weekly
Encryption	All backups encrypted with AES-256; stored in access-controlled vaults
Storage Locations	Primary: NIC Cloud Vault, Secondary: Statewise DR Zones

Flow Diagram of system Performance Design



7.5 Internal Communications Detailed Design

The eMaap platform is composed of a multi-tiered, distributed architecture hosted on NIC Cloud infrastructure. It enables communication between frontend clients, backend service components, data layers, enforcement modules, and monitoring systems. Internal communication is critical to manage service orchestration, maintain low latency, and ensure seamless end-to-end transactions.

Number of servers and clients on Each area Network

Network Segment	Components	Count
Client LAN (Front-end)	State User Devices (Stakeholders, Inspectors)	~10,000 concurrent clients
Web Tier LAN	Load Balancer, Web Servers (Nginx/Apache)	1 LB + 2 Web Servers
App Tier LAN	Application Servers (Spring Boot APIs, Controllers)	3–5 horizontally scalable
DB & Storage LAN	Database Cluster, Document Storage, Backup Vault	2–3 DB nodes + 2 file storage
Monitoring LAN	Prometheus, Grafana, ELK Stack	2-3 monitoring nodes
DR & Archive LAN	DR Site Components, Backup Archive	1 hot standby (sync), cold storage

Specifications for Bus Timing Requirements and Bus Control

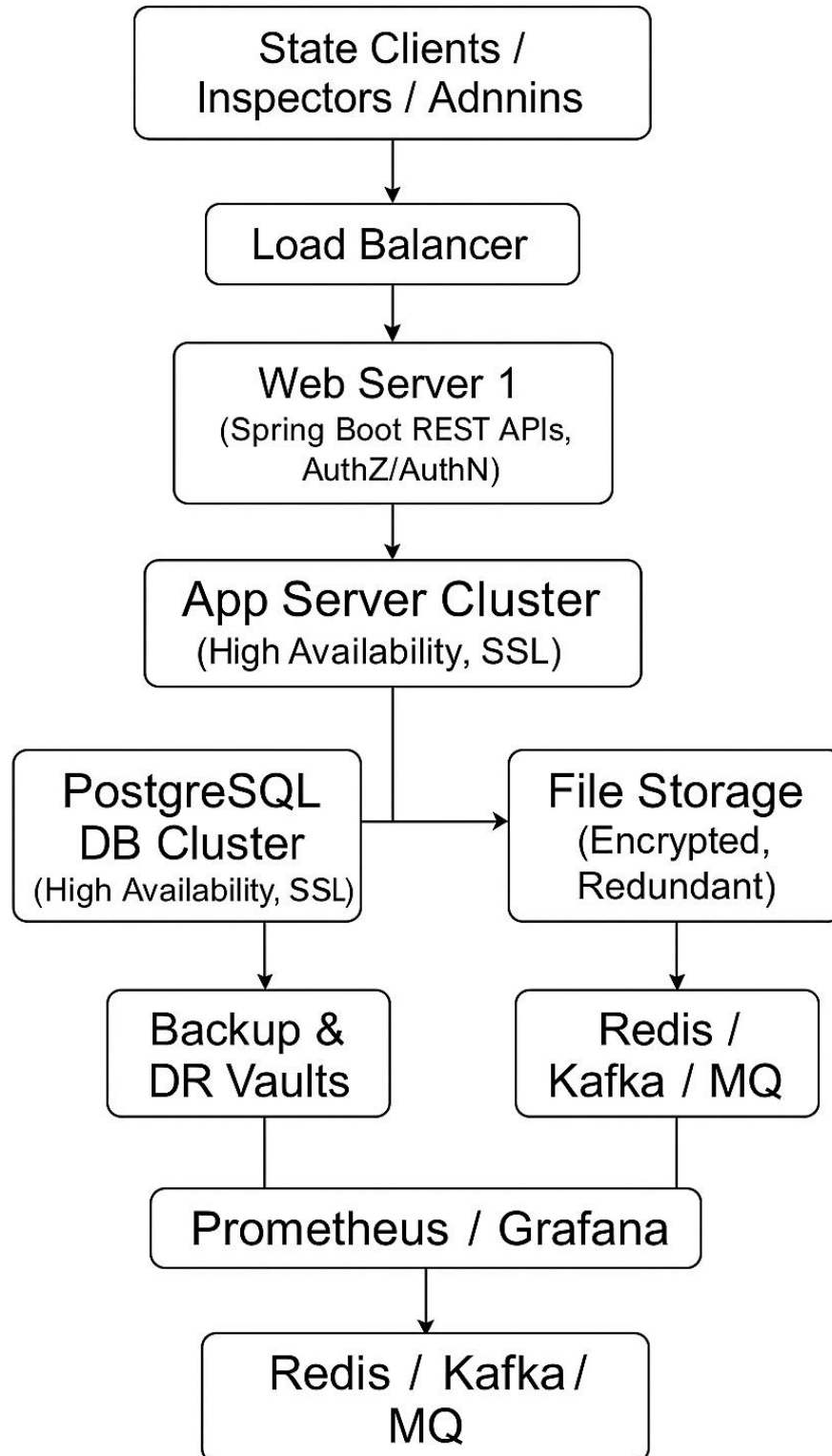
Communication Bus	Protocol	Timings	Control Mechanism
Web <-> App Tier	HTTP/HTTPS (TLS 1.3)	~10–100 ms latency under load	Load Balancer & API Gateway
App <-> DB Tier	JDBC over SSL	< 100 ms per query (optimized)	Query Optimizer, Read Replica

App <-> Storage	REST APIs (JSON/XML), S3-compatible	< 1 sec for file read/write	Token-based auth, ACLs
App <-> Monitoring	Prometheus exporters	10 sec scrape interval	Alert Manager
Internal Messaging	Kafka/RabbitMQ	ms-level event triggers	Producer/Consumer Acks
DR Replication	rsync, PostgreSQL streaming	Every 5 min (async) or real-time (sync mode)	CRON + DB trigger

Format for Data Being Exchanged Between Components

Component Pair	Data Format	Serialization	Encryption
Web ↔ App	JSON (RESTful APIs)	UTF-8 JSON	TLS 1.3 (HTTPS)
App ↔ DB	SQL Queries/Stored Procs	ANSI SQL	SSL
App ↔ Storage	Multipart/Form, Binary (PDFs)	Base64/Raw Streams	AES-256
App ↔ Notification	JSON Payloads	JSON via POST	TLS
App ↔ Monitoring	Time Series Metrics	Prometheus Exporters	Secured Channels
App ↔ Message Bus (Async)	JSON / Avro / XML	Serialized Queues	Encrypted Queue

Graphical Representation of Connectivity and Data Flow



LAN Topology

LAN Segment	Topology	Justification
Client LAN	Star	End-users access centralized load balancer
Web/App Tier LAN	Full Mesh	Redundant routing between horizontally scaled nodes
App ↔ DB / Storage	Point-to-Point	Direct encrypted connections for performance
Monitoring LAN	Star + Collector	Centralized metrics aggregation with alerts
DR LAN	Peer-to-Peer	Replication and fallback across DR zones

8. System Integrity Controls

The eMaap platform ensures robust system integrity and security through a layered approach incorporating access restrictions, auditability, and verification mechanisms. The system design includes the following minimum levels of control:

Internal Security Controls

To maintain internal system integrity, role-based access control (RBAC) is implemented:

- User roles such as Admin, Inspector, Legal Officer, Stakeholder (Dealer, Manufacturer, Repairer, Importer) are each granted permissions only necessary for their functions.
- Sensitive data items (such as user identification details, license certificates, inspection records) are stored securely and accessed only by authenticated and authorized users.
- Access to configuration settings, master data, and user management functions is strictly restricted to Admins and Legal Officers.
- All forms and modules validate user roles before rendering or processing any data.

Audit Procedures for Operational and Management Reports

To ensure compliance with retention and reporting standards:

- All reports (application status, license issuance logs, inspection records, etc.) are stored securely and maintained per regulatory retention periods.
- The audit log system captures key actions across modules including registration, submissions, verifications, approvals, and modifications.
- The system includes functionality for exporting, printing, and archiving reports for review and audits by the Legal Metrology Division (DoCA) and State Controllers.

Application Audit Trail Mechanism

The application includes a dynamic and tamper-proof audit trail system:

- Every interaction with designated critical data (license applications, inspection notes, controller decisions) is logged.
- Records include details such as:
 - User ID
 - Action performed (Create / Read / Update / Delete)
 - Timestamp (Date and Time)
 - Data element accessed or modified
- Audit trails are immutable, stored securely, and accessible only to authorized admin users for security reviews and legal compliance.

Standard Tables for Data Validation

To ensure accuracy and consistency of input data:

- Master data tables are used for validating dropdowns and text inputs across modules. These include:
 - Business Category
 - License Type
 - States/Districts (linked to LGD codes)
 - Inspection Status
 - Document Types
- Standard validation rules are enforced, such as PAN format, GSTIN rules, OTP verification standards, and mandatory field checks.

Verification Process for Additions, Deletions, or Updates

All additions, updates, and deletions of critical data undergo verification processes:

- Dual OTP Verification for registration (mobile + email).
- Admin approval is mandatory before any user gains active system access.
- Version control is implemented for key documents to track changes and maintain records of updates.
- Inspection-based verifications are mandatory before license issuance.
- Queries raised during inspections or reviews must be resolved before the system permits further progression.

Detailed Audit Identification Capabilities

The audit system is capable of tracking and identifying all system access and data modification using:

- User Identification (User ID, Role)
- Device/Terminal Information (IP address, location metadata if available)
- Date and Time Stamps
- Affected Data Fields (e.g., modified inspection status, uploaded license, deletion of draft applications)
- These logs are encrypted and stored for compliance with the Information Technology Act, 2000 and RTI Act, 2005 and can be retrieved on demand for internal or external audit.

9. External Interfaces

The **eMaap platform** interacts with several external systems to ensure smooth service delivery, legal compliance, real-time verification, and administrative efficiency. These interfaces are primarily API-based, using secure protocols and identity/authentication mechanisms, and are critical for supporting functionalities such as user registration, identity verification, license validation, document authentication, and complaint redressal.

Overview of External Interfaces

S. No.	External System	Purpose of Integration	Interface Type	Remarks
1	UIDAI (Aadhaar Services)	Identity verification of users	RESTful API over HTTPS	Used during registration
2	CBDT PAN Verification System	Validate PAN during user onboarding	REST API	Required for dealers, manufacturers
3	GSTN (Goods & Services Tax Network)	GSTIN verification	Secure API	Required for business validation
4	DigiLocker/e-Sign	Access and verify uploaded documents and sign certificates	Web service (e-Sign API)	Required for legal digital authentication
5	SMS Gateway	OTP and notification delivery	REST API (SMS push)	Used for dual OTP verification and alerts

6	Email Gateway (NIC/MeitY)	System-generated emails	SMTP (Secured)	For alerts and workflow notifications
7	State Legal Metrology Portals	Exchange of licensing and enforcement data	Secure REST API with mutual authentication	Bi-directional integration
8	Geo-location/Mapping APIs	Geo-tagging of inspections, business premises	Web-based GIS APIs	Required for real-time mapping
9	CPGRAMS or State Grievance Portals	Grievance redressal alignment	Webhook/API	Optional but planned for complaint syncing

UIDAI Aadhaar Verification Interface

- Purpose: Verify the identity of individuals during registration (Inspectors, Legal Officers, Stakeholders).
- Type: RESTful API, secured by UIDAI guidelines.
- Functionality Supported:
 - Demographic verification
 - OTP-based Aadhaar authentication
- Security Protocol: eKYC or UID Token-based flow.

PAN Verification System (CBDT)

- Purpose: Validate PAN for business entities and legal officers.
- Type: Encrypted REST API
- Functionality Supported:
 - Format and record validation
 - Entity verification
- Security: IP-whitelisted integration with CBDT node.

GSTN Interface

- Purpose: To validate GST numbers for dealers and manufacturers.
- Interface: API-based, with secure handshakes and tokens.
- Functionality Supported:
 - GSTIN structure validation
 - Entity legal name and registration status retrieval
- Security Compliance: As per GSTN protocols (OAuth 2.0 / IP restrictions).

Digi Locker and e-Sign Services

- Purpose:
 - Digitally sign applications, licenses, and inspection reports.
 - Pull documents securely from DigiLocker accounts.
- Interface Type: e-Sign API and DigiLocker fetch services.
- Functions Supported:
 - e-KYC document retrieval
 - Certificate signing (per Section 3 of IT Act)
- Security: Aadhaar-linked OTP, SSL/TLS, token authentication.

SMS Gateway

- Purpose: OTP delivery, notifications for registration, queries, approvals.
- Interface: JSON-based POST requests via REST API
- Gateway Provider: NIC SMS Gateway or state-authorized telecom vendor
- Failure Handling:
 - Retry mechanisms
 - Logging of delivery receipts

Email Notification System

- Purpose: Deliver user alerts, application status changes, approvals.
- Type: SMTP secured with STARTTLS
- Scope:
 - Dual OTP (email)
 - Workflow stage changes
 - Rejection notices
- Provider: NIC Mail Gateway ([Gov.in](https://gov.in))

State Legal Metrology Portals

- **Purpose:** Share registration, license, inspection, and enforcement data.
- **Interface:** Bi-directional REST API
- **Data Exchange:**
 - License records
 - Verification requests
 - Inspection outcomes
- **Authentication:** OAuth2 + IP restriction
- **Error Handling:** Defined by state-level SLAs

Geo-Location and Mapping APIs

- **Purpose:** Real-time geotagging of inspections, map-based tracking.
- **Interface:** JavaScript/REST API (NIC/NICSI GeoPlatform or Google Maps equivalent).
- **Features:**
 - Location capture during inspection
 - Premises mapping
 - MIS dashboard plotting

CPGRAMS/Grievance Redressal

- **Purpose:** Optional future integration to align user complaints raised through CPGRAMS.
- **Interface:** Webhook push or API poll (when implemented)
- **Expected Fields:** Ticket ID, status, action logs
- **ICD Reference:** *ICD-GRIEVANCE-EMAAP-2025-09*

General Integration Standards

- **Data Format:** JSON over HTTPS
- **Security:**
 - OAuth 2.0 tokens for authentication
 - TLS 1.2+ encryption
 - Role-based request access
- **Logging:**
 - All API requests and responses logged
 - Failure and retry attempts tracked

9.1 Interface Architecture

The interface architecture for the eMaap Portal illustrates the interaction and data exchange between the system and various external systems essential for identity verification, document validation, communication, and regulatory compliance. These interfaces are implemented primarily via RESTful APIs over secure HTTPS connections. The system is designed with modular connectivity, allowing secure and scalable integration across central and state agencies.

Interface Description and Mechanism

1. UIDAI (Aadhaar Services):

- Mechanism: RESTful API
- Purpose: Identity authentication during registration
- Direction: Outbound request from eMaap to UIDAI servers
- Security: Token-based access and encrypted Aadhaar data

2. CBDT PAN Verification System:

- Mechanism: REST API
- Purpose: Verify PAN information for stakeholder legitimacy
- Location: Hosted by CBDT; accessed securely by eMaap
- Interface Control: Token authentication with IP whitelisting

3. GSTN:

- Mechanism: Encrypted REST API
- Purpose: Validate GST numbers for businesses
- Interface Architecture: Tokenized access using GSTN credentials
- Data Flow: License application ↔ GST validation

4. DigiLocker / eSign:

- Mechanism: API-based document retrieval and digital signing
- Used for: Signing inspection reports and licenses
- Compliance: Section 3 of IT Act, 2000
- Direction: Two-way – eMaap fetches documents and pushes signed data

5. SMS Gateway:

- Mechanism: REST API
- Used for: OTP verification, alerts, license issuance messages
- Hosted by: NIC SMS Gateway or authorized telecom provider
- Direction: Outbound from eMaap to gateway

6. Email Gateway:

- Mechanism: Secured SMTP integration
- Used for: Application status updates, workflow notifications

- Provider: NIC / MeitY
- Direction: Outbound

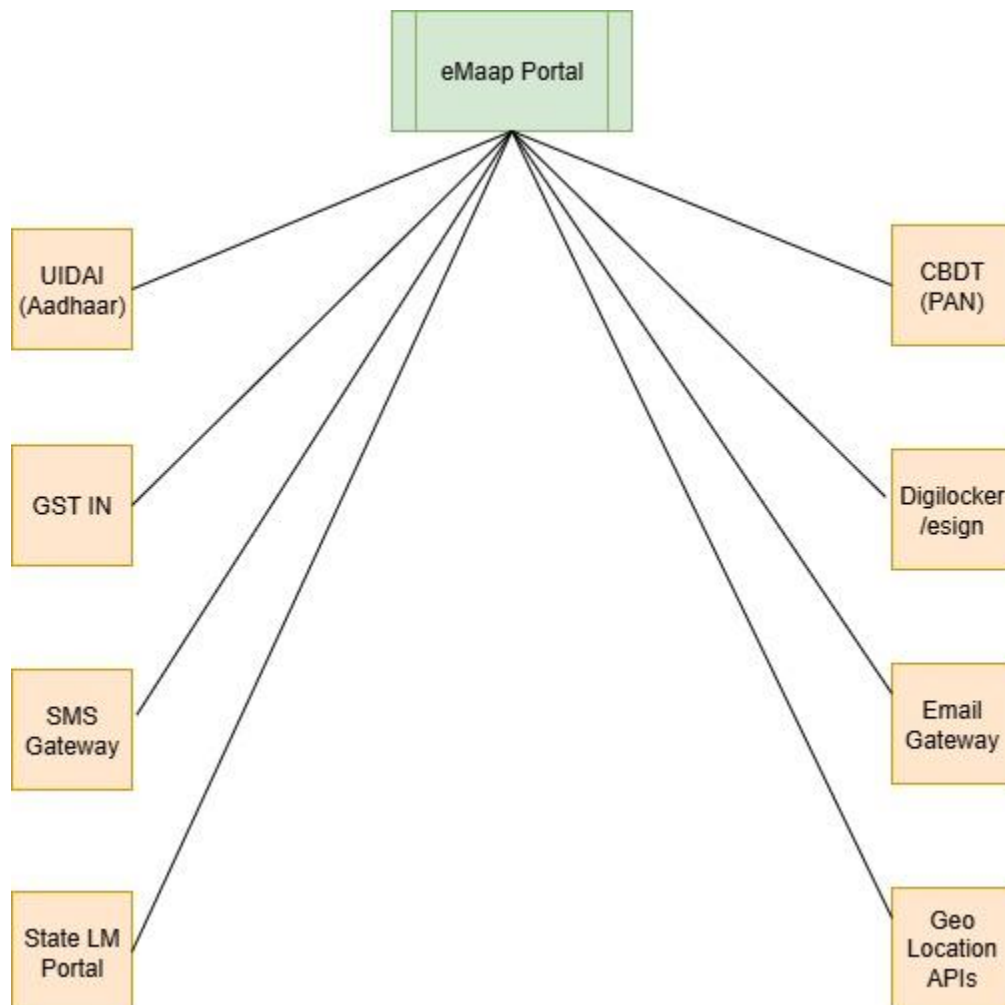
7. State Legal Metrology Portals:

- Mechanism: API-based integration
- Used for: Sharing license, inspection, and enforcement data
- Nature: Bi-directional exchange
- Authenticated via: OAuth and IP filtering

8. Geolocation APIs:

- Mechanism: Web-based APIs for mapping and geotagging
- Purpose: Real-time inspection tracking, map-based dashboards
- Integration with: NIC GeoPlatform or third-party GIS tools

Interface Architecture of eMaap Portal



9.2 Interface Detailed Design

The **eMaap platform** is a centralized, API-driven portal that interfaces with multiple external government and cloud-based systems to ensure real-time validation, secure data exchange, and seamless end-to-end service delivery. Each interface is governed by well-defined standards and control mechanisms to ensure interoperability, availability, and security.

Interface Matrix Summary

Interface Name	External System	Purpose	Direction	Format
UIDAI e-KYC API	UIDAI Aadhaar System	Identity verification using Aadhaar number	Bidirectional	XML/JSON
GSTIN Validation API	GST Network (GSTN)	Business identity verification using GSTIN	Unidirectional	REST JSON
PAN Verification API	CBDT / NSDL	Personal/Entity PAN validation	Unidirectional	REST JSON
DigiLocker Integration	MeitY DigiLocker	Document fetch/upload from verified repository	Bidirectional	XML/JSON (signed)
eSign Gateway	CCA Certified eSign Provider	Digital signing of forms/certificates	Unidirectional	XML (PKCS#7)
State LM API	State Legal Metrology Portals	License status, inspection data, enforcement sync	Bidirectional	REST/JSON
SMS Gateway	NIC SMS Gateway	Alert and OTP delivery via SMS	Outbound	REST
Email Gateway	NIC Email Relay Service	Notification/confirmation via email	Outbound	SMTP/REST
Geo-tagging & Location API	NIC Location Service	GPS verification during field inspections	Bidirectional	JSON
Compounding Fee Treasury Interface	State Treasury / PFMS	Payment status validation and reconciliation	Bidirectional	REST/JSON/XML

Interface Communication Rules and Control

Parameter	Specification
Authentication	OAuth 2.0 (Token-based); eSign uses CCA-compliant PKI certificates
Authorization	Role-based access control (RBAC) mapped to eMaap user roles
Rate Limits	API requests limited to 1000 requests/minute per source (throttle control)
Timeout	Request timeout configured at 30 seconds per API call
Retries	Automatic retries (up to 3) on gateway or timeout failures
Message Logging	All outbound and inbound interactions logged with timestamp and transaction ID
Failure Alerts	Failures logged to ELK stack and alerted via Prometheus/Grafana system monitors
Encryption	HTTPS (TLS 1.3); Payload-level AES-256 encryption where applicable
Validation Protocols	Schema validation (XSD/JSON Schema), signature verification (for DigiLocker/eSign)

Interface Data Format & Mapping Standards

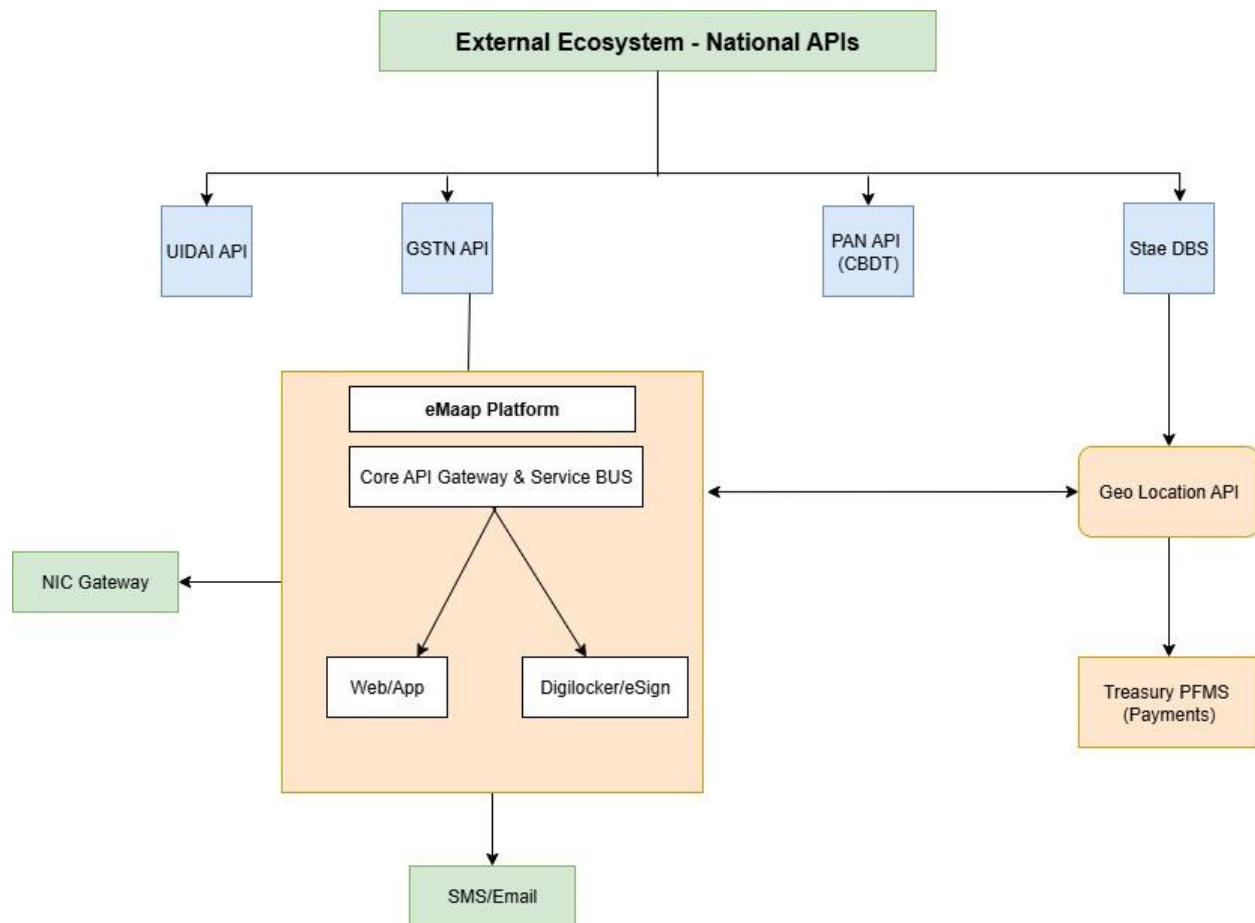
Standard	Usage Area	Format
JSON (REST API)	UIDAI, GSTIN, State API, Location APIs	Key-value pairs, UTF-8
XML	DigiLocker, eSign, PAN	XSD-based validated
PDF (Document Upload)	Certificates, Licenses	Base64 encoded via API
CSV (Export Reports)	MIS Dashboards	UTF-8 formatted

QR Code (Verification)

Generated for each
certificate/license

ISO/IEC 18004

Graphical Representation of Interface Connectivity



Interface Control Document (ICD) References

External System	ICD Reference	Responsible Agency
UIDAI Aadhaar Interface	ICD-UIDAI-eMaap-INTF-2025-V1	NIC Aadhaar Team
GSTIN Verification API	ICD-GSTN-eMaap-V1.1	NIC GST Cell

PAN/CBDT Interface	ICD-CBDT-PAN-INTF-2025	NIC PAN Team
DigiLocker Integration	ICD-DLocker-eMaap-V2.0	MeitY DigiLocker Division
State LM API	ICD-StateLM-Connect-eMaap-2025	NIC + State Nodal LM Departments
eSign Digital Signing Service	ICD-eSign-eMaap-INTF-V1.3	eMudhra / CCA-accredited vendor
SMS/Email Gateway	NIC-GW-INTF-SPEC-2024	NIC Messaging Division
Geo Location API	ICD-GIS-Location-eMaap	NIC GIS Division
Treasury Payment Integration	ICD-Treasury-Finance-eMaap	State IFMS/PFMS

Appendix A: Acronyms

Acronym	Full Form / Literal Translation
ABAC	Attribute-Based Access Control
API	Application Programming Interface
BIS	Bureau of Indian Standards
BPMN	Business Process Model and Notation
CBDT	Central Board of Direct Taxes
CI/CD	Continuous Integration / Continuous Deployment
DR	Disaster Recovery
DSC	Digital Signature Certificate
ELK	Elasticsearch, Logstash, and Kibana
ERD	Entity-Relationship Diagram
FRS	Functional Requirement Specification
GIS	Geographic Information System
Gol	Government of India
GPS	Global Positioning System
GST	Goods and Services Tax
GSTIN	Goods and Services Taxpayer Identification Number
HLD	High-Level Design

HSM	Hardware Security Module
HTML	Hypertext Markup Language
HTTP(S)	Hypertext Transfer Protocol (Secure)
IAM	Identity and Access Management
ICD	Interface Control Document
IDS/IPS	Intrusion Detection System / Intrusion Prevention System
IT	Information Technology
JSON	JavaScript Object Notation
JWT	JSON Web Token
K8s	Kubernetes
LAN	Local Area Network
LM	Legal Metrology
MFA	Multi-Factor Authentication
MIS	Management Information System
NIC	National Informatics Centre
OTP	One-Time Password
PAN	Permanent Account Number
PDF	Portable Document Format
PFMS	Public Financial Management System
PII	Personally Identifiable Information
PKI	Public Key Infrastructure
QoS	Quality of Service
RBAC	Role-Based Access Control
RDBMS	Relational Database Management System
REST	Representational State Transfer
RPO	Recovery Point Objective
RTO	Recovery Time Objective
SIEM	Security Information and Event Management
SLA	Service Level Agreement
SMS	Short Message Service
SPOF	Single Point of Failure
SWAN	State Wide Area Network
TDE	Transparent Data Encryption
TLS	Transport Layer Security
UI	User Interface
UIDAI	Unique Identification Authority of India
URL	Uniform Resource Locator
UAT	User Acceptance Testing
WAF	Web Application Firewall

WAN	Wide Area Network
XML	extensible Markup Language

Appendix B: Glossary

Term	Definition
2FA (Two-Factor Authentication)	A security mechanism requiring two types of credentials for authentication—typically a password and an OTP.
API (Application Programming Interface)	A set of protocols and tools for building software and enabling communication between systems.
Audit Trail	A secure and tamper-proof record of all user and system actions for compliance and traceability.
Blockchain Certificate	Digitally issued certificate recorded on a blockchain ledger to ensure integrity and non-repudiation.
Controller (Legal Metrology)	Senior officer overseeing state-level implementation of Legal Metrology laws.
Dashboard	A role-based UI panel displaying relevant metrics, tasks, and notifications.
DigiLocker	A digital document storage service provided by MeitY for storing verified digital certificates.
Disaster Recovery (DR)	A system design strategy ensuring service continuity during catastrophic failure.
eSign	A digital signature service that allows users to sign documents electronically using Aadhaar or other methods.
Geo-tagging	The process of adding geographic information (latitude, longitude) to digital records such as inspection logs.
Grievance Redressal	Module for submitting, tracking, and resolving user complaints or system-related issues.
Inspector	An authorized official who conducts field inspections and verifications under Legal Metrology Act.
License Management	Workflow enabling application, approval, renewal, or cancellation of stakeholder licenses.
Metrology	The science of measurement; Legal Metrology refers to measurements used for trade and regulation.
OTP (One-Time Password)	A time-sensitive, system-generated numeric code used for verifying user identity.
RBAC (Role-Based Access Control)	A security model that restricts system access to authorized users based on assigned roles.

Stakeholder	Any individual or organization applying for, holding, or interacting with legal metrology services.
UIDAI (Unique Identification Authority of India)	Authority responsible for Aadhaar-based identity verification services.
Workflow Engine	A backend system that automates and tracks the flow of processes such as approvals, inspections, and rejections.

Appendix C: Referenced Documents

This design document was developed with reference to the following key documents and frameworks. These sources guided the development of requirements, architecture, security models, and interoperability standards.

Document Title	Reference/Version	Purpose/Relation
Legal Metrology Act, 2009	Government of India	Primary statutory framework defining compliance requirements for weights and measures.
FRS for eMaap Platform	NIC, Version 5.0, July 2025	Served as the source for functional and non-functional requirements.
UIDAI API Integration Guidelines	UIDAI Developer Portal	Used for Aadhaar-based eKYC integration specifications.
GSTN Verification API Documentation	GST Network	Provided standards for validating GSTIN and related business identifiers.
PAN Validation API Specification	CBDT/NSDL	Used for PAN-based business and individual identity verification.
NIC SMS & Email Gateway Specifications	NIC, 2024	Defined integration methods for sending OTPs and alerts through national gateways.
DigiLocker Integration Guide	MeitY	Provided standards for secure storage and retrieval of official documents.
eSign Integration Guide (CCA-Compliant)	MeitY/NIC	Specified signing mechanism and authentication flows for digital documents.

OWASP Top 10 Security Standards	OWASP Foundation	Used for identifying and mitigating security vulnerabilities in the application.
ISO/IEC 27001 Standards	International Organization for Standardization	Guided the design of secure information systems and audit mechanisms.
NIC Cloud Hosting Guidelines	NIC, MeghRaj Framework	Provided compliance requirements for cloud deployment and infrastructure security.
IEEE 26514 Software Design Documentation Standard	IEEE	Used for structuring this LLD document in compliance with industry documentation standards.